CCC Course By Syv Solution

Document Information

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Chapter 1: Introduction to Computer

1.0 Introduction

A computer is an electronic device that processes raw data to produce meaningful information. It can perform calculations, store data, retrieve information, and process tasks at incredible speed with high accuracy. In today's digital age, understanding computers is essential for every citizen, especially for government job aspirants

This chapter lays the foundation for the entire CCC course by covering the evolution of computers, basic hardware and software concepts, and modern IT gadgets.

1.1 Objectives

After studying this chapter, you will be able to:

- Understand the evolution and generations of computers
- Identify various IT gadgets and their applications
- Differentiate between hardware and software components
- Explain the functions of CPU, input, output, and memory devices
- Classify different types of software
- Distinguish between open source and proprietary software

1.2 Computer and Latest IT Gadgets

1.2.1 Evolution of Computers & Its Applications

Characteristics of a Computer

Characteristic	Description
Speed	Can process millions of instructions per second
Accuracy	Produces error-free results if input is correct
Diligence	Never gets tired; can work continuously
Storage	Can store vast amounts of data for long periods
Versatility	Can perform different types of tasks
Automation	Can perform tasks automatically once programmed

Generations of Computers

Generation	Period	Technology Used	Features & Examples
First Generation	1940-1956	Vacuum Tubes	Large size, high power consumption, generated heat. e.g., ENIAC, UNIVAC
Second Generation	1956-1963	Transistors	Smaller, faster, more reliable. e.g., IBM 1401, IBM 1620
Third Generation	1964-1971	Integrated Circuits (ICs)	Multiple transistors on a single chip. e.g., IBM 360, PDP-8

Generation	Period	Technology Used	Features & Examples
Fourth Generation	1971-1980	Microprocessors	Personal computers (PCs) emerged. e.g., Apple II, IBM PC
Fifth Generation	1980-Present	ULSI, AI, Parallel Processing	Artificial Intelligence, Supercomputers, Laptops, Smartphones

Classification of Computers by Size

Type	Description	Examples
Supercomputer	Fastest and most expensive; used for complex scientific calculations	PARAM, CRAY
Mainframe Computer	Large capacity; used by banks, government for centralized processing	IBM Z-series
Minicomputer	Medium-sized; used by medium organizations	PDP-11
Microcomputer	Personal computers; used by individuals	Desktop, Laptop, Tablet

Applications of Computers

Area	Applications
Education	Online learning, digital classrooms, e-books
Healthcare	Patient records, diagnostic tools, telemedicine

Area	Applications
Banking	ATM, online banking, UPI transactions
Government	E-governance, digital services (DigiLocker, UMANG)
Defense	Missile systems, communication, surveillance
Entertainment	Streaming, gaming, digital media

Gadget	Description	Common Applications
Smartphone	Mobile phone with advanced computing capabilities	Calls, messaging, internet browsing, UPI payments, apps
Tablet	Touchscreen device larger than phone but smaller than laptop	Reading, browsing, basic productivity
Laptop	Portable personal computer	Office work, education, programming
Smartwatch	Wearable device with computing features	Fitness tracking, notifications, health monitoring
Digital Camera	Device for capturing images	Photography, video recording
IoT Devices	Internet-connected smart devices	Smart home, smart city applications

Scanner	Device that converts physical documents to digital	Document digitization
Printer	Output device for producing physical copies	Printing documents, photos

1.2.2 IT Gadgets and Their Applications

1.3 Basics of Hardware and Software

1.3.1 Hardware

Hardware refers to the physical components of a computer system that you can see and touch.

1.3.1.1 Central Processing Unit (CPU)

The CPU is the **brain of the computer**. It performs all processing tasks.

```text
┌──────────────────────────────────────┐
│           CPU (Processor)            │
│  ┌──────────────────────────────┐    │
│  │         Control Unit (CU)    │    │
│  │   - Directs operations       │    │
│  │   - Fetches instructions     │    │
│  └──────────────────────────────┘    │
│                                      │
│  ┌──────────────────────────────┐    │
│  │      Arithmetic Logic Unit   │    │
│  │          (ALU)               │    │
│  │   - Performs calculations    │    │
│  │   - Logical comparisons      │    │
│  └──────────────────────────────┘    │
│                                      │
│  ┌──────────────────────────────┐    │
│  │         Memory Unit (MU)     │    │
│  │   - Registers (L1, L2 Cache) │    │
│  └──────────────────────────────┘    │
└──────────────────────────────────────┘
```

Components of CPU:

- **Control Unit (CU):** Directs the flow of data and instructions. It acts like a traffic controller.
- **Arithmetic Logic Unit (ALU):** Performs arithmetic operations (+, -, ×, ÷) and logical operations (>, <, =, AND, OR, NOT).
- **Memory Unit (Registers):** Small, high-speed storage locations inside the CPU.

How CPU Works (Machine Cycle):

1. **Fetch** – Retrieves instruction from memory
2. **Decode** – Translates instruction
3. **Execute** – Performs the operation
4. **Store** – Writes result back to memory

1.3.1.2 Input Devices

Input devices are used to enter data and instructions into the computer.

Device	Description	Common Use
Keyboard	Most common input device with alphanumeric keys	Typing text, commanc
Mouse	Pointing device with buttons	Navigation, selection
Scanner	Converts physical documents to digital images	Scanning photos, docu
Microphone	Captures audio input	Voice recording, calls
Webcam	Captures video input	Video conferencing
Touchscreen	Input by touching the screen	Smartphones, tablets

Device	Description	Common Use
Barcode Reader	Reads barcodes	Retail billing
Biometric Device	Reads fingerprints, iris	Attendance, security

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1.3.1.3 Output Devices

Output devices display or present processed data to the user.

Device	Description	Type
Monitor (VDU)	Displays visual output	Soft Copy Output
Printer	Produces printed output	Hard Copy Output
Speaker	Produces audio output	Soft Copy Output
Projector	Projects display onto a large screen	Soft Copy Output
Plotter	Large-format printer for drawings	Hard Copy Output

Types of Monitors:

- **CRT (Cathode Ray Tube):** Old, bulky technology
- **LCD (Liquid Crystal Display):** Thin, energy-efficient
- **LED (Light Emitting Diode):** Better than LCD, brighter
- **OLED:** Advanced, better color quality

Types of Printers:

Type	Example	Characteristics
Impact	Dot Matrix	Noisy, used for multipart forms
Non-Impact	Inkjet	Quiet, good quality, affordable
Non-Impact	Laser	Fast, high quality, expensive

1.3.1.4 Computer Memory & Storage

```text
                    COMPUTER MEMORY HIERARCHY
                          ┌──────────────┐
                          │  Registers   │ ← Fastest, smallest, most expensive
                          │   (CPU)      │
                          └──────┬───────┘
                                 │
                          ┌──────┴───────┐
                          │   Cache      │
                          │  (L1, L2)    │
                          └──────┬───────┘
                                 │
                          ┌──────┴───────┐
                          │   RAM        │ ← Primary Memory (Volatile)
                          │ (Main Mem)   │
                          └──────┬───────┘
                                 │
                          ┌──────┴───────┐
                          │  Secondary   │ ← Non-Volatile
                          │  Storage     │   (HDD, SSD, etc.)
                          └──────────────┘
```

Primary Memory (Volatile – Data lost when power is off)

Type	Description	Characteristics
RAM (Random Access Memory)	Temporary storage for running programs and data	Volatile, read/write, faster
ROM (Read Only Memory)	Permanent storage for boot instructions	Non-volatile, read-only
Cache Memory	High-speed memory between CPU and RAM	Very fast, expensive

Types of RAM:

- **DRAM (Dynamic RAM):** Requires constant refreshing; used as main memory
- **SRAM (Static RAM):** Faster, used for cache; does not need refreshing

Types of ROM:

- **PROM:** Programmable once
- **EPROM:** Erasable with UV light
- **EEPROM:** Electrically erasable

Secondary Memory (Non-Volatile – Data retained permanently)

Device	Description	Capacity	Use
Hard Disk Drive (HDD)	Magnetic storage	500 GB – 10 TB	Main storage in PCs
Solid State Drive (SSD)	Flash-based storage	128 GB – 4 TB	Faster than HDD
Pen Drive / USB Flash Drive	Portable flash storage	8 GB – 1 TB	Data transfer

Device	Description	Capacity	Use
Memory Card	Small flash storage	4 GB – 512 GB	Cameras, phones
Optical Discs	CD, DVD, Blu-ray	700 MB – 50 GB	Movies, software

Units of Memory:

Unit	Size
1 Bit	0 or 1
1 Nibble	4 Bits
1 Byte	8 Bits
1 Kilobyte (KB)	1024 Bytes
1 Megabyte (MB)	1024 KB
1 Gigabyte (GB)	1024 MB
1 Terabyte (TB)	1024 GB
1 Petabyte (PB)	1024 TB

1.3.2 Software

Software is a set of instructions that tells the hardware what to do. It is intangible.

```text
                        SOFTWARE
                           │
           ┌───────────────┼───────────────┐
           │               │               │
    System Software   Application     Utility
                       Software       Software
           │               │               │
    ┌──────┴──────┐  ┌──────┴──────┐  ┌──────┴──────┐
    │ OS, Drivers │  │ MS Word,    │  │ Antivirus,  │
    │ Compilers   │  │ Excel, etc  │  │ Disk Clean  │
    └─────────────┘  └─────────────┘  └─────────────┘
```

1.3.2.1 Application Software

Category	Examples	Purpose
Word Processing	MS Word, LibreOffice Writer	Creating documents
Spreadsheet	MS Excel, LibreOffice Calc	Calculations, data analysis
Presentation	MS PowerPoint, LibreOffice Impress	Creating slideshows
Database	MS Access, MySQL	Managing data
Web Browser	Chrome, Firefox, Edge	Browsing internet
Email Client	Outlook, Thunderbird	Managing emails

Graphics	Photoshop, GIMP	Image editing
Accounting	Tally, QuickBooks	Financial management

Software designed to perform specific tasks for the user.

1.3.2.2 Systems Software

Software that manages and controls hardware resources.

Type	Description	Examples
Operating System (OS)	Manages hardware and provides interface	Windows, Linux, macOS, Android, iOS
Device Drivers	Enables OS to communicate with hardware	Printer driver, graphics driver
Language Processors	Converts programming code to machine language	Compiler, Interpreter, Assembler

Functions of Operating System:

1. Process Management
2. Memory Management
3. File Management
4. Device Management
5. Security and Access Control
6. User Interface (GUI/CLI)

1.3.2.3 Utility Software

System software that helps maintain and optimize the computer.

Utility	Purpose	Examples
Antivirus	Protects from malware, viruses	Quick Heal, Kaspersky, Avast
Disk Cleanup	Removes temporary files	Windows Disk Cleanup, CCleane
Disk Defragmenter	Organizes fragmented files	Windows Defragmenter
Backup Software	Creates data backups	Windows Backup, Acronis
File Compression	Reduces file size	WinRAR, 7-Zip
Firewall	Monitors network traffic	Windows Firewall

1.3.2.4 Open Source and Proprietary Software

Feature	Open Source Software	Proprietary Software
Source Code	Available to users	Kept secret
Cost	Usually free	Paid (license required)
Modification	Can be modified	Cannot be modified
Support	Community-based	Vendor support

Feature	Open Source Software	Proprietary Software
Examples	Linux, LibreOffice, Firefox, GIMP, Android	Windows, MS Office, Adobe Photoshop macOS
Licensing	GPL, MIT, Apache	EULA (End User License Agreement)

Important for UP Exams: CCC exam focuses on **LibreOffice (Open Source)** and **Ubuntu Linux**, not MS Office/Windows exclusively.

1.3.2.5 Mobile Apps

Software applications designed for mobile devices.

App Type	Description	Examples
Native Apps	Built for specific OS	Android Apps (Java/Kotlin), iOS Apps (Swift)
Web Apps	Accessed via browser	Gmail, Facebook mobile site
Hybrid Apps	Combination of both	Instagram, Uber

App Distribution:

- **Google Play Store:** For Android devices
- **Apple App Store:** For iOS devices
- **Other:** Amazon Appstore, Samsung Galaxy Store

Important Government Apps for UP Candidates:

- **UMANG:** Unified Mobile App for New-age Governance

- **DigiLocker:** Digital document storage
- **BHIM:** UPI payments
- **Aarogya Setu:** Health monitoring
- **ePathshala:** Educational resources

1.4 Summary

- A computer is an electronic device that processes data into information with speed, accuracy, and storage capabilities.
- Computers have evolved through five generations from vacuum tubes to modern AI-based systems.
- **Hardware** includes CPU (Control Unit, ALU, Memory Unit), input devices (keyboard, mouse, scanner), output devices (monitor, printer), and memory (RAM, ROM, HDD, SSD).
- **Software** is classified into System Software (OS, drivers), Application Software (word processors, spreadsheets), and Utility Software (antivirus, disk tools).
- **Open Source Software** provides source code and is usually free (Linux, LibreOffice), while **Proprietary Software** requires licensing (Windows, MS Office).
- Modern IT gadgets include smartphones, tablets, smartwatches, and IoT devices.

1.5 Model Questions and Answers

Multiple Choice Questions (MCQs)

1. Which of the following is the brain of the computer?

- a) Monitor
- b) Keyboard
- c) CPU

- d) Printer

Answer: c) CPU

---

2. Which generation of computers used transistors?

- a) First Generation
- b) Second Generation
- c) Third Generation
- d) Fourth Generation

Answer: b) Second Generation

---

3. Which memory is volatile?

- a) ROM
- b) HDD
- c) RAM
- d) Pen Drive

Answer: c) RAM

---

4. Which of the following is an example of system software?

- a) MS Word
- b) Windows 11
- c) Tally
- d) Photoshop

Answer: b) Windows 11

---

5. 1 GB is equal to:

- a) 1000 KB
- b) 1024 KB
- c) 1024 MB
- d) 1000 MB

Answer: c) 1024 MB

---

6. Which device is used for hard copy output?

- a) Monitor
- b) Printer
- c) Speaker
- d) Projector

Answer: b) Printer

---

7. Which of the following is open source software?

- a) Microsoft Windows
- b) Adobe Photoshop
- c) LibreOffice
- d) MS Office

Answer: c) LibreOffice

---

8. The full form of RAM is:

- a) Random Access Memory
- b) Readily Available Memory
- c) Read Access Memory
- d) Random Available Memory

Answer: a) Random Access Memory

---

9. Which component of CPU performs calculations?

- a) CU
- b) MU
- c) ALU
- d) Cache

Answer: c) ALU

---

10. Which of the following is an input device?

- a) Monitor
- b) Printer
- c) Scanner
- d) Speaker

Answer: c) Scanner

True/False Questions

Statement	Answer
ROM is volatile memory.	**False** (ROM is non-volatile)
The Control Unit (CU) manages the operations of the computer.	**True**
Linux is an example of proprietary software.	**False** (Linux is open source)
A printer produces soft copy output.	**False** (Printer produces hard c
1 KB equals 1024 bytes.	**True**

Short Answer Questions

Q1: What are the characteristics of a computer?
Ans: Speed, Accuracy, Diligence, Storage, Versatility, Automation.

Q2: Differentiate between RAM and ROM.
Ans: RAM is volatile, read/write, temporary storage. ROM is non-volatile, read-only, permanent storage.

Q3: What is the difference between system software and application software?
Ans: System software manages hardware resources (e.g., OS). Application software performs specific user tasks (e.g., MS Word).

Q4: Define open source software with examples.
Ans: Software with source code available for modification and distribution, usually free. Examples: Linux, LibreOffice, Firefox.

Q5: Name four input and four output devices.

Ans: Input: Keyboard, Mouse, Scanner, Microphone. Output: Monitor, Printer, Speaker, Projector.

Practical Lab Exercises

Exercise No.	Task	Purpose
1	Identify and label all hardware components on a desktop computer	Recognize CPU, RAM, HDD, motherboard, input/output devices
2	Identify and label all ports (USB, HDMI, VGA, Ethernet, audio) on a computer	Understand connectivity
3	Check system properties (Windows: System Information; Linux: System Monitor)	View installed RAM, processor type, OS version
4	Identify software installed on a lab computer (categorize as system/application/utility)	Differentiate software types
5	Identify open source software from the installed list	Recognize open source vs proprietary

---

Important Points for UP CCC Exam

Topic	Key Points to Remember
Generations	1st: Vacuum Tubes, 2nd: Transistors, 3rd: IC, 4th: Microprocessor, 5th: AI
CPU Components	CU (Control), ALU (Arithmetic/Logical), MU (Memory)
Memory Units	1 Byte = 8 Bits, 1 KB = 1024 Bytes, 1 MB = 1024 KB, 1 GB = 1024 MB
Open Source	Linux, Ubuntu, LibreOffice, Firefox, GIMP, Android
Proprietary	Windows, MS Office, Adobe, macOS
Input Devices	Keyboard, Mouse, Scanner, Microphone, Webcam, Barcode Reader
Output Devices	Monitor, Printer, Speaker, Projector, Plotter

Chapter 2: Introduction to Operating System

2.0 Introduction

An Operating System (OS) is the most essential software that acts as an intermediary between the computer hardware and the user. Without an operating system, a computer is just a collection of electronic components that cannot perform any useful task. The OS manages all hardware resources, provides a user interface, and allows applications to run smoothly.

For CCC exam candidates, it is crucial to understand both **Windows** and **Ubuntu Linux** operating systems, as the exam covers both.

2.1 Objectives

After studying this chapter, you will be able to:

- Understand the basic concepts and functions of an operating system
- Differentiate between desktop/laptop and mobile operating systems
- Navigate the user interface including taskbar, icons, and shortcuts
- Perform basic system settings like date/time, display, and printer configuration
- Manage files and folders efficiently
- Identify common file extensions and their associated programs

---

2.2 Operating System

2.2.1 Basics of Operating System

An **Operating System (OS)** is system software that manages computer hardware and software resources and provides common services for computer programs.

Functions of an Operating System

Function	Description
Process Management	Manages execution of programs, scheduling, and multitasking
Memory Management	Allocates and deallocates memory space to programs
File Management	Organizes, stores, and retrieves files on storage devices
Device Management	Controls input/output devices through drivers
Security & Access Control	Protects data with user authentication and permissions
User Interface	Provides interaction between user and computer (GUI/CLI)
Error Detection	Handles errors and displays appropriate messages

Types of Operating Systems

Type	Description	Examples
Single-User	Designed for one user at a time	Windows, macOS
Multi-User	Allows multiple users to use system simultaneously	Linux, Unix
Multitasking	Runs multiple programs simultaneously	All modern OS
Real-Time	Responds to input instantly	RTOS, VxWorks
Distributed	Manages multiple computers as one system	Linux clusters
Network OS	Manages network resources	Windows Server

2.2.2 Operating Systems for Desktop and Laptop

OS Name	Developer	Type	Key Features
Microsoft Windows	Microsoft	Proprietary	Most popular, user-friendly GUI, extensive software support
Linux (Ubuntu, Fedora, etc.)	Open Source Community	Open Source	Free, secure, customizable, command-line powerful
macOS	Apple	Proprietary	Sleek design, exclusive to Apple hardware

OS Name	Developer	Type	Key Features
Chrome OS	Google	Proprietary	Lightweight, cloud-based, Chromebooks

For CCC Exam: Focus on **Windows 10/11** and **Ubuntu Linux** (as NIELIT uses Ubuntu in labs).

2.2.3 Operating Systems for Mobile Phone and Tablets

OS Name	Developer	Key Features
Android	Google (Open Source)	Most widely used, customizable, Google Play Store
iOS	Apple	Exclusive to Apple devices, highly secure, App Store
HarmonyOS	Huawei	Emerging alternative to Android
KaiOS	KaiOS Technologies	For feature phones with smart capabilities

---

2.3 User Interface for Desktop and Laptop

The **User Interface (UI)** is how users interact with the operating system. There are two types:

Type	Full Form	Description	Examples
GUI	Graphical User Interface	Uses icons, windows, menus	Windows, Ubuntu Desktop, macOS
CLI	Command Line Interface	Uses text commands	Command Prompt, Terminal, PowerShell

---

2.3.1 Task Bar

The **Taskbar** is a horizontal bar typically located at the bottom of the screen in Windows and at the top in Ubuntu.

Windows Taskbar Components

```text
┌──────────────────────────────────────────────────────────────────────────────┐
│ [Start] [Search] [Task View] [Pin Icons] ... [System Tray] [Date/Time] │
│   Button   Box                 [Apps]                      [Notification] │
└──────────────────────────────────────────────────────────────────────────────┘
```

Component	Description
Start Button	Opens Start Menu to access programs, settings, and power options
Search Box/Cortana	Search for files, apps, or web content
Task View	Shows all open windows and virtual desktops
Pinned Applications	Quick access to frequently used apps
Running Apps	Displays currently open applications

Component	Description
System Tray	Icons for background apps (antivirus, network, volume)
Notification Area	Shows notifications and system alerts
Date & Time	Displays current date and time

Ubuntu Taskbar (GNOME Desktop)

```text
┌──────────────────────────────────────────────────────────────────────────────┐
│ [Activities] [Dock Icons] ... [System Menu] [Network] [Volume] [Date/Time]│
└──────────────────────────────────────────────────────────────────────────────┘
```

Component	Description
Activities	Opens overview to search, view windows, and launch apps
Dock	Icons for favorite and running applications
System Menu	Access settings, lock screen, power options
Top Bar	Contains system indicators and date/time

2.3.2 Icons & Shortcuts

Icons

Icons are small graphical representations of files, folders, applications, or system functions.

Icon Type	Description	Examples
Application Icons	Launch programs	Word, Chrome, Firefox
Folder Icons	Represent directories	Documents, Downloads
File Icons	Represent specific files	.txt, .jpg, .pdf
Shortcut Icons	Point to original file location (has arrow)	Desktop shortcuts
System Icons	System functions	Recycle Bin, Computer

Keyboard Shortcuts (Important for Lab)

Shortcut	Windows	Ubuntu	Purpose
Ctrl + C	Copy	Copy	Copy selected item
Ctrl + X	Cut	Cut	Cut selected item
Ctrl + V	Paste	Paste	Paste copied/cut item
Ctrl + Z	Undo	Undo	Undo last action
Ctrl + A	Select All	Select All	Select all items
Ctrl + S	Save	Save	Save current file
Ctrl + P	Print	Print	Print document
Alt + F4	Close Window	Close Window	Close active window

Shortcut	Windows	Ubuntu	Purpose
Alt + Tab	Switch Apps	Switch Apps	Switch between open apps
Windows/Super Key	Open Start Menu	Open Activities	Open application menu
F2	Rename	Rename	Rename selected item
Delete	Delete	Move to Trash	Delete selected item
Shift + Delete	Permanent Delete	(Different)	Permanently delete (skip recycle bin)

2.3.3 Running an Application

Methods to Start an Application

Method	Windows	Ubuntu
Desktop Icon	Double-click icon	Double-click icon
Start Menu	Click Start → Select app	Click Activities → Search/Select app
Taskbar/Dock	Click pinned icon	Click icon in dock
Run Dialog	Windows + R → Type program name	Alt + F2 → Type program name
Command Line	Open Command Prompt → type program	Open Terminal → type program

Common Applications to Practice

Application	Windows	Ubuntu (LibreOffice)
Word Processor	Microsoft Word	LibreOffice Writer
Spreadsheet	Microsoft Excel	LibreOffice Calc
Presentation	Microsoft PowerPoint	LibreOffice Impress
Web Browser	Microsoft Edge / Chrome	Firefox / Chrome
File Manager	File Explorer	Files (Nautilus)

2.4 Operating System Simple Settings

2.4.1 Using Mouse and Changing its Properties

Mouse Operations

Operation	Description
Click	Press and release left button once (select)
Double-Click	Two quick clicks (open file/folder)
Right-Click	Press right button (context menu)
Drag & Drop	Press, hold, move, release (move/copy)

Operation	Description
Scroll	Use scroll wheel to move up/down
Hover	Point without clicking (tooltip appears)

Change Mouse Properties in Windows

Step	Action
1	Open **Settings** (Windows + I)
2	Click **Bluetooth & Devices** → **Mouse**
3	Adjust: Primary button (left/right), scroll speed, pointer size, pointer speed
4	For advanced: Control Panel → Hardware and Sound → Devices and Printers → Mouse

Change Mouse Properties in Ubuntu

Step	Action
1	Click **Activities** → Search **Settings**
2	Click **Mouse & Touchpad**
3	Adjust: Primary button, mouse speed, natural scrolling, acceleration

---

2.4.2 Changing System Date and Time

Windows

Step	Action
1	Right-click on **Date & Time** in taskbar → Select **Adjust date/time**
2	OR Open **Settings** → **Time & Language** → **Date & time**
3	**Set time automatically** – Toggle ON (recommended)
4	**Set time zone automatically** – Toggle ON
5	To set manually: Toggle OFF automatic → Click **Change** under "Set the date and time manuall
6	Click **Sync now** to synchronize with internet time server

Ubuntu

Step	Action
1	Click on **Date & Time** in top bar → Click **Date & Time Settings**
2	OR Open **Settings** → **Date & Time**
3	**Automatic Date & Time** – Toggle ON
4	**Automatic Time Zone** – Toggle ON
5	To set manually: Toggle OFF automatic → Click date/time to change

2.4.3 Changing Display Properties

Windows

Setting	Steps
Background/Wallpaper	Settings → Personalization → Background → Choose picture/solid color/slideshow
Display Resolution	Settings → System → Display → Display resolution → Select recommended
Screen Brightness	Settings → System → Display → Brightness slider
Themes	Settings → Personalization → Themes → Select or create theme
Text Size	Settings → Accessibility → Text size → Adjust slider
Night Light	Settings → System → Display → Night light → Schedule

Ubuntu

Setting	Steps
Background/Wallpaper	Settings → Appearance → Background → Choose image/color
Display Resolution	Settings → Displays → Resolution → Select
Screen Brightness	Settings → Power → Brightness slider (laptop)
Themes	Settings → Appearance → Style (Light/Dark)

Setting	Steps
Night Light	Settings → Displays → Night Light → Schedule

2.4.4 To Add or Remove Program and Features

Windows – Add/Remove Programs

Method	Steps
Settings Method	Settings → Apps → Apps & features → Search for app → Click Uninstall
Control Panel Method	Control Panel → Programs → Programs and Features → Select app → Unin

Windows – Install Programs

Method	Steps
Microsoft Store	Open Microsoft Store → Search app → Click Install
Installer File	Download .exe/.msi file → Double-click → Follow wizard

Ubuntu – Add/Remove Programs

Method	Steps
Ubuntu Software (GUI)	Open Ubuntu Software → Search app → Click Install/Remove

Method	Steps
Terminal (apt)	`sudo apt update` (refresh package list) → `sudo apt install <package-name>` → `apt remove <package-name>`

2.4.5 Adding, Removing & Sharing Printers

Windows – Add Printer

Step	Action
1	Open **Settings** → **Bluetooth & devices** → **Printers & scanners**
2	Click **Add device**
3	Windows searches for available printers
4	If not found, click **Add manually** → Follow wizard (USB, Network, Wireless)

Windows – Remove Printer

Step	Action
1	Settings → Bluetooth & devices → Printers & scanners
2	Click on printer → Click **Remove** → Confirm

Windows – Share Printer

Step	Action
1	Settings → Bluetooth & devices → Printers & scanners
2	Click on printer → Click **Printer properties**
3	Click **Sharing** tab
4	Check **Share this printer** → Enter share name → Click OK

Ubuntu – Add Printer

Step	Action
1	Open **Settings** → **Printers**
2	Click **Add Printer**
3	Select detected printer or enter network printer address
4	Click **Add** → Install driver if prompted

2.5 File and Folder Management

Basic Concepts

Term	Definition
File	A collection of related data stored on a storage device (document, image, progran
Folder	A container that holds files and other folders (also called directory)
Root Directory	The top-most directory (C:\ in Windows, / in Linux)
Path	The location of a file/folder (e.g., C:\Users\Name\Documents)
Parent Folder	The folder that contains the current folder
Subfolder	A folder inside another folder

File Explorer / Files Manager

Windows	Ubuntu
File Explorer (Windows + E)	Files (Nautilus)
Displays drives: C:, D:, etc.	Displays: Home, Documents, Downloads, etc.
This PC shows all drives	Computer shows all drives

File and Folder Operations

Operation	Windows Steps	Ubuntu Steps
Create Folder	Right-click → New → Folder → Name it	Right-click → New Folder → Name it
Rename	Right-click → Rename → Type new name	Right-click → Rename → Type new name
Copy	Right-click → Copy (Ctrl+C)	Right-click → Copy (Ctrl+C)
Cut	Right-click → Cut (Ctrl+X)	Right-click → Cut (Ctrl+X)
Paste	Right-click → Paste (Ctrl+V)	Right-click → Paste (Ctrl+V)
Delete	Right-click → Delete (Delete key)	Right-click → Move to Trash (Delete)
Permanent Delete	Shift + Delete	(Use Terminal: `rm` command)
Restore from Trash	Open Recycle Bin → Right-click → Restore	Open Trash → Right-click → Restore
Select All	Ctrl + A	Ctrl + A
Select Multiple	Ctrl + Click (individual), Shift + Click (range)	Ctrl + Click, Shift + Click
Properties	Right-click → Properties	Right-click → Properties
Search	Search box in File Explorer	Search icon or Ctrl + F

File Organization Best Practices

```text
Recommended Folder Structure:
C:\Users\YourName\
    ├── Documents\
    │   ├── Office\
    │   ├── Personal\
    │   └── Projects\
    ├── Downloads\
    ├── Desktop\
    ├── Music\
    ├── Pictures\
    └── Videos\
```

2.6 Types of File Extensions

A **file extension** is a suffix at the end of a filename (after a dot) that indicates the file type and which program can open it.

Common File Extensions

Category	Extension	File Type	Associated Program
Text/Document	.txt	Plain Text	Notepad, gedit
	.doc, .docx	Word Document	MS Word, LibreOffice Writer
	.pdf	Portable Document Format	Adobe Reader, PDF viewer
	.odt	OpenDocument Text	LibreOffice Writer
	.rtf	Rich Text Format	WordPad, Writer

Category	Extension	File Type	Associated Program
Spreadsheet	.xls, .xlsx	Excel Spreadsheet	MS Excel, LibreOffice Calc
	.ods	OpenDocument Spreadsheet	LibreOffice Calc
	.csv	Comma Separated Values	Excel, Calc
Presentation	.ppt, .pptx	PowerPoint Presentation	MS PowerPoint, LibreOffice Impress
	.odp	OpenDocument Presentation	LibreOffice Impress
Image	.jpg, .jpeg	JPEG Image	Photo viewer, browsers
	.png	PNG Image	Photo viewer
	.gif	GIF Image	Photo viewer
	.bmp	Bitmap Image	Paint
Audio	.mp3	MP3 Audio	Music player
	.wav	Wave Audio	Windows Media Player
Video	.mp4	MP4 Video	Video player
	.avi	AVI Video	Video player

Category	Extension	File Type	Associated Program
Compressed	.zip	ZIP Archive	WinRAR, 7-Zip
	.rar	RAR Archive	WinRAR
Executable	.exe	Executable File	Windows application
	.msi	Windows Installer	Installer package
	.deb	Debian Package	Ubuntu software installer
System	.dll	Dynamic Link Library	System component
	.sys	System File	Windows system file

File Extensions in Ubuntu (Linux)

Extension	Purpose
.sh	Shell script
.py	Python script
.deb	Debian package (installer)
.conf	Configuration file
.log	Log file

2.7 Summary

- An **Operating System** is system software that manages hardware resources and provides user interface.
- **Desktop OS:** Windows, Ubuntu Linux, macOS; **Mobile OS:** Android, iOS.
- **GUI** uses icons and windows; **CLI** uses text commands.
- **Taskbar** (Windows) and **Dock/Top Bar** (Ubuntu) provide quick access to applications.
- **Keyboard shortcuts** improve productivity (Ctrl+C, Ctrl+V, Alt+Tab, etc.).
- **System settings** include mouse properties, date/time, display, programs, and printers.
- **File management** includes creating, copying, moving, renaming, and deleting files/folders.
- **File extensions** identify file types and associated programs (.docx, .pdf, .jpg, .exe, etc.).

2.8 Model Questions and Answers

Multiple Choice Questions (MCQs)

1. Which of the following is not an operating system?

- a) Windows 11
- b) Ubuntu Linux
- c) Microsoft Office
- d) Android

Answer: c) Microsoft Office (It is application software)

---

2. The shortcut key to copy a selected file is:

- a) Ctrl + X
- b) Ctrl + C
- c) Ctrl + V
- d) Ctrl + Z

Answer: b) Ctrl + C

---

3. Which component of the taskbar displays background applications?

- a) Start Button
- b) System Tray
- c) Task View
- d) Search Box

Answer: b) System Tray

---

4. What is the file extension for a Microsoft Word document?

- a) .xlsx
- b) .pptx
- c) .docx
- d) .pdf

Answer: c) .docx

---

5. Which key is used to rename a selected file in Windows?

- a) F1
- b) F2
- c) F3
- d) F4

Answer: b) F2

---

6. The shortcut to switch between open applications is:

- a) Ctrl + Tab
- b) Alt + Tab
- c) Windows + Tab
- d) Shift + Tab

Answer: b) Alt + Tab

---

7. Which of the following is an open source operating system?

- a) Windows 11
- b) macOS
- c) Ubuntu Linux
- d) iOS

Answer: c) Ubuntu Linux

---

8. The permanent deletion of a file (skipping Recycle Bin) in Windows is done by:

- a) Delete key
- b) Ctrl + Delete
- c) Shift + Delete
- d) Alt + Delete

Answer: c) Shift + Delete

---

9. Which file extension is used for compressed files?

- a) .exe
- b) .zip
- c) .dll
- d) .sys

Answer: b) .zip

---

10. In Ubuntu, which application is used to manage files?

- a) File Explorer
- b) Finder
- c) Files (Nautilus)
- d) Dolphin

Answer: c) Files (Nautilus)

True/False Questions

Statement	Answer
The operating system is application software.	**False** (It is system software)
Android is an operating system for mobile devices.	**True**
Ctrl + V is used to paste copied content.	**True**
.jpg files contain audio data.	**False** (Images)
The Recycle Bin permanently deletes files.	**False** (Temporary deletion)

Short Answer Questions

Q1: What are the main functions of an operating system?
Ans: Process management, memory management, file management, device management, security, user interface, error detection.

Q2: Differentiate between GUI and CLI.
Ans: GUI uses graphical icons and windows; CLI uses text commands. GUI is easier for beginners; CLI is faster for advanced users.

Q3: How do you change the desktop wallpaper in Windows 11?
Ans: Settings → Personalization → Background → Choose picture, solid color, or slideshow.

Q4: What is the difference between cut+paste and copy+paste?
Ans: Cut removes the original file and moves it to new location. Copy creates a duplicate and keeps the original.

Q5: Name four common file extensions and their associated programs.

Ans: .docx (MS Word), .xlsx (MS Excel), .pdf (Adobe Reader), .jpg (Photo Viewer)

Practical Lab Exercises

Exercise No.	Task	Purpose
1	Identify and explore Taskbar/Desktop components in Windows and Ubuntu	Understand UI navigation
2	Practice all keyboard shortcuts listed in the chapter	Increase efficiency
3	Launch applications using different methods (Start Menu, Search, Run dialog)	Master app launching
4	Change date/time, mouse properties, display settings	System configuration
5	Install a program (e.g., VLC Media Player) and uninstall it	Software management
6	Add a network printer (if available)	Printer configuration
7	Create folder structure: Documents > Office > CCC; create 5 files in different formats	File management
8	Perform copy, move, rename, delete operations on files/folders	Master file operations
9	Check file properties for different file types	Understand file information
10	Search for a file by name in File Explorer/Files	Searching skills

Important Points for UP CCC Exam

Topic	Key Points to Remember
OS Functions	Process, Memory, File, Device Management, Security
Desktop OS	Windows, Ubuntu, macOS
Mobile OS	Android, iOS
Taskbar	Start, Search, System Tray, Date/Time
Common Shortcuts	Ctrl+C, Ctrl+V, Ctrl+X, Ctrl+Z, Alt+Tab, F2, Shift+Delete
File Extensions	.docx (Word), .xlsx (Excel), .pptx (PowerPoint), .pdf, .jpg, .mp3, .zip
Settings	Date/Time, Display, Programs, Printers
File Management	Create, Copy, Move, Rename, Delete, Restore

Chapter 3: Word Processing

Complete Theory & Lab Notes

3.0 Introduction

Word processing is the creation of text documents using a computer . A word processor is a software application designed to assist with the production of a wide variety of documents, including letters, reports, memoranda, and manuals, rapidly and at relatively low cost . It has long since replaced the electric typewriter, offering unparalleled flexibility in editing, formatting, and document management .

For the CCC exam, you need to be familiar with both **Microsoft Word** (proprietary) and **LibreOffice Writer** (open source), as NIELIT exams often use Ubuntu Linux with LibreOffice .

3.1 Objectives

After studying this chapter, you will be able to:

- Open, create, save, and close word processing documents
- Understand the word processor interface (Title Bar, Menu Bar, Toolbars, Sidebar)
- Perform text creation, editing, and manipulation
- Format text with fonts, colors, styles, and alignment
- Use AutoCorrect, spell check, and grammar tools
- Find and replace text
- Format paragraphs with indentation, bullets, and numbering
- Insert and format tables
- Work with headers, footers, and page setup

- Use mail merge for form letters
- Export documents as PDF files

3.2 Word Processing Basics

3.2.1 Opening Word Processing Package

Microsoft Word

Method	Steps
Start Menu	Click Start → Type "Word" → Click Microsoft Word
Desktop Shortcut	Double-click Word icon on desktop
Taskbar	Click pinned Word icon on taskbar
Run Dialog	Windows + R → Type "winword" → Enter

LibreOffice Writer (Ubuntu/Linux)

Method	Steps
Activities	Click Activities → Type "Writer" → Click LibreOffice Writer
Terminal	Type `libreoffice --writer` and press Enter
Application Menu	Applications → Office → LibreOffice Writer

3.2.2 Title Bar, Menu Bar, Toolbars & Sidebar

```text

┌──────────────────────────────────────────────────────────────────────────────┐
│ [Document1 - LibreOffice Writer]                                    – □ × │ ← Ti
tle Bar
├──────────────────────────────────────────────────────────────────────────────┤
│ File  Edit  View  Insert  Format  Styles  Table  Tools  Window  Help       │ ← M
enu Bar
├──────────────────────────────────────────────────────────────────────────────┤
│ [New] [Open] [Save] [Print] [Cut] [Copy] [Paste] [Undo] [Redo] [B] [I] [U] │ ← S
tandard/Formatting Toolbar
├──────────────────────────────────────────────────────────────────────────────┤
│                                                                              │
│                                                                              │
│                          Document Editing Area                               │
│                                                                              │
│                                                                              │
├──────────────────────────────────────────────────────────────────────────────┤
│ Page: 1 / 1    Words: 0    Language: English (USA)    Insert    STD        │ ← S
tatus Bar
└──────────────────────────────────────────────────────────────────────────────┘
│ [Properties] [Styles] [Gallery] [Navigator] [Page] [Style Inspector]       │ ← S
idebar (right side)
└──────────────────────────────────────────────────────────────────────────────┘
```

Title Bar

- Displays the name of the current document
- Shows "Untitled" or "Unnamed X" for new unsaved documents
- Contains window control buttons (Minimize, Maximize/Restore, Close)

Menu Bar

Contains dropdown menus with commands:

Menu	**Key Functions**
File	New, Open, Save, Save As, Print, Export as PDF, Close

Menu	Key Functions
Edit	Undo, Redo, Cut, Copy, Paste, Find, Replace, Select All
View	Zoom, Toolbars, Sidebar, Print Layout, Web Layout
Insert	Page Break, Table, Picture, Header, Footer, Page Number
Format	Character, Paragraph, Bullets, Page Style, Alignment
Table	Insert Table, Delete, Merge Cells, Split Cells
Tools	Spell Check, AutoCorrect, Word Count, Options
Help	Help documentation, About

Toolbars

- **Standard Toolbar**: Common file operations (New, Open, Save, Print)
- **Formatting Toolbar**: Text formatting options (Font, Size, Bold, Italic, Alignment)

Sidebar (LibreOffice Writer)

The Sidebar is docked to the right side of the Writer window by default.
Press **Ctrl+F5** to display it if hidden .

Deck	Purpose
Properties	Direct formatting of content (character, paragraph, table properties)
Styles	Manage and apply paragraph, character, and page styles
Gallery	Images and diagrams included in Gallery themes

Deck	Purpose
Navigator	Navigate document by headings, tables, frames, images
Page	Modify page size, margins, headers, footers
Style Inspector	Display attributes of applied styles

Status Bar

The Status bar at the bottom provides document information :

Element	Information
Page Number	Current page / Total pages
Word Count	Number of words and characters in document
Language	Current language for spell checking
Insert/Overwrite	Indicates Insert mode (default) or Overwrite mode
Zoom Slider	Adjust document magnification

3.2.3 Creating a New Document

Microsoft Word

Method	Steps
Blank Document	File → New → Blank Document
Template	File → New → Choose from available templates (resumes, reports, letters)
Keyboard Shortcut	Ctrl + N
Quick Access Toolbar	Click New icon (blank page)

LibreOffice Writer

Method	Steps
Blank Document	File → New → Text Document
Template	File → New → Templates → Select template
Wizard	File → Wizards → Choose wizard type (letter, fax, etc.)
Keyboard Shortcut	Ctrl + N
From Scratch	File → New → Text Document

3.3 Opening and Closing Documents

3.3.1 Opening Documents

Microsoft Word

Method	Steps
File Menu	File → Open → Browse to location → Select file → Open
Recent Documents	File → Open → Recent → Select from list
Keyboard Shortcut	Ctrl + O
Double-click File	Double-click document file in File Explorer

LibreOffice Writer

Method	Steps
File Menu	File → Open → Browse → Select file → Open
Recent Documents	File → Recent Documents → Select file
Keyboard Shortcut	Ctrl + O

3.3.2 Save and Save As

Save (First Time / Existing Document)

Operation	Microsoft Word	LibreOffice Writer
Save New Document	File → Save As → Choose location → Enter filename → Save	File → Save As → Choose location → Enter filename → Save
Save Existing	File → Save (or Ctrl+S)	File → Save (or Ctrl+S)
Save Icon	Click disk icon on Quick Access Toolbar	Click disk icon on Standard Toolbar

Save As (Create Copy or Change Format)

Use **Save As** when you want to:

- Save a document with a different name
- Save in a different location
- Save in a different file format

Format	Extension	When to Use
Word Document	.docx	Default Word format (Microsoft 365/2019/2016)
Word 97-2003 Document	.doc	For compatibility with older Word versions
OpenDocument Text	.odt	Default LibreOffice Writer format
PDF	.pdf	For sharing final documents (cannot be easily edited)

3.3.3 Closing Document

Method	Steps
File Menu	File → Close
Close Button	Click × on document tab or window
Keyboard Shortcut	Ctrl + W (Word) or Ctrl+F4 (Writer)
Alt + F4	Close entire application

3.3.4 Using The Help

Microsoft Word Help

Method	Steps
F1 Key	Press F1 → Search for topic
Tell Me Box	Type question in "Tell me what you want to do" box on ribbon
Help Menu	File → Help → Help

LibreOffice Writer Help

Method	Steps
F1 Key	Press F1 → Open Help window
Help Menu	Help → LibreOffice Help

Method	Steps
What's This?	Help → What's This? (Shift+F1) then click on any item

---

3.3.5 Page Setup

Page setup controls how your document appears on printed pages.

Page Setup Options

Option	Description
Margins	Space between text and edge of paper (Top, Bottom, Left, Right)
Orientation	Portrait (vertical) or Landscape (horizontal)
Paper Size	A4, Letter, Legal, etc.
Columns	Number of columns for text layout

Microsoft Word

Steps

Layout → Margins → Select preset or Custom Margins

Layout → Orientation → Portrait or Landscape

Layout → Size → Select paper size

LibreOffice Writer

Steps

Format → Page Style → Page tab

Set margins, orientation, paper size

Format → Columns for multiple column layout

3.3.6 Print Preview

Print Preview shows how the document will look when printed before actually printing .

Microsoft Word

Method	**Steps**
File Menu	File → Print (Print Preview appears on right side)
Keyboard	Ctrl + F2 or Ctrl + P
Quick Access	Add Print Preview to Quick Access Toolbar

LibreOffice Writer

Method	**Steps**
File Menu	File → Print Preview

Method	Steps
Toolbar	Click Print Preview icon
Keyboard	Ctrl + Shift + O

3.3.7 Printing of Documents

Print Settings

Setting	Purpose
Printer	Select which printer to use
Copies	Number of copies to print
Page Range	All, Current page, Pages (e.g., 1-3, 5, 7-9)
Collate	Print multiple copies in order
One-sided/Duplex	Print on one side or both sides

Microsoft Word

Steps

File → Print (or Ctrl + P)

Select printer, set copies, page range

Steps

Click Print

LibreOffice Writer

Steps

File → Print (or Ctrl + P)

Select printer, set copies, page range

Click Print

3.3.8 PDF File and Saving a Document as PDF File

PDF (Portable Document Format) is a file format that preserves document formatting and can be viewed on any device without the original software.

Why Save as PDF?

Advantage	Description
Universal	Can be opened on any computer/smartphone
Fixed Layout	Formatting remains consistent regardless of device
Security	Can be password protected; not easily editable
Professional	Preferred for resumes, official documents, forms

Save as PDF in Microsoft Word

Steps

File → Save As

Choose location → Enter filename

In "Save as type" dropdown, select **PDF (*.pdf)**

Click Save

Export as PDF in LibreOffice Writer

Steps

File → Export As → Export as PDF (or click PDF icon)

PDF Options dialog appears → Set options

Click Export → Enter filename → Save

PDF Export Options (LibreOffice Writer)

Option	**Purpose**
Lossless compression	Higher quality, larger file
JPEG compression	Smaller file, suitable for images
PDF/A-1a	Archival format for long-term storage

Option	Purpose
Export bookmarks	Include headings as navigation bookmarks

3.4 Text Creation and Manipulation

3.4.1 Document Creation

To start typing, simply click in the document area. The cursor (blinking vertical line) indicates where text will appear.

Word Wrap: When you reach the end of a line, the word processor automatically moves the cursor to the next line. You do not need to press Enter at the end of each line—only at the end of paragraphs .

3.4.2 Editing Text

Action	Method
Insert Text	Click where you want to add text and start typing
Delete Character	Press Backspace (left) or Delete (right)
Delete Word	Ctrl + Backspace (delete word left) or Ctrl + Delete (delete word right)
Delete Line	Select line and press Delete

Action	Method
Overtype Mode	Press Insert key to toggle Overwrite mode (replaces existing text)

3.4.3 Text Selection

Selection	Mouse Method	Keyboard Method
Single Character	Click and drag over character	Shift + Arrow key
Word	Double-click the word	Ctrl + Shift + Arrow
Line	Click in left margin next to line	Home, then Shift + End
Paragraph	Triple-click in paragraph	Ctrl + Shift + Down, then Up
Entire Document	Triple-click in left margin	Ctrl + A

3.4.4 Cut, Copy and Paste

These operations use the **Clipboard**, a temporary storage area in computer memory .

Operation	Purpose	Shortcut
Copy	Duplicate selected text; original remains	Ctrl + C

Operation	Purpose	Shortcut
Cut	Remove selected text; stores it on Clipboard	Ctrl + X
Paste	Insert Clipboard contents at cursor location	Ctrl + V

Difference Between Cut and Copy

Action	Result
Copy	Selected text stays in place; a copy is placed on Clipboard
Cut	Selected text is removed from document; placed on Clipboard

Advanced Paste Options

Option	Purpose
Keep Source Formatting	Preserve original font, color, etc.
Merge Formatting	Match formatting of destination document
Keep Text Only	Remove all formatting, paste plain text

3.4.5 Font, Color, Style and Size Selection

Font is a character style that determines how text appears on screen and in print. **Size** is measured in points (72 points = 1 inch) .

Font Formatting Options

Option	Description	Shortcut
Bold	Makes text darker and thicker	Ctrl + B
Italic	Slants text to the right	Ctrl + I
Underline	Adds line under text	Ctrl + U
Font Type	Changes typeface (Arial, Times New Roman, etc.)	-
Font Size	Changes text size (e.g., 12pt, 14pt, 18pt)	-
Font Color	Changes text color	-
Highlight	Adds background color	-

Change Font Formatting in Microsoft Word

Steps

Select the text you want to format

Home tab → Font group

Click font name dropdown to change typeface

Click font size dropdown to change size

Click B, I, or U for style

Steps

Click font color (A) to change color

Change Font Formatting in LibreOffice Writer

Steps

Select text

Format → Character (or right-click → Character)

Or use Formatting Toolbar: Font, Size, B, I, U, Color

3.4.6 Alignment of Text

Alignment controls how text is positioned between left and right margins .

Alignment	Icon	Description	Shortcut
Left Align	⬅	Text aligns to left margin; right margin ragged	Ctrl + L
Center Align	↔	Text centered between margins	Ctrl + E
Right Align	➡	Text aligns to right margin; left margin ragged	Ctrl + R
Justify	↔↔	Text aligns to both left and right margins	Ctrl + J

Apply Alignment

Microsoft Word	**LibreOffice Writer**
Home → Paragraph group → Alignment icons	Formatting toolbar → Alignment icons
Or use keyboard shortcuts	Or Format → Paragraph → Alignment tab

---

3.4.7 Undo & Redo

Operation	**Purpose**	**Shortcut**
Undo	Reverses the last action	Ctrl + Z
Redo	Reverses the last Undo action	Ctrl + Y (Word) or Ctrl + Shift + Z (Writer)

3.4.8 AutoCorrect, Spelling & Grammar

AutoCorrect

AutoCorrect automatically corrects common typos and formatting errors as you type .

AutoCorrect Feature	**Example**
Capitalize first letter of sentence	"hello" → "Hello"

AutoCorrect Feature	Example
Correct two initial capitals	"THis" → "This"
Replace text	"teh" → "the"
Smart quotes	"quote" → “quote”

Configure AutoCorrect

Microsoft Word	LibreOffice Writer
File → Options → Proofing → AutoCorrect Options	Tools → AutoCorrect → AutoCorrect Options

Spell Check

Spell Check compares words in your document against an online dictionary and highlights words not found .

Feature	Microsoft Word	LibreOffice Writer
On-the-fly Spelling	Red wavy underline under misspelled words	Red wavy underline under misspelled words
Spell Check Dialog	Review → Spelling & Grammar (F7)	Tools → Spellcheck (F7)
Right-click Options	Right-click underlined word → suggestions	Right-click underlined word → suggestions

Grammar Check

Grammar Check analyzes sentence structure and suggests corrections for grammar errors.

Feature	Microsoft Word	LibreOffice Writer
On-the-fly Grammar	Blue double underline	Tools → Options → Language Settings → Grammar checking
Grammar Check	Review → Spelling & Grammar (included)	Tools → Language → Grammar Check

---

3.4.9 Find and Replace

Find and Replace allows you to locate specific text and optionally replace it with something else .

Find Text

Method	Steps	Shortcut
Find	Home → Find	Ctrl + F

Find and Replace

Method	Steps	Shortcut
Replace	Home → Replace	Ctrl + H

Find and Replace Options

Option	Purpose
Match case	Finds only text with exact capitalization
Find whole words only	Prevents finding text within longer words
Use wildcards	Use * for multiple characters, ? for single character

LibreOffice Writer Find Options

Press **Ctrl+F** to open Find toolbar at bottom. Click dropdown arrow for:

- Match case
- Whole words only
- Regular expressions
- Find All, Find Previous, Find Next

3.5 Formatting the Text

3.5.1 Paragraph Indentation

Indentation temporarily alters the margins for specific paragraphs .

Indent Type	Description
Left Indent	Moves paragraph left margin inward
Right Indent	Moves paragraph right margin inward

Indent Type	Description
First Line Indent	Indents only the first line of paragraph
Hanging Indent	Indents all lines except the first line

Set Indentation in Microsoft Word

Method	Steps
Ruler	Drag markers on ruler (Top: first line indent, Bottom: hanging indent, Square: left indent)
Paragraph Dialog	Home → Paragraph launcher → Indentation section
Increase/Decrease	Home → Increase Indent / Decrease Indent

Set Indentation in LibreOffice Writer

Method	Steps
Ruler	Drag indent markers on ruler
Paragraph Dialog	Format → Paragraph → Indents & Spacing tab

3.5.2 Bullets and Numbering

Bulleted and numbered lists help organize information and make documents easier to read .

Bulleted Lists (unordered)

Use for items that don't require a specific order.

Bullet Style	Examples
Symbol	• , ◦ , ▪
Picture	Custom image bullets
Numbered	1, 2, 3 or A, B, C

Numbered Lists (ordered)

Use for sequential steps, rankings, or items with specific order.

Numbering Style	Examples
Arabic Numbers	1, 2, 3
Roman Numerals	I, II, III or i, ii, iii
Letters	A, B, C or a, b, c

Create Bulleted/Numbered Lists

Microsoft Word	LibreOffice Writer
Home → Paragraph → Bullets icon	Formatting toolbar → Bullets icon
Home → Paragraph → Numbering icon	Formatting toolbar → Numbering icon
Type 1. and press Space to start numbered list	Type 1. and press Space

Multilevel Lists

Create outlines with multiple levels (e.g., 1 → a → i)

Microsoft Word	**LibreOffice Writer**
Home → Multilevel List	Formatting toolbar → Multilevel List button
Tab to demote list level	Tab to demote; Shift+Tab to promote
Shift+Tab to promote	

3.5.3 Change Case

Change the capitalization of selected text without retyping.

Case Option	**Result**
Sentence case	First letter of first word capitalized
lowercase	All letters lowercase
UPPERCASE	ALL LETTERS CAPITAL
Capitalize Each Word	First Letter Of Each Word Capitalized
tOGGLE cASE	Reverses current capitalization

Change Case in Microsoft Word

Steps

Select text

Home → Font group → Change Case button (Aa)

Choose desired option

Change Case in LibreOffice Writer

Steps

Select text

Format → Text → Change Case → Choose option

3.5.4 Header & Footer

Headers appear at the top of every page; **Footers** appear at the bottom of every page .

Common Uses

Header	**Footer**
Document title	Page numbers
Chapter name	Date

Header	**Footer**
Company logo	Copyright information
Author name	Confidentiality notice

Insert Header/Footer in Microsoft Word

Steps

Insert → Header → Choose style → Type content

Insert → Footer → Choose style → Type content

Double-click header/footer area to edit

Close Header and Footer (red X) when done

Insert Header/Footer in LibreOffice Writer

Steps

Insert → Header → Default Page Style (or select style)

Insert → Footer → Default Page Style

Type content in header/footer area

Click outside to exit

Insert Page Numbers

Microsoft Word	**LibreOffice Writer**
Insert → Page Number → Choose position	Insert → Page Number
Or Insert → Footer → Edit Footer → Page Number	Or Insert → Field → Page Number

Different First Page

To have a different header/footer on the first page (common for title pages):

Microsoft Word	**LibreOffice Writer**
Insert → Header → Edit Header → Check "Different First Page"	Format → Page Style → Header/Footer tab → Check "Same content left/right" → Use First Page style

---

3.6 Table Manipulation

Tables organize data into rows and columns .

3.6.1 Insert & Draw Table

Insert Table (Grid Method)

Microsoft Word	**LibreOffice Writer**
Insert → Table → Move mouse over grid → Click	Table → Insert → Table → Enter rows and columns

Insert Table (Dialog Method)

Microsoft Word	**LibreOffice Writer**
Insert → Table → Insert Table → Enter rows and columns	Table → Insert → Table → Enter rows and columns
Also set AutoFit behavior	Also set AutoFit behavior

Draw Table (Word Only)

Steps

Insert → Table → Draw Table

Use pencil cursor to draw table outline, rows, columns

3.6.2 Changing Cell Width and Height

Adjust Column Width

Method	**Steps**
Mouse Drag	Hover over column border → Double-headed arrow appears → Drag left/right
Double-click	Double-click column border to AutoFit to content
Table Properties	Right-click column → Table Properties → Column tab → Enter width

Adjust Row Height

Method	Steps
Mouse Drag	Hover over row border → Double-headed arrow appears → Drag up/down
Table Properties	Right-click row → Table Properties → Row tab → Enter height
Distribute Rows	Table Tools → Layout → Distribute Rows Evenly

---

3.6.3 Alignment of Text in Cell

Text within table cells can be aligned horizontally and vertically .

Horizontal Alignment (left, center, right, justify)

Method	Steps
Toolbar	Select cells → Use alignment icons in toolbar
Keyboard	Ctrl+L, Ctrl+E, Ctrl+R, Ctrl+J

Vertical Alignment (top, center, bottom)

Microsoft Word	LibreOffice Writer
Table Tools → Layout → Alignment group → Choose Top/Center/Bottom	Right-click table → Table Properties → Text Flow tab → Vertical Alignment

Cell Margins

Microsoft Word	**LibreOffice Writer**
Table Tools → Layout → Cell Margins	Right-click table → Table Properties → Borders & Background tab → Spacing to contents

---

3.6.4 Delete/Insertion of Row, Column and Merging & Splitting of Cells

Insert Rows or Columns

Microsoft Word	**LibreOffice Writer**
Right-click → Insert → Insert Rows Above/Below or Insert Columns Left/Right	Right-click → Row → Insert Row Above/Below
Table Tools → Layout → Rows & Columns group	Right-click → Column → Insert Column Before/After

Delete Rows or Columns

Microsoft Word	**LibreOffice Writer**
Right-click → Delete Rows or Delete Columns	Right-click → Row → Delete Row
	Right-click → Column → Delete Column

Merge Cells

Combine multiple cells into one cell.

Microsoft Word

Select cells → Right-click → Merge Cells

Table Tools → Layout → Merge Cells

LibreOffice Writer

Select cells → Right-click → Merge Cells

Table → Merge Cells

Split Cells

Divide one cell into multiple rows/columns.

Microsoft Word

Select cell → Right-click → Split Cells → Enter rows and columns

Table Tools → Layout → Split Cells

LibreOffice Writer

Select cell → Right-click → Split Cell → Enter rows and columns

Table → Split Cells

3.6.5 Border and Shading

Table Borders

Microsoft Word

Select table/cells → Table Design tab → Borders → Choose border style

Choose border type (All, Outside, Inside, etc.)

LibreOffice Writer

Select table/cells → Format → Page (or right-click) → Borders tab

Choose line arrangement, style, color, width

Common Border Options

Option	Result
All Borders	All cells have borders
Outside Borders	Only outer edges have borders
Inside Borders	Only internal cell divisions have borders
No Border	Remove all borders
Left/Right/Top/Bottom	Border only on selected side

Table Shading (Background Color)

Microsoft Word	LibreOffice Writer
Select table/cells → Table Design tab → Shading → Choose color	Select table/cells → Format → Page → Background tab → Choose color Right-click → Table Properties → Background tab

Table Styles

Quickly apply pre-designed formatting (borders, shading, font) to tables.

Microsoft Word	LibreOffice Writer
Table Design tab → Table Styles gallery	Sidebar → Styles deck → Table Styles

---

3.7 Mail Merge

Mail Merge creates customized letters from a form letter and a list of names and addresses . It allows you to send the same document to multiple recipients with personalized information.

Mail Merge Process

```text
┌──────────────────┐     ┌──────────────────┐     ┌──────────────────┐
│   Main Document  │     │   Data Source    │     │   Merged         │
│   (Form Letter)  │  +  │   (Recipient     │  =  │   Documents      │
│                  │     │    List)         │     │   (Personalized  │
│ Dear «Name»,     │     │ Name, Address,   │     │   Letters)       │
│                  │     │ City, etc.       │     │                  │
└──────────────────┘     └──────────────────┘     └──────────────────┘
```

Components

Component	Description
Main Document	Contains text that remains the same for all recipients, plus merge fields (placeholde
Data Source	File containing variable information (names, addresses, etc.)—can be Excel spreadsl Access database, or Word table
Merge Fields	Placeholders like «FirstName», «City» that get replaced with actual data

Steps for Mail Merge in Microsoft Word

Step	Action
1. Start	Mailings → Start Mail Merge → Step-by-Step Mail Merge Wizard
2. Select Document Type	Choose Letters, E-mail, Envelopes, or Labels
3. Select Document	Use current document, start from template, or existing document
4. Select Recipients	Type new list (create in Word), use existing list (Excel), or Outlook contacts
5. Write Letter	Type letter content → Insert Merge Fields where personalization needed
6. Preview Results	Check preview to verify data appears correctly
7. Complete Merge	Print directly or Edit Individual Documents to create separate files

Mail Merge in LibreOffice Writer

Step	Action
1. Start	Tools → Mail Merge Wizard
2. Select Document	Choose current document or create new
3. Select Address List	Use existing spreadsheet, database, or create new
4. Insert Address Block/Fields	Insert → Fields → Other → Database tab
5. Preview	View merged data
6. Print/Save	Print merged documents or save as individual files

3.8 Shortcut Keys

General Operations

Shortcut	Microsoft Word	LibreOffice Writer	Purpose
Ctrl + N	New document	New document	Create new document
Ctrl + O	Open document	Open document	Open existing document
Ctrl + S	Save	Save	Save current document
Ctrl + P	Print	Print	Print document
Ctrl + W	Close document	Close document	Close current document
Alt + F4	Exit program	Exit program	Close application
Ctrl + Z	Undo	Undo	Undo last action
Ctrl + Y	Redo	-	Redo last undone action
Ctrl + Shift + Z	-	Redo	Redo last undone action
F1	Help	Help	Open help

Text Editing

Shortcut	Microsoft Word	LibreOffice Writer	Purpose
Ctrl + C	Copy	Copy	Copy selected text
Ctrl + X	Cut	Cut	Cut selected text
Ctrl + V	Paste	Paste	Paste from clipboard
Ctrl + A	Select All	Select All	Select entire document
Delete	Delete right	Delete right	Delete character to right
Backspace	Delete left	Delete left	Delete character to left
Ctrl + Backspace	Delete word left	Delete word left	Delete word to left
Ctrl + Delete	Delete word right	Delete word right	Delete word to right

Text Formatting

Shortcut	Microsoft Word	LibreOffice Writer	Purpose
Ctrl + B	Bold	Bold	Apply/remove bold

Shortcut	Microsoft Word	LibreOffice Writer	Purpose
Ctrl + I	Italic	Italic	Apply/remove italic
Ctrl + U	Underline	Underline	Apply/remove underline
Ctrl + [	Decrease font size	Decrease font size	Decrease font by 1 point
Ctrl +]	Increase font size	Increase font size	Increase font by 1 point
Ctrl + D	Font dialog	-	Open font formatting dialog

Alignment

Shortcut	Microsoft Word	LibreOffice Writer	Purpose
Ctrl + L	Left align	Left align	Align text left
Ctrl + E	Center	Center	Center text
Ctrl + R	Right align	Right align	Align text right
Ctrl + J	Justify	Justify	Justify text

Navigation and Selection

Shortcut	Microsoft Word	LibreOffice Writer	Purpose
Ctrl + Home	Go to beginning	Go to beginning	Move to document start
Ctrl + End	Go to end	Go to end	Move to document end
Shift + Arrow	Select characters	Select characters	Extend selection
Ctrl + Shift + Arrow	Select words	Select words	Select entire words
F7	Spell check	Spell check	Open spell check dialog
Ctrl + F	Find	Find	Open find dialog
Ctrl + H	Replace	Find & Replace	Open replace dialog

3.9 Summary

- **Word Processing** is the creation and editing of text documents using computer software .
- **Microsoft Word** and **LibreOffice Writer** are the two main word processors covered in CCC exam.
- **Key Interface Components**: Title Bar, Menu Bar, Toolbars, Sidebar, Status Bar .

- **File Operations**: New (Ctrl+N), Open (Ctrl+O), Save (Ctrl+S), Save As, Close, Print (Ctrl+P).
- **PDF Export**: Save documents as PDF for universal sharing and fixed formatting.
- **Text Editing**: Insert, delete, cut (Ctrl+X), copy (Ctrl+C), paste (Ctrl+V), select text.
- **Text Formatting**: Font type, size, bold (Ctrl+B), italic (Ctrl+I), underline (Ctrl+U), font color, alignment (Ctrl+L/E/R/J).
- **Spelling & Grammar**: AutoCorrect, on-the-fly spell checking (red underline), F7 for spell check dialog.
- **Find and Replace (Ctrl+H)** : Locate and replace text throughout document.
- **Paragraph Formatting**: Indentation (left, right, first line, hanging), bullets, numbering, multilevel lists.
- **Headers & Footers**: Text at top/bottom of every page; page numbers.
- **Tables**: Insert, modify rows/columns, merge/split cells, borders, shading.
- **Mail Merge**: Combine form letter with data source to create personalized documents .

3.10 Model Questions and Answers

Multiple Choice Questions (MCQs)

1. Which of the following is a word processing software?

- a) Microsoft Excel
- b) Microsoft PowerPoint
- c) Microsoft Word
- d) Microsoft Access

Answer: c) Microsoft Word

---

2. The shortcut key to save a document is:

- a) Ctrl + O
- b) Ctrl + S
- c) Ctrl + P
- d) Ctrl + N

Answer: b) Ctrl + S

3. Which alignment makes text flush with both left and right margins?

- a) Left align
- b) Right align
- c) Center align
- d) Justify

Answer: d) Justify

4. The shortcut to paste copied text is:

- a) Ctrl + C
- b) Ctrl + X
- c) Ctrl + V
- d) Ctrl + Z

Answer: c) Ctrl + V

5. Which feature automatically corrects common typing errors?

- a) Spell Check
- b) AutoCorrect
- c) Grammar Check
- d) Find and Replace

Answer: b) AutoCorrect

6. What is the default file extension of a Microsoft Word document?

- a) .doc
- b) .docx
- c) .odt
- d) .txt

Answer: b) .docx

---

7. Which shortcut key is used to open the Help window?

- a) F1
- b) F2
- c) F5
- d) F7

Answer: a) F1

---

8. Which operation removes selected text and stores it on the Clipboard?

- a) Copy
- b) Paste
- c) Cut
- d) Delete

Answer: c) Cut

---

9. To insert a table in a document, which menu is used?

- a) File

- b) Edit
- c) Insert
- d) Format

Answer: c) Insert

---

10. Which of the following is the default page orientation in word processors?

- a) Landscape
- b) Portrait
- c) Square
- d) Custom

Answer: b) Portrait

True/False Questions

Statement	Answer
Word processing software can be used to create letters and reports.	**True**
Ctrl + Z is used to redo the last action.	**False** (Ctrl+Z is Undo; Ctrl+Y is Redo)
Headers appear at the bottom of each page.	**False** (Headers appear at top; Footers at bottom)
A table can be inserted using the Insert menu.	**True**

Statement	Answer
LibreOffice Writer is an open source word processor.	**True**

Short Answer Questions

Q1: What is word processing? List any four features of a word processor.
Ans: Word processing is the creation, editing, and formatting of text documents using computer software .

Four features:

1. Text editing (cut, copy, paste)
2. Text formatting (bold, italic, underline)
3. Spell check and grammar check
4. Table insertion and formatting

Q2: Differentiate between Save and Save As.
Ans:

- **Save** stores changes to the currently open document (first save opens Save As dialog; subsequent saves use same name/location) .
- **Save As** creates a copy with a new name or in a different location/format, keeping the original unchanged.

Q3: What is the difference between Cut and Copy?
Ans:

- **Cut** removes selected text from the document and stores it on Clipboard (Ctrl+X) .
- **Copy** duplicates selected text, leaving original in place, and stores copy on Clipboard (Ctrl+C).

Q4: Explain the use of Find and Replace feature.
Ans: Find and Replace locates specific text in a document and optionally replaces it with different text . It's useful for correcting repeated errors, updating names, or changing terminology throughout a document.

Q5: What is Mail Merge? Name its components.
Ans: Mail Merge creates personalized documents (like form letters) for multiple recipients using a data source .

Components:

1. Main Document (form letter with merge fields)
2. Data Source (list of recipient information)

Practical Lab Exercises

Exercise No.	Task	Purpose
1	Open Writer/Word, create new document, type a paragraph about yourself	Understand interface and basic typing

2	Save document as .odt and .docx formats; also export as PDF	Practice Save, Save As, PDF export
3	Type 5 sentences; practice Cut, Copy, Paste operations	Master Clipboard operations
4	Format text with different fonts, sizes, colors, bold, italic, underline	Apply character formatting
5	Create document with left, center, right, and justified text paragraphs	Practice alignment
6	Type a paragraph with spelling errors; use spell check to correct	Spell checking practice
7	Use Find and Replace to change a word throughout document	Find and replace practice
8	Create bulleted and numbered lists with multiple levels	List formatting
9	Insert header with document title and footer with page number	Headers and footers
10	Create a 4×4 table; insert data; merge cells; apply borders and shading	Table manipulation
11	Create a mail merge document: form letter with recipient list	Mail merge practice

---

Important Points for UP CCC Exam

Topic	Key Points to Remember
Word Processors	MS Word (proprietary), LibreOffice Writer (open source)
File Formats	.docx (Word), .odt (Writer), .pdf (universal)
Common Shortcuts	Ctrl+N (New), Ctrl+O (Open), Ctrl+S (Save), Ctrl+P (Print), Ctrl+Z (Undo), Ctrl+C (Ctrl+X (Cut), Ctrl+V (Paste)
Alignment	Left (Ctrl+L), Center (Ctrl+E), Right (Ctrl+R), Justify (Ctrl+J)
Text Formatting	Bold (Ctrl+B), Italic (Ctrl+I), Underline (Ctrl+U)
Page Setup	Margins, Orientation (Portrait/Landscape), Paper Size (A4)
Tables	Insert via Insert menu; merge/split cells; borders and shading
Mail Merge	Main document + Data source = Personalized documents

Chapter 4: Spreadsheet

4.0 Introduction

A **spreadsheet** is a digital worksheet that organizes data in rows and columns, allowing users to perform calculations, analyze data, and create charts . It is widely used for financial analysis, budgeting, data management, and reporting. Spreadsheets are indispensable in offices, schools, and businesses for tasks ranging from simple calculations to complex data analysis.

For the CCC exam, you need to be proficient in both **Microsoft Excel** (proprietary) and **LibreOffice Calc** (open source), as NIELIT examinations often use Ubuntu Linux with LibreOffice.

4.1 Objectives

After studying this chapter, you will be able to:

- Understand spreadsheet concepts: cells, rows, columns, and cell addresses
- Create, open, save, and close spreadsheet documents
- Enter and edit different types of data (text, numbers, dates)
- Format cells and worksheets
- Perform cut, copy, paste, and paste special operations
- Adjust row height and column width
- Insert and delete rows and columns
- Use AutoFill for data series
- Sort and filter data
- Freeze panes for easier navigation
- Create formulas using basic arithmetic operations
- Use built-in functions (SUM, COUNT, MAX, MIN, AVERAGE)
- Create and format charts (Bar, Pie, Line)

- Set up page layout and print spreadsheets

4.2 Elements of Spreadsheet

A spreadsheet consists of a grid of **rows** and **columns**. The intersection of a row and a column is called a **cell**.

4.2.1 Creating a Spreadsheet

Microsoft Excel

Method	Steps
Blank Workbook	Click Excel icon → Select Blank Workbook
Start Menu	Start → Type "Excel" → Click Microsoft Excel
Template	File → New → Choose from available templates (budget, calendar, etc.)
Keyboard Shortcut	Ctrl + N

LibreOffice Calc

Method	Steps
Blank Spreadsheet	Click LibreOffice Calc icon → New Spreadsheet
Activities	Activities → Type "Calc" → Click LibreOffice Calc
Template	File → New → Templates → Select template

Method	Steps
Keyboard Shortcut	Ctrl + N

4.2.2 Concept of Cell Address [Row and Column] and Selecting a Cell

Spreadsheet Structure

```text
        A           B           C           D           E
    ┌───────────┬───────────┬───────────┬───────────┬───────────┐
 1  │    A1     │    B1     │    C1     │    D1     │    E1     │
    ├───────────┼───────────┼───────────┼───────────┼───────────┤
 2  │    A2     │    B2     │    C2     │    D2     │    E2     │
    ├───────────┼───────────┼───────────┼───────────┼───────────┤
 3  │    A3     │    B3     │    C3     │    D3     │    E3     │
    ├───────────┼───────────┼───────────┼───────────┼───────────┤
 4  │    A4     │    B4     │    C4     │    D4     │    E4     │
    ├───────────┼───────────┼───────────┼───────────┼───────────┤
 5  │    A5     │    B5     │    C5     │    D5     │    E5     │
    │           │           │           │           │           │
```

.

Term	Description
Row	Horizontal line of cells; identified by numbers (1, 2, 3, ...)
Column	Vertical line of cells; identified by letters (A, B, C, ...)

Term	Description
Cell	Intersection of a row and a column
Cell Address/Reference	Unique identifier combining column letter and row number (e.g., A1, B5, C10)
Active Cell	Currently selected cell; indicated by a dark border
Cell Range	Group of cells; denoted by starting and ending addresses with colon (e.g., A1:C5)

Selecting Cells

Selection	Mouse Method	Keyboard Method
Single Cell	Click on the cell	Use arrow keys to navigate
Range of Cells	Click and drag across cells	Click first cell, hold Shift, click last cell
Entire Row	Click row number (1, 2, 3, etc.)	Shift + Spacebar
Entire Column	Click column letter (A, B, C, etc.)	Ctrl + Spacebar
Entire Sheet	Click top-left corner (between A and 1)	Ctrl + A (twice)
Non-adjacent Cells	Ctrl + Click on each cell	Ctrl + Click

4.2.3 Entering Data [Text, Number, Date] in Cells

Types of Data

Data Type	Description	Examples	Default Alignment
Text/Label	Alphanumeric data not used in calculations	Name, Address, "ABC123"	Left-aligned
Number	Numeric data used in calculations	100, 25.5, -50	Right-aligned
Date/Time	Calendar dates and times	25/03/2026, 10:30 AM	Right-aligned
Formula	Expression starting with =	=A1+B1, =SUM(A1:A5)	Formula result

Entering Data

Action	Steps
Select cell	Click the desired cell
Type data	Start typing (text, number, date, or formula starting with =)
Confirm entry	Press Enter (moves down), Tab (moves right), or click ✓ on formula bar
Cancel entry	Press Esc or click ✗ on formula bar

Entering Dates

Format	Example	Notes
DD/MM/YYYY	25/03/2026	Common in India
MM/DD/YYYY	03/25/2026	US format
DD-MMM-YYYY	25-Mar-2026	Text format
Date shortcut	Ctrl + ; (semicolon)	Enters current date

Entering Numbers as Text

To enter a number that should be treated as text (e.g., phone number, PIN code):

- Type an apostrophe (') before the number: `'0123456789`
- Or format cell as Text before entering

4.2.4 Page Setup

Page Setup controls how the spreadsheet appears when printed.

Page Setup Options

Option	Description
Margins	Space between data and edge of paper (Top, Bottom, Left, Right)
Orientation	Portrait (vertical) or Landscape (horizontal)

Option	Description
Paper Size	A4, Letter, Legal, etc.
Scaling	Fit sheet to one page, fit to specific number of pages
Print Area	Specify which cells to print
Print Titles	Repeat row/column headers on every page
Header/Footer	Add text at top/bottom of each printed page
Gridlines	Show/hide cell borders when printing

Page Setup in Microsoft Excel

Steps

Page Layout tab → Page Setup group

Margins → Select preset or Custom Margins

Orientation → Portrait or Landscape

Size → Select paper size (A4 recommended)

Print Area → Set Print Area

Print Titles → Rows to repeat at top, Columns to repeat at left

Page Setup in LibreOffice Calc

Steps

Format → Page (or right-click on sheet tab → Page Setup)

Page tab → Set orientation, paper size, margins

Sheet tab → Set print area, repeat rows/columns, gridlines

4.2.5 Printing of Sheet

Print Settings

Setting	Purpose
Printer	Select which printer to use
Copies	Number of copies to print
Print Range	All sheets, current sheet, selected cells, or specific pages
Collate	Print multiple copies in order
Print Selection	Print only selected cells
Print Active Sheets	Print currently selected sheet(s)
Print Entire Workbook	Print all sheets in the workbook

Print Preview

Always use **Print Preview** before printing to see how the document will appear on paper .

Microsoft Excel

Steps

File → Print (or Ctrl + P)

Print Preview appears on right side

Adjust settings as needed

Click Print

LibreOffice Calc

Steps

File → Print Preview (or Ctrl + Shift + O)

Preview shows how sheet will print

Click Print (or Ctrl + P) to open print dialog

4.2.6 Saving Spreadsheet

Save (First Time / Existing Document)

Operation	Microsoft Excel	LibreOffice Calc
Save New	File → Save As → Choose location → Enter filename → Save	File → Save As → Choose location → Enter filename → Save
Save Existing	File → Save (or Ctrl + S)	File → Save (or Ctrl + S)
Save Icon	Click disk icon on Quick Access Toolbar	Click disk icon on Standard Toolbar

File Formats

Format	Extension	When to Use
Excel Workbook	.xlsx	Default Excel format
Excel 97-2003	.xls	For older Excel versions
OpenDocument Spreadsheet	.ods	Default LibreOffice Calc format
CSV (Comma Separated Values)	.csv	For data exchange with other applications
PDF	.pdf	For sharing final spreadsheets

4.2.7 Opening and Closing

Opening a Spreadsheet

Method	Steps
File Menu	File → Open → Browse to location → Select file → Open
Recent Documents	File → Open → Recent → Select from list
Keyboard Shortcut	Ctrl + O
Double-click File	Double-click spreadsheet file in File Explorer/Files

Closing a Spreadsheet

Method	Steps
File Menu	File → Close
Close Button	Click × on document tab or window
Keyboard Shortcut	Ctrl + W (Excel) or Ctrl+F4 (Calc)
Alt + F4	Close entire application

4.3 Manipulation of Cells & Sheet

4.3.1 Modifying / Editing Cell Content

Edit Cell Content

Method	Steps
Double-click	Double-click the cell → Edit directly in cell
F2 Key	Select cell → Press F2 → Edit in cell
Formula Bar	Select cell → Click in formula bar → Edit
Overwrite	Select cell → Type new content → Overwrites existing

Clear Cell Content

Action	Steps
Delete Content Only	Select cell → Press Delete
Clear All	Select cell → Home → Clear → Clear All
Clear Formats	Select cell → Home → Clear → Clear Formats
Clear Contents	Select cell → Home → Clear → Clear Contents

4.3.2 Formatting Cell (Font, Alignment, Style)

Font Formatting

Option	Description	Shortcut
Font Type	Change typeface (Arial, Calibri, etc.)	-
Font Size	Change text size	Ctrl + Shift + > (increase) / < (decrease)
Bold	Make text bold	Ctrl + B
Italic	Italicize text	Ctrl + I
Underline	Underline text	Ctrl + U
Font Color	Change text color	-
Fill Color	Change cell background color	-

Number Formatting

Format	Description	Example
General	Default format	1234.567
Number	With decimal places	1,234.57
Currency	Money format	₹1,234.57
Accounting	Aligns currency symbols	₹ 1,234.57
Date	Date formats	25-Mar-2026

Format	Description	Example
Percentage	Multiplies by 100 and adds %	50%
Text	Treats numbers as text	'01234

Alignment

Alignment	Description	Shortcut
Left Align	Text aligns to left	Ctrl + L
Center	Text centered	Ctrl + E
Right Align	Text aligns to right	Ctrl + R
Merge & Center	Merge cells and center content	-
Wrap Text	Display multiple lines in cell	-
Vertical Alignment	Top, Center, Bottom	-

Apply Formatting

Microsoft Excel	LibreOffice Calc
Home tab → Font/Alignment/Number groups	Formatting toolbar or Format → Cells
Or right-click → Format Cells	Or right-click → Format Cells

4.3.3 Cut, Copy, Paste & Paste Special

Basic Operations

Operation	Purpose	Shortcut
Copy	Duplicate selected cells	Ctrl + C
Cut	Remove selected cells (to move)	Ctrl + X
Paste	Insert copied/cut cells	Ctrl + V

Paste Special

Paste Special allows you to control what aspects of copied data are pasted .

Paste Special Option	Purpose
All	Pastes everything (default)
Formulas	Pastes only formulas, not formatting
Values	Pastes only calculated results, not formulas
Formats	Pastes only formatting, not data
Transpose	Switches rows to columns and columns to rows
Operation (Add/Subtract/etc.)	Performs arithmetic with existing values

Using Paste Special

Microsoft Excel	**LibreOffice Calc**
Copy cells → Right-click → Paste Special	Copy cells → Edit → Paste Special
Or Home → Paste → Paste Special	Or right-click → Paste Special

---

4.3.4 Changing Cell Height and Width

Adjust Column Width

Method	**Steps**
Mouse Drag	Hover over column border (between letters) → Double-headed arrow → Drag left/right
AutoFit	Double-click column border → Automatically adjusts to fit content
Exact Width	Right-click column letter → Column Width → Enter value
Multiple Columns	Select multiple columns → Adjust any one → All adjust

Adjust Row Height

Method	**Steps**
Mouse Drag	Hover over row border (between numbers) → Double-headed arrow → Drag up/down
AutoFit	Double-click row border → Automatically adjusts to fit content

Method	Steps
Exact Height	Right-click row number → Row Height → Enter value

---

4.3.5 Inserting and Deleting Rows, Columns

Insert Rows or Columns

Action	Microsoft Excel	LibreOffice Calc
Insert Row	Right-click row number → Insert	Right-click row number → Insert Rows
Insert Column	Right-click column letter → Insert	Right-click column letter → Insert Columns
Insert Multiple	Select multiple rows/columns → Right-click → Insert	Same method

Delete Rows or Columns

Action	Microsoft Excel	LibreOffice Calc
Delete Row	Right-click row number → Delete	Right-click row number → Delete Rows
Delete Column	Right-click column letter → Delete	Right-click column letter → Delete Columns

Insert/Delete Cells (Shift Cells)

Action	Purpose
Shift cells right	Insert new cells; existing cells move right
Shift cells down	Insert new cells; existing cells move down
Entire row	Insert entire row
Entire column	Insert entire column

4.3.6 AutoFill

AutoFill automatically fills a series of cells based on a pattern from selected cells.

Using AutoFill

Steps

1. Select the cell(s) containing the starting value(s)

2. Locate the **fill handle** (small square at bottom-right corner of selected cell)

3. Drag the fill handle down, right, up, or left

4. Release mouse button to fill

AutoFill Examples

Initial Value(s)	Drag Direction	Result
1	Down	1, 2, 3, 4, ...
Jan	Down	Jan, Feb, Mar, Apr, ...
Monday	Right	Monday, Tuesday, Wednesday, ...
10, 20	Down	10, 20, 30, 40, ... (continues pattern)
Product 1	Down	Product 1, Product 2, Product 3, ...
5	Down with Ctrl	5, 5, 5, 5 (copy value)

AutoFill Options Button

After dragging, a small **AutoFill Options** button appears, allowing you to choose:

- **Copy Cells** (repeat same value)
- **Fill Series** (create sequential series)
- **Fill Formatting Only** (copy format only)
- **Fill Without Formatting** (copy values only)

4.3.7 Sorting & Filtering

Sorting

Sorting arranges data in a specific order (ascending or descending).

Sort Type	Description
Ascending	A to Z (text), Smallest to Largest (numbers), Oldest to Newest (dates)
Descending	Z to A, Largest to Smallest, Newest to Oldest

Sorting in Microsoft Excel

Steps

Select the column to sort

Data tab → Sort A to Z (ascending) or Sort Z to A (descending)

For multiple columns: Data → Sort → Add Level

Sorting in LibreOffice Calc

Steps

Select the data range

Data → Sort → Select column and order

Click OK

Filtering

Filtering displays only rows that meet specific criteria; hides other rows temporarily.

Apply Filter

Microsoft Excel	**LibreOffice Calc**
Select data → Data → Filter	Select data → Data → AutoFilter
Dropdown arrows appear on headers	Dropdown arrows appear on headers
Click arrow → Select criteria (text, number, date filters)	Click arrow → Select Standard Filter or values

Filter Options

Option	**Purpose**
Text Filters	Contains, Begins With, Ends With, Equals, etc.
Number Filters	Greater Than, Less Than, Between, etc.
Date Filters	Today, This Week, Last Month, etc.
Clear Filter	Remove filter to show all data

Remove Filter

Microsoft Excel	**LibreOffice Calc**
Data → Clear Filter	Data → AutoFilter (toggle off)

4.3.8 Freezing Panes

Freezing Panes keeps specific rows or columns visible while scrolling through the rest of the sheet .

Common Uses

Freeze Type	Purpose
Freeze Top Row	Keep column headers visible when scrolling down
Freeze First Column	Keep row labels visible when scrolling right
Freeze Panes	Keep both rows and columns (select cell below and right of rows/columns to freeze)

Freeze Panes in Microsoft Excel

Action	Steps
Freeze Top Row	View → Freeze Panes → Freeze Top Row
Freeze First Column	View → Freeze Panes → Freeze First Column
Freeze Panes	Select cell below the row and to the right of column → View → Freeze Panes → Freeze Panes
Unfreeze	View → Freeze Panes → Unfreeze Panes

Freeze Panes in LibreOffice Calc

Action	Steps
Freeze Rows/Columns	Select cell below and right of area to freeze → Window → Freeze
Unfreeze	Window → Freeze (toggle off)

4.4 Formulas, Functions and Charts

4.4.1 Using Formulas for Numbers (Addition, Subtraction, Multiplication & Division)

A **formula** is an expression that performs calculations on values in cells. All formulas begin with an equals sign (=).

Arithmetic Operators

Operator	Operation	Example	Result
+	Addition	=10+5	15
-	Subtraction	=10-5	5
*	Multiplication	=10*5	50
/	Division	=10/5	2
^	Exponentiation	=10^2	100

Using Cell References in Formulas

Formula	Meaning
=A1+A2	Adds values in cells A1 and A2
=B1-B2	Subtracts B2 from B1
=C1*C2	Multiplies C1 and C2
=D1/D2	Divides D1 by D2
=(A1+A2+A3)/3	Sum of A1, A2, A3 divided by 3

Order of Operations (BODMAS)

Excel follows standard mathematical order of operations:

Priority	Operators	Example
1	Parentheses ( )	= (A1+B1)*C1
2	Exponents ^	= A1^2
3	Multiplication * and Division /	= A1*B1/C1
4	Addition + and Subtraction -	= A1+B1-C1

Example: `=10+5*2` equals 20 (multiplication first: 5*2=10, then 10+10=20)

4.4.2 AutoSum

AutoSum is a shortcut for adding a range of numbers quickly .

Using AutoSum

Steps

1. Select the cell where the sum should appear (usually below or to the right of numbers)

2. Click AutoSum button (Σ symbol)

3. Excel/Calc automatically selects the range

4. Press Enter to accept

AutoSum Options

Microsoft Excel	**LibreOffice Calc**
Home → AutoSum (Σ)	Formula → AutoSum (Σ)
Dropdown includes: Sum, Average, Count, Max, Min	Same functions available

---

4.4.3 Functions (Sum, Count, MAX, MIN, AVERAGE)

A **function** is a predefined formula that performs calculations using specific values called arguments .

Common Functions

Function	Syntax	Purpose	Example
SUM	=SUM(range)	Adds all numbers in a range	=SUM(A1:A5)
AVERAGE	=AVERAGE(range)	Calculates arithmetic mean	=AVERAGE(B1:B10)
COUNT	=COUNT(range)	Counts cells containing numbers	=COUNT(C1:C20)
COUNTA	=COUNTA(range)	Counts non-empty cells	=COUNTA(D1:D20)
MAX	=MAX(range)	Returns largest value	=MAX(E1:E50)
MIN	=MIN(range)	Returns smallest value	=MIN(E1:E50)
SUMIF	=SUMIF(range, criteria)	Sums cells meeting criteria	=SUMIF(A1:A10,">100")
COUNTIF	=COUNTIF(range, criteria)	Counts cells meeting criteria	=COUNTIF(B1:B10,"Yes")

Function Syntax

```text
= FUNCTION_NAME ( argument1, argument2, ... )

Examples:
```

```
=SUM(A1:A10)           (single range)
=SUM(A1:A10, B1:B10)   (multiple ranges)
=AVERAGE(A1, B1, C1)   (individual cells)
```

Using Functions

Method	Steps
Type Directly	Type = followed by function name and arguments
Function Wizard	Click Insert Function (fx) → Search function → Enter arguments
Formula Tab	Formula tab → Select function → Select range

4.4.4 Charts (Bar, Pie, Line)

Charts are graphical representations of data that make it easier to understand patterns and trends .

Types of Charts

Chart Type	Purpose	Best Used For
Bar/Column Chart	Compare values across categories	Sales by month, marks by subject
Pie Chart	Show parts of a whole (percentages)	Market share, budget allocation
Line Chart	Show trends over time	Stock prices, temperature changes
Area Chart	Emphasize magnitude of change	Cumulative data over time

Creating a Chart

Step	Action
1	Select the data range (include headers)
2	Click Insert → Chart
3	Choose chart type (Bar, Pie, Line, etc.)
4	Chart appears in worksheet
5	Customize using Chart Tools/Design tabs

Chart Components

Component	Description
Chart Title	Describes what the chart represents
X-Axis (Horizontal)	Category axis (e.g., months, products)
Y-Axis (Vertical)	Value axis (e.g., sales, marks)
Legend	Identifies data series
Data Labels	Shows exact values on bars/pie slices
Gridlines	Helps read values

Creating Chart in Microsoft Excel

Steps

Select data range

Insert tab → Chart group → Choose chart type (Column, Bar, Pie, Line)

Chart appears → Use Chart Design and Format tabs to customize

Creating Chart in LibreOffice Calc

Steps

Select data range

Insert → Chart (or click Chart icon on toolbar)

Chart Wizard appears → Choose chart type → Customize options

Click Finish

Formatting Charts

Action	Excel	Calc
Change Chart Type	Chart Design → Change Chart Type	Right-click chart → Chart Type
Add Chart Title	Chart Design → Add Chart Element → Chart Title	Insert → Titles

Action	Excel	Calc
Add Data Labels	Chart Design → Add Chart Element → Data Labels	Right-click data series → Insert Data Labels
Format Colors	Double-click chart element → Format pane	Right-click element → Format

---

4.5 Summary

- A **spreadsheet** organizes data in rows and columns; **cell address** is column letter + row number (e.g., A1) .
- **LibreOffice Calc** (open source) and **Microsoft Excel** (proprietary) are covered in CCC exam.
- **Data Types**: Text (left-aligned), Numbers (right-aligned), Dates, Formulas (begin with =).
- **File Formats**: .xlsx (Excel), .ods (Calc), .pdf (export), .csv (data exchange).
- **Cell Operations**: Edit (F2), Cut (Ctrl+X), Copy (Ctrl+C), Paste (Ctrl+V), Paste Special (control what is pasted).
- **AutoFill** creates series by dragging fill handle (numbers, dates, patterns).
- **Sorting** arranges data A-Z or smallest-largest; **Filtering** shows only data meeting criteria.
- **Freeze Panes** keeps headers visible while scrolling.
- **Formulas** begin with = and use arithmetic operators (+, -, *, /).
- **Functions**: SUM (adds), AVERAGE (mean), COUNT (counts numbers), MAX (largest), MIN (smallest).
- **Charts**: Bar (compare), Pie (parts of whole), Line (trends over time).

4.6 Model Questions and Answers

Multiple Choice Questions (MCQs)

1. In a spreadsheet, the intersection of a row and column is called:

- a) Cell
- b) Field
- c) Record
- d) Table

Answer: a) Cell

---

2. Which of the following is the correct cell address?

- a) 1A
- b) A1
- c) A-1
- d) 1-A

Answer: b) A1

---

3. The formula in a spreadsheet always begins with:

- a) @
- b) #
- c) &
- d) =

Answer: d) =

---

4. Which function is used to add a range of cells?

- a) ADD
- b) TOTAL
- c) SUM
- d) PLUS

Answer: c) SUM

---

5. The shortcut key for AutoSum is:

- a) Alt + =
- b) Ctrl + =
- c) Shift + =
- d) Alt + S

Answer: a) Alt + = (Excel) / Ctrl + F9 (Calc)

---

6. Which chart type is best for showing trends over time?

- a) Pie Chart
- b) Bar Chart
- c) Line Chart
- d) Column Chart

Answer: c) Line Chart

---

7. Which function returns the largest value in a range?

- a) LARGE
- b) MAX
- c) HIGHEST
- d) TOP

Answer: b) MAX

---

8. To keep the header row visible while scrolling, which feature is used?

- a) Freeze Panes
- b) Split
- c) Lock Cells
- d) Protect Sheet

Answer: a) Freeze Panes

---

9. Which feature automatically fills cells with a series?

- a) AutoSum
- b) AutoFill
- c) AutoFormat
- d) AutoCorrect

Answer: b) AutoFill

---

10. The default file extension of LibreOffice Calc is:

- a) .xlsx
- b) .xls
- c) .ods
- d) .odt

Answer: c) .ods

True/False Questions

Statement	Answer
A cell address is identified by row letter and column number.	**False** (Column letter + Row number)
The SUM function adds all numbers in a range.	**True**
A pie chart is used to show trends over time.	**False** (Line chart is for trends; pie chart shows parts of a whole)
Filtering permanently deletes data that does not meet criteria.	**False** (Hides temporarily; data remains)
Formulas in Calc and Excel always begin with an equals sign (=).	**True**

Short Answer Questions

Q1: What is a cell address? Give an example.

Ans: A cell address is the unique identifier of a cell, combining the column letter and row number . Example: A1, B5, C10.

Q2: Differentiate between sorting and filtering.

Ans:

- **Sorting** arranges data in a specific order (ascending or descending) .
- **Filtering** displays only rows that meet certain criteria and hides the rest temporarily.

Q3: What is AutoFill? Give two examples.

Ans: AutoFill automatically fills a series of cells based on a pattern . Examples:

- Enter "Jan" and drag → Feb, Mar, Apr, ...
- Enter "1" and drag → 2, 3, 4, 5, ...

Q4: Write the syntax and purpose of SUM and AVERAGE functions.

Ans:

- **SUM**: `=SUM(range)` – Adds all numbers in the specified range .
- **AVERAGE**: `=AVERAGE(range)` – Calculates the arithmetic mean of numbers in the range.

Q5: What are the three main types of charts? When is each used?

Ans:

- **Bar/Column Chart**: Used to compare values across categories .
- **Pie Chart**: Used to show parts of a whole (percentages).
- **Line Chart**: Used to show trends over time.

Practical Lab Exercises

Exercise No.	Task	Purpose
1	Create a new spreadsheet; enter text, numbers, and dates in different cells	Data entry practice
2	Format cells with different fonts, colors, borders, and number formats	Formatting skills
3	Create a mark sheet with 5 subjects; calculate total (SUM) and average (AVERAGE); find highest (MAX) and lowest (MIN) marks	Functions practice
4	Use AutoFill to create: (a) months of year (b) numbers 1-50 (c) series 5,10,15,20	AutoFill practice
5	Create a salary table; apply formatting; adjust column widths and row heights	Layout skills
6	Insert and delete rows and columns in the table	Table manipulation
7	Sort the mark sheet by total marks in descending order	Sorting practice
8	Apply filter to show only students with marks above 80 in a subject	Filtering practice
9	Freeze the header row; scroll down to verify	Freeze panes practice
10	Create a sales data table; create Bar chart, Pie chart, and Line chart	Chart creation
11	Set page margins, orientation to Landscape; add header with title; print preview	Page setup practice

Exercise No.	Task	Purpose
12	Save spreadsheet as .xlsx, .ods, and .pdf formats	File format practice

---

Important Points for UP CCC Exam

Topic	Key Points to Remember
Cell Address	Column letter + Row number (A1, B5, C10)
Formulas	Begin with = (equals sign)
AutoSum	Alt + = (Excel) or Ctrl + F9 (Calc)
Functions	SUM, AVERAGE, COUNT, MAX, MIN
AutoFill	Drag fill handle to create series
Sorting	Ascending (A-Z, 0-9), Descending (Z-A, 9-0)
Filtering	Shows only rows meeting criteria; hides others
Freeze Panes	Keeps rows/columns visible while scrolling
File Extensions	.xlsx (Excel), .ods (Calc)
Chart Types	Bar (compare), Pie (percentages), Line (trends)

Chapter 5: Presentation

Complete Theory & Lab Notes

5.0 Introduction

A **presentation** is a collection of slides that conveys information to an audience in an organized and visually appealing manner. Presentation software is used to create slide shows for business meetings, educational lectures, project proposals, and training sessions. It allows users to combine text, images, charts, animations, and multimedia elements to communicate ideas effectively.

For the CCC exam, you need to be proficient in both **Microsoft PowerPoint** (proprietary) and **LibreOffice Impress** (open source), as NIELIT examinations often use Ubuntu Linux with LibreOffice.

5.1 Objectives

After studying this chapter, you will be able to:

- Create presentations using templates and blank presentations
- Insert, edit, and format text on slides
- Insert, delete, and rearrange slides
- Insert tables, pictures, and other objects
- Resize and scale objects
- Create and use Master Slides for consistent formatting
- Set up and run slide shows
- Apply transitions and set slide timings
- Automate slide shows

- Enhance text presentation with formatting
- Work with colors and line styles
- Add movies and sound to presentations
- Add headers, footers, and speaker notes
- Print slides, handouts, and notes pages

5.2 Creation of Presentation

5.2.1 Creating a Presentation Using a Template

A **template** is a pre-designed presentation with predefined layouts, colors, fonts, and backgrounds that provide a consistent look throughout the presentation .

Microsoft PowerPoint

Method	Steps
Start Screen	Open PowerPoint → Select a template from gallery (e.g., "Business", "Education")
File Menu	File → New → Search for templates online or browse featured templates
Custom Template	File → New → Personal → Select saved template

LibreOffice Impress

Method	Steps
Start Center	Open Impress → Select "From Template" → Choose template
File Menu	File → New → Templates → Select template

Method	Steps
Wizard	File → Wizards → Presentation → Follow wizard steps

5.2.2 Creating a Blank Presentation

A **blank presentation** starts with a single, blank slide where you can design everything from scratch.

Microsoft PowerPoint

Method	Steps
Start Screen	Open PowerPoint → Click "Blank Presentation"
File Menu	File → New → Blank Presentation
Keyboard Shortcut	Ctrl + N

LibreOffice Impress

Method	Steps
Start Center	Open Impress → Select "New Presentation"
File Menu	File → New → Presentation
Keyboard Shortcut	Ctrl + N

5.2.3 Inserting & Editing Text on Slides

Text in presentations is typically entered in **placeholders** – pre-defined boxes on slides.

Adding Text

Method	Steps
Click Placeholder	Click inside a text placeholder (e.g., "Click to add title") → Start typing
Insert Text Box	Insert → Text Box → Click on slide → Type text
Copy/Paste	Copy text from another document → Paste into placeholder or text box

Editing Text

Action	Steps
Select Text	Click and drag over text, or double-click to select word, triple-click for paragraph
Delete Text	Select text → Press Delete or Backspace
Replace Text	Select text → Start typing new text
Move Text	Select → Drag to new location
Format Text	Select → Use Home tab formatting options (Font, Size, Bold, Italic, etc.)

5.2.4 Inserting and Deleting Slides in a Presentation

Inserting New Slides

Method	Microsoft PowerPoint	LibreOffice Impress
New Slide Button	Home → New Slide	Slide → New Slide (or Insert → Slide)
Shortcut	Ctrl + M	Ctrl + M
Right-click	Right-click in slide pane → New Slide	Right-click in slide pane → New Slide
Duplicate Slide	Right-click slide thumbnail → Duplicate Slide	Slide → Duplicate Slide

Slide Layouts

When inserting a new slide, you can choose a **layout** that determines the arrangement of placeholders:

Layout	Description
Title Slide	Title and subtitle placeholders (first slide)
Title and Content	Title and content area (for text, tables, charts)
Section Header	Section title and description
Two Content	Two content areas side by side
Comparison	Two content areas with titles
Blank	No placeholders – complete freedom

Deleting Slides

Method	Steps
Select and Delete	Click slide thumbnail → Press Delete key
Right-click	Right-click slide thumbnail → Delete Slide
Multiple Slides	Ctrl + Click to select multiple → Press Delete

---

5.2.5 Saving a Presentation

Save (First Time / Existing)

Operation	Microsoft PowerPoint	LibreOffice Impress
Save New	File → Save As → Choose location → Enter filename → Save	File → Save As → Choose location → Enter filename → Save
Save Existing	File → Save (or Ctrl + S)	File → Save (or Ctrl + S)

File Formats

Format	Extension	When to Use
PowerPoint Presentation	.pptx	Default PowerPoint format
PowerPoint 97-2003	.ppt	For older PowerPoint versions

Format	Extension	When to Use
OpenDocument Presentation	.odp	Default LibreOffice Impress format
PDF	.pdf	For sharing final presentations
PowerPoint Show	.ppsx	Opens directly in slideshow mode
Video	.mp4	Export as video

5.3 Manipulating Slides

5.3.1 Inserting Table

Tables organize data into rows and columns within a slide.

Insert Table in Microsoft PowerPoint

Method	Steps
Insert Table	Insert → Table → Select number of rows and columns from grid
Insert Table Dialog	Insert → Table → Insert Table → Enter rows and columns
Draw Table	Insert → Table → Draw Table → Draw with pencil cursor
Table from Content Layout	Click table icon in content placeholder

Insert Table in LibreOffice Impress

Method	Steps
Insert Menu	Insert → Table → Enter rows and columns → OK
Table Icon	Click Table icon on toolbar → Drag to select size

Formatting Tables

Feature	Purpose
Table Style	Apply pre-designed formatting to table
Cell Shading	Fill background color of cells
Borders	Add or remove borders around cells
Merge/Split	Combine or divide cells
Text Alignment	Align text horizontally and vertically

5.3.2 Adding ClipArt Pictures

ClipArt refers to pre-made images and illustrations available within the software.

Microsoft PowerPoint

Method	Steps
Insert ClipArt	Insert → Pictures → Stock Images → Icons/Illustrations
Online Pictures	Insert → Pictures → Online Pictures → Search
Icons	Insert → Icons → Browse categories

LibreOffice Impress

Method	Steps
Gallery	View → Gallery (or press Ctrl+F5) → Drag image to slide
Insert Image	Insert → Image → Select from file

5.3.3 Inserting Other Objects

Insert Picture from File

Microsoft PowerPoint	LibreOffice Impress
Insert → Pictures → This Device → Select file → Insert	Insert → Image → Select file → Open

Insert Shapes

Shape Type	Examples
Lines	Arrow, straight line, curved line
Rectangles	Square, rounded rectangle
Basic Shapes	Circle, triangle, heart, cloud
Block Arrows	Up arrow, down arrow, circular arrow
Flowchart	Process, decision, data shapes
Callouts	Speech bubble, thought bubble

Insert SmartArt / Flowchart

SmartArt (PowerPoint) and **Flowcharts** (Impress) create visual diagrams:

Type	Purpose
List	Show non-sequential information
Process	Show steps in a process
Cycle	Show repeating process
Hierarchy	Show organizational structure
Relationship	Show connections
Matrix	Show parts of a whole

Insert Chart

Microsoft PowerPoint	**LibreOffice Impress**
Insert → Chart → Select chart type	Insert → Chart → Select chart type
Data opens in Excel window	Data opens in embedded spreadsheet

Insert Text Box

Microsoft PowerPoint	**LibreOffice Impress**
Insert → Text Box → Click and drag	Insert → Text Box → Click and drag

---

5.3.4 Resizing and Scaling an Object

Resize an Object

Method	**Steps**
Drag Corners	Select object → Click and drag corner handle (maintains proportions)
Drag Sides	Select object → Click and drag side handle (stretches)
Size Dialog	Right-click object → Size and Position → Enter exact dimensions

Scaling

Scaling refers to resizing while maintaining or altering proportions.

Option	Result
Corner Handle	Maintains aspect ratio (width and height proportionally)
Side Handle	Changes only width or only height
Lock Aspect Ratio	Enable to maintain proportions when resizing

Rotate an Object

Method	Steps
Rotation Handle	Select object → Drag green rotation handle (circular arrow)
Rotate Menu	Format → Rotate → Select rotation option
Flip	Format → Rotate → Flip Horizontal or Vertical

5.3.5 Creating & Using Master Slide

The **Master Slide** (or Slide Master) controls the overall appearance of all slides in a presentation. Changes made to the master slide apply to all slides that use that master, ensuring consistency .

What Can Be Set on Master Slide

Element	Description
Background	Color, gradient, picture, or texture for all slides

Element	Description
Fonts	Default title and body fonts
Colors	Theme colors for text, shapes, and backgrounds
Placeholders	Position of title, content, footer, date, slide number
Logos	Company logo appears on every slide
Headers/Footers	Consistent header and footer text
Bullet Styles	Default bullet appearance

Access Slide Master

Microsoft PowerPoint	LibreOffice Impress
View → Slide Master	View → Master Slide
Or Shift + Click Normal View button	Or Master View button on status bar

Slide Master Structure

```text
┌──────────────────────────────────────────────────────────────────┐
│                      SLIDE MASTER                                 │
│   ┌──────────────────────────────────────────────────────────┐   │
│   │  Click to edit Master title style                         │   │
│   │  ────────────────────────────────────────────────────────│   │
│   │  Click to edit Master text styles                         │   │
│   │    • Second level                                         │   │
│   │        • Third level                                      │   │
│   │            • Fourth level                                 │   │
│   │                • Fifth level                              │   │
│   └──────────────────────────────────────────────────────────┘   │
│   ┌──────────────────────────┐  ┌──────────────────────────┐    │
│   │      Footer area         │  │  Date / Slide Number     │    │
│   └──────────────────────────┘  └──────────────────────────┘    │
```

```
└──────────────────────────────────────────────────────────────┘
        │
        ▼
┌──────────────────────────────────────────────────────────────┐
│                LAYOUT MASTERS (Child Slides)                 │
│  Title Layout │ Title & Content │ Two Content │ Blank, etc.  │
└──────────────────────────────────────────────────────────────┘
```

Making Changes to Master Slide

Step	Action
1	Switch to Slide Master view
2	Select the topmost master slide (affects all layouts)
3	Make changes (background, font, logo, etc.)
4	Optionally modify individual layout masters
5	Click "Close Master View" to return to normal view

Benefits of Using Master Slide

Benefit	Description
Consistency	Ensures uniform appearance across all slides
Efficiency	Make one change to affect all slides
Branding	Easy to add company logos and colors
Professionalism	Creates polished, cohesive presentations

5.4 Presentation of Slides

5.4.1 Choosing a Set Up for Presentation

Set Up Slide Show

Setting	Microsoft PowerPoint	LibreOffice Impress
Access	Slide Show → Set Up Slide Show	Slide Show → Slide Show Settings
Show Type	Presented by speaker, Browsed by individual, Browsed at kiosk	Full screen, Window

Presentation Options

Option	Description
Presented by Speaker	Full screen, presenter controls, most common
Browsed by Individual	Windowed mode, user controls navigation
Browsed at Kiosk	Continuous loop, no user controls, for unattended displays
Loop Continuously	Restarts after last slide
Show Without Narration	Skip recorded narration
Show Without Animation	Skip animations

5.4.2 Running a Slide Show

Start Slide Show

Method	Microsoft PowerPoint	LibreOffice Impress
From First Slide	Slide Show → From Beginning (F5)	Slide Show → Start from First (F5)
From Current Slide	Slide Show → From Current Slide (Shift+F5)	Slide Show → Start from Current (Shift+F5)
Status Bar	Click Slide Show icon on status bar	Click Slide Show icon on status bar

Navigation During Slide Show

Action	Keyboard	Mouse
Next Slide	Right Arrow, Down Arrow, Spacebar, N, Enter	Left Click
Previous Slide	Left Arrow, Up Arrow, P, Backspace	Right Click → Previous
Go to Specific Slide	Slide number + Enter	Right Click → Go to Slide
End Show	Esc	Right Click → End Show
Black Screen	B (B for Black)	Right Click → Screen → Black Screen
White Screen	W (W for White)	Right Click → Screen → White Screen

Action	Keyboard	Mouse
Pen/Laser Pointer	Ctrl + P (pen), Ctrl + L (laser)	Right Click → Pointer Options

---

5.4.3 Transition and Slide Timings

Slide Transitions

Transitions are visual effects that occur when moving from one slide to the next during a slide show .

Transition Category	Examples
Subtle	Fade, Push, Wipe, Split
Exciting	Curtains, Doors, Vortex, Ripple
Dynamic Content	Pan, Zoom, Ferris Wheel

Apply Transition in Microsoft PowerPoint

Steps

Select slide(s) in thumbnail pane

Transitions tab → Choose transition effect

Set Effect Options for direction/variation

Adjust Duration (slower/faster)

Steps

Set Advance Slide: On Mouse Click or After (timed)

Click Apply to All for same transition on all slides

Apply Transition in LibreOffice Impress

Steps

Select slide(s)

Slide → Slide Transition (or Sidebar → Slide Transition)

Choose transition from list

Set Speed (Slow, Medium, Fast)

Set Advance slide: On mouse click or Automatically after

Click Apply to All if needed

Transition Settings

Setting	Purpose
Duration	How long the transition takes (e.g., 1.00 seconds)
Sound	Optional audio during transition
On Mouse Click	Advance to next slide when clicked

Setting	Purpose
After (Timing)	Automatically advance after specified time
Apply to All	Apply same transition to every slide

5.4.4 Automating a Slide Show

An **automated slide show** advances slides automatically based on set timings, without manual clicks .

Setting Automatic Timings

Microsoft PowerPoint	LibreOffice Impress
Transitions tab → Advance Slide → "After" checkbox → Set time	Sidebar → Slide Transition → "Automatically after" → Set time
Apply to All for consistent timings	Apply to All for consistent timings

Rehearse Timings (PowerPoint)

Steps

Slide Show → Rehearse Timings

Presentation opens in full screen with recording toolbar

Click through slides at desired pace

Steps

Timings are automatically saved

Recording Narration (PowerPoint)

Steps

Slide Show → Record Slide Show

Record narration and slide timings

Audio is saved with each slide

Loop Continuously

Microsoft PowerPoint	LibreOffice Impress
Slide Show → Set Up Slide Show → Check "Loop continuously until 'Esc'"	Slide Show → Slide Show Settings → Select "Auto" and "Repeat until Esc"

Self-Running Presentation

To create a presentation that runs unattended (e.g., at a kiosk):

Setting	Value
Show Type	Browsed at a kiosk (full screen)
Loop	Continuously until Esc
Slide Advance	Automatic timings set on all slides

Setting	Value
Animations	Set to start automatically

5.5 Providing Aesthetics to Slides & Printing

5.5.1 Enhancing Text Presentation

Text Formatting Options

Format	Description	Shortcut
Font	Change typeface	-
Font Size	Increase/decrease text size	Ctrl+Shift+> / <
Bold	Thicker text	Ctrl+B
Italic	Slanted text	Ctrl+I
Underline	Underlined text	Ctrl+U
Text Shadow	Adds shadow effect	-
WordArt	Stylized text with effects	-
Text Fill	Color of text	-
Text Outline	Border around text	-

Format	Description	Shortcut
Text Effects	Shadow, reflection, glow, bevel, 3D rotation	-

Bullets and Numbering

Option	Purpose
Bulleted List	For unordered items (•, ◦, ▪)
Numbered List	For sequential steps (1, 2, 3)
Multilevel List	Create sub-bullets (Tab to indent)

5.5.2 Working with Color and Line Style

Shape Fill (Background Color)

Option	Description
Solid Fill	Single uniform color
Gradient Fill	Blend of two or more colors
Picture or Texture Fill	Image or pattern as background
No Fill	Transparent background

Shape Outline (Line Style)

Option	Description
Color	Line color
Weight	Thickness of line (e.g., 1 pt, 3 pt)
Dash Type	Solid, dashed, dotted
Arrow Type	Arrowheads for lines

Shape Effects

Effect	Description
Shadow	Adds depth with shadow
Reflection	Mirror image below shape
Glow	Soft colored glow around shape
Bevel	3D edge effect
3D Rotation	Rotate shape in 3D space

5.5.3 Adding Movie and Sound

Insert Video

Microsoft PowerPoint	**LibreOffice Impress**
Insert → Media → Video → This Device	Insert → Audio/Video → Select file
Online Video → Paste embed code	Insert → Media → Video from File
Supported: .mp4, .mov, .avi, .wmv	Supported: .mp4, .avi, .mov (requires codecs)

Insert Audio

Microsoft PowerPoint	**LibreOffice Impress**
Insert → Media → Audio → This Device	Insert → Audio/Video → Select audio file
Record Audio → Record narration	Insert → Media → Audio from File
Supported: .mp3, .wav, .m4a	Supported: .mp3, .wav

Playback Options

Option	**Description**
Automatically	Media plays when slide appears
On Click	Media plays when clicked
Loop Until Stopped	Media repeats continuously

Option	Description
Play Across Slides	Media continues playing across multiple slides
Hide During Show	Media icon hidden during slideshow

5.5.4 Adding Headers, Footers and Notes

Headers and Footers

Headers appear at the top of slides; **Footers** appear at the bottom .

Element	Purpose
Date and Time	Current date, fixed date, or slide update date
Slide Number	Auto-numbered slide numbers
Footer	Custom text (e.g., company name, presentation title)

Add Header/Footer in Microsoft PowerPoint

Steps

Insert → Header & Footer

Slide tab: Check Date and time, Slide number, Footer

Notes and Handouts tab: For printed notes pages

Steps

Click "Apply" for selected slide or "Apply to All"

Add Header/Footer in LibreOffice Impress

Steps

Insert → Header and Footer

Slide tab: Check Date/Time, Slide number, Footer text

Notes tab: For notes pages

Click Apply to All

Speaker Notes

Speaker Notes are private notes visible to the presenter during the slide show (not shown to audience).

Add Notes

Microsoft PowerPoint	**LibreOffice Impress**
Click "Notes" below slide	Click "Notes" view
Type notes in notes pane	View → Notes Page
Notes visible in Presenter View	Type notes in notes area

Presenter View

Presenter View shows the current slide, next slide, speaker notes, and slide controls on the presenter's screen while the audience sees only the current slide.

Microsoft PowerPoint	LibreOffice Impress
Slide Show → Presenter View	Slide Show → Presenter Console

5.5.5 Printing Slides and Handouts

Print Options

Print Layout	Description
Full Page Slides	One slide per page (standard print)
Handouts	Multiple slides per page (2, 3, 4, 6, or 9)
Notes Pages	Slide at top, speaker notes below
Outline	Only text content, no graphics

Print Settings

Setting	Description
Printer	Select printer
Copies	Number of copies

Setting	Description
Print Range	All slides, selection, current slide, or custom range
Collate	Print copies in order
Color	Color, Grayscale, or Black and White

Print Handouts

Handouts are useful for distributing to audience members .

Microsoft PowerPoint	LibreOffice Impress
File → Print → Under Settings, select "Handouts"	File → Print → Under "Print" section, select "Handouts"
Choose slides per page (2, 3, 4, 6, 9)	Choose slides per page
Options: Frame slides, Print comments	Options: Print hidden slides, Print in color/greyscale

Handout Layout Options

Slides per Page	Best Used For
2 slides	Detailed slides with large text
3 slides	Includes lines for audience notes (most popular)
4 slides	More slides per page, smaller text
6 slides	Overview/reference

Slides per Page	Best Used For
9 slides	Summary view

5.6 Summary

- **Presentation software** creates slide shows for communicating information visually (Microsoft PowerPoint, LibreOffice Impress) .
- **Templates** provide pre-designed layouts for consistent styling; **blank presentations** start from scratch .
- **Slides** can be inserted, deleted, duplicated, and rearranged; each slide has a **layout** determining placeholder positions .
- **Master Slide** controls overall design (background, fonts, logos) for all slides, ensuring consistency .
- **Objects** include tables, pictures, shapes, SmartArt, charts, and media; can be resized, rotated, and formatted .
- **Slide transitions** are effects when moving between slides; can be set to advance on click or automatically after timing .
- **Automated shows** use rehearsed timings and can loop continuously for kiosk displays .
- **Text enhancements** include fonts, colors, WordArt, bullets, and numbering .
- **Media** (video, audio) can be inserted and configured to play automatically or on click .
- **Headers, footers, slide numbers, and speaker notes** add professionalism and presenter aids .
- **Print options**: full-page slides, handouts (multiple slides per page), notes pages, and outline view .

5.7 Model Questions and Answers

Multiple Choice Questions (MCQs)

1. Which of the following is a presentation software?

- a) Microsoft Word
- b) Microsoft Excel
- c) Microsoft PowerPoint
- d) Microsoft Access

Answer: c) Microsoft PowerPoint

---

2. The shortcut key to start a slide show from the first slide is:

- a) F1
- b) F5
- c) F7
- d) F12

Answer: b) F5

---

3. Which view is used to make changes that affect all slides?

- a) Normal View
- b) Slide Sorter View
- c) Slide Master View
- d) Outline View

Answer: c) Slide Master View

---

4. Which feature controls how one slide moves to the next during a presentation?

- a) Animation
- b) Transition
- c) Timing
- d) Effect

Answer: b) Transition

---

5. The default file extension of LibreOffice Impress is:

- a) .pptx
- b) .odp
- c) .ppt
- d) .odt

Answer: b) .odp

---

6. Which print layout prints multiple slides per page for audience distribution?

- a) Full Page Slides
- b) Notes Pages
- c) Handouts
- d) Outline

Answer: c) Handouts

---

7. Which key is used to end a slide show?

- a) Enter
- b) Spacebar
- c) Esc
- d) Tab

Answer: c) Esc

---

8. Speaker notes are visible to:

- a) Audience only
- b) Presenter only
- c) Both presenter and audience
- d) Neither

Answer: b) Presenter only

---

9. Which option is used to insert a pre-designed diagram like an organization chart?

- a) SmartArt
- b) ClipArt
- c) Shape
- d) Chart

Answer: a) SmartArt

---

10. The shortcut to insert a new slide is:

- a) Ctrl + N
- b) Ctrl + M
- c) Ctrl + S
- d) Ctrl + P

Answer: b) Ctrl + M

True/False Questions

Statement	Answer
A template can be used to create a presentation with a consistent look.	**True**
Slide transitions control how individual objects appear on a slide.	**False** (Transitions control slide-to-slide movement; animations control objects on a slide)
The Master Slide is used to make changes that apply to all slides.	**True**
Handouts print one slide per page.	**False** (Handouts print multiple slides per page)
F5 starts the slide show from the current slide.	**False** (F5 starts from beginning; Shift+F5 starts from current)

Short Answer Questions

Q1: What is the difference between a template and a blank presentation?
Ans:

- **Template**: Pre-designed presentation with layouts, colors, fonts, and backgrounds already set .
- **Blank Presentation**: Starts with a single empty slide; user designs everything from scratch.

Q2: What is a Master Slide? What is its purpose?
Ans: The Master Slide controls the overall appearance of all slides in a presentation . Changes made to the master slide (background, fonts, logos, placeholders) automatically apply to all slides, ensuring consistency and saving time.

Q3: Differentiate between transition and animation.
Ans:

- **Transition**: Visual effect that occurs when moving from one slide to the next .
- **Animation**: Visual effect applied to individual objects (text, images, shapes) on a single slide.

Q4: How do you set a slide show to run automatically without mouse clicks?
Ans:

1. Set transitions to "After" with specific timings
2. Apply to all slides

3. For PowerPoint: Set Up Slide Show → Browsed at a kiosk
4. For Impress: Slide Show Settings → Auto and Repeat until Esc

---

Q5: What are the different ways to print a presentation?
Ans:

- **Full Page Slides**: One slide per page
- **Handouts**: Multiple slides per page (2, 3, 4, 6, or 9)
- **Notes Pages**: Slide with speaker notes below
- **Outline**: Only text content, no graphics

Practical Lab Exercises

Exercise No.	Task	Purpose
1	Create a new blank presentation with 5 slides (Title, 3 content slides, Conclusion)	Basic presentation creation
2	Create a presentation using a template; customize colors and fonts	Template usage
3	Insert, delete, duplicate, and rearrange slides	Slide management
4	Add text, images, shapes, and SmartArt to slides	Object insertion

Exercise No.	Task	Purpose
5	Insert a table showing monthly sales data	Table insertion
6	Modify the Master Slide: add logo, change background, set footer	Master Slide usage
7	Apply different transitions to slides; set automatic timings	Transitions and timings
8	Add speaker notes to each slide	Notes practice
9	Insert a video and audio clip; set playback options	Media insertion
10	Add headers, footers, and slide numbers to all slides	Headers and footers
11	Run the slide show in Presenter View	Presentation practice
12	Print handouts with 3 slides per page	Printing practice

Important Points for UP CCC Exam

Topic	Key Points to Remember
Software	Microsoft PowerPoint (proprietary), LibreOffice Impress (open source)
File Extensions	.pptx (PowerPoint), .odp (Impress)
Shortcuts	F5 (Start show), Shift+F5 (From current), Ctrl+M (New slide), Esc (End show)

Topic	Key Points to Remember
Master Slide	View → Slide Master (PowerPoint), View → Master Slide (Impress)
Transition	Effect between slides; set On Click or After timing
Animation	Effect on individual objects
Speaker Notes	Visible to presenter; not to audience
Handouts	Multiple slides per page; 3 slides per page includes note lines
Media	Insert video/audio; set to Auto or On Click

Chapter 6: Introduction to Internet and WWW

6.0 Introduction

The **Internet** is a global network of interconnected computers that communicate with each other using standardized protocols. It has revolutionized the way we communicate, access information, conduct business, and entertain ourselves.
The **World Wide Web (WWW)** is one of the most widely used services on the Internet, providing access to billions of web pages containing text, images, videos, and interactive content.

For CCC exam candidates, understanding internet fundamentals is essential as digital literacy is a key requirement for government jobs in Uttar Pradesh.

6.1 Objectives

After studying this chapter, you will be able to:

- Understand basic computer network concepts (LAN, WAN, topology)
- Explain the concept of Internet and World Wide Web (WWW)
- Identify applications of the Internet
- Understand website addresses (URL) and IP addresses
- Explain ISP and its role
- Understand different modes of connecting to the Internet
- Identify IP/MAC/IMEI addresses of various devices
- Use popular web browsers
- Surf the web and use search engines effectively
- Download and print web pages

6.2 Basics of Computer Networks

A **computer network** is a collection of interconnected computers and devices that can communicate and share resources with each other.

6.2.1 Local Area Network (LAN)

LAN is a network that connects computers within a limited geographical area such as a building, office, school, or campus.

Feature	Description
Geographical Area	Small (within a building or campus)
Speed	High (100 Mbps to 10 Gbps)
Ownership	Private (owned by organization)
Cost	Relatively low
Examples	Office network, school computer lab, home network
Media	Ethernet cables, Wi-Fi

Characteristics of LAN:

- High data transfer rate
- Low error rate
- Limited geographical coverage
- Typically uses Ethernet technology
- Can be wired (UTP cables) or wireless (Wi-Fi)

6.2.2 Wide Area Network (WAN)

WAN is a network that covers a large geographical area, often spanning cities, countries, or even continents.

Feature	Description
Geographical Area	Large (city to global)
Speed	Variable (slower than LAN)
Ownership	Public or private
Cost	High
Examples	Internet, Bank ATM network, Railway reservation system

Comparison: LAN vs WAN

Parameter	LAN	WAN
Area	Small (building/campus)	Large (city/country/global)
Speed	Very high (Gbps)	Lower (Mbps)
Cost	Low	High
Ownership	Private	Public/Shared
Technology	Ethernet, Wi-Fi	Leased lines, Satellite, Fiber

6.2.3 Network Topology

Network topology refers to the physical or logical arrangement of computers and devices in a network.

Topology	Description	Advantages	Disadvantages
Bus Topology	All devices connected to a single central cable (backbone)	Simple, less cable	If backbone fails, entire network fails
Star Topology	All devices connected to a central hub/switch	Easy to manage, failure of one device doesn't affect others	Central hub is single point of failure
Ring Topology	Devices connected in a circular loop	Data flows in one direction, predictable	Failure of one device affects whole network
Mesh Topology	Each device connected to multiple other devices	Highly reliable, multiple paths	Expensive, complex wiring
Tree Topology	Hierarchical structure combining bus and star	Scalable, easy to manage	Central root failure affects entire network

```text

Star Topology:
        ┌─────┐
        │     │
     ┌─┤ Hub ├─┐
     │ └─────┘ │
   ┌─┴─┐     ┌─┴─┐
```

```
  │ PC1 │    │ PC2 │
  └─────┘    └─────┘


Bus Topology:
  ┌─────┐    ┌─────┐    ┌─────┐
  │ PC1 │    │ PC2 │    │ PC3 │
  └──┬──┘    └──┬──┘    └──┬──┘
     └──────────┼──────────┘
                │
            Backbone Cable
```

6.3 Internet

6.3.1 Concept of Internet & WWW

Internet

The **Internet** is a global network of billions of computers and other electronic devices connected together, allowing communication and data exchange. It is a **network of networks** that connects millions of private, public, academic, business, and government networks.

Key Characteristics:

- Global reach
- Decentralized structure
- Uses TCP/IP protocol suite
- No single owner or controller

World Wide Web (WWW)

The **World Wide Web (WWW)** is a system of interlinked hypertext documents accessed via the Internet. It is one of the services running on the Internet (other services include email, FTP, etc.).

Feature	Description
Inventor	Tim Berners-Lee (1989)
Protocol	HTTP/HTTPS
Content	Web pages, images, videos, etc.
Access	Through Web Browsers

Difference between Internet and WWW:

Internet	World Wide Web (WWW)
Physical infrastructure (hardware)	Service running on the Internet (software)
Includes cables, routers, servers	Includes web pages, websites, hyperlinks
Enables various services (email, FTP, WWW)	Primarily for accessing information via browsers
Existed before WWW	Developed in 1989

6.3.2 Applications of Internet

Application	Description	Examples
Communication	Exchange messages and information	Email, WhatsApp, Zoom, Skype
Information Search	Access knowledge and data	Google, Wikipedia, YouTube

Application	Description	Examples
E-Commerce	Buying and selling online	Amazon, Flipkart, OLX
E-Governance	Government services online	UMANG, DigiLocker, e-District
Banking	Financial transactions	Net Banking, UPI, Mobile Banking
Education	Online learning	MOOCs, ePathshala, SWAYAM
Entertainment	Streaming media	YouTube, Netflix, Spotify
Social Networking	Connect with people	Facebook, Instagram, Twitter
Job Search	Employment opportunities	Naukri.com, LinkedIn
Travel	Book tickets, hotels	IRCTC, MakeMyTrip
Healthcare	Medical services	Telemedicine, e-Hospital

6.3.3 Website Address and URL

URL (Uniform Resource Locator) is the address of a web page or resource on the Internet.

Structure of URL

```text
https://www.google.com/search?q=ccc+exam
  |       |         |     |   |
Protocol Domain    Path  ?  Query String
         Name
```

Component	Description	Example
Protocol	Method of communication	http://, https://, ftp://
Domain Name	Unique name of website	www.google.com
Path	Specific location on website	/search
Query String	Parameters (after ?)	?q=ccc+exam

Parts of Domain Name

```text
www.google.co.in
|  |      |  |
|  |      |  └── Country Code Top-Level Domain (ccTLD)
|  |      └────── Top-Level Domain (TLD)
|  └─────────────── Second-Level Domain
└────────────────── Subdomain
```

TLD	Purpose
.com	Commercial organizations
.org	Non-profit organizations
.edu	Educational institutions
.gov	Government entities
.in	India
.uk	United Kingdom

6.3.4 Introduction to IP Address

An **IP Address (Internet Protocol Address)** is a unique numerical identifier assigned to each device connected to a network. It serves two main functions: host identification and location addressing.

Types of IP Address

Type	Description	Example
IPv4	32-bit address (4 numbers separated by dots)	192.168.1.1
IPv6	128-bit address (8 groups of hex digits)	2001:0db8:85a3::8a2e:0370:7334

IP Address Classes (IPv4)

Class	Range	Purpose
A	1.0.0.0 to 126.255.255.255	Large networks
B	128.0.0.0 to 191.255.255.255	Medium networks
C	192.0.0.0 to 223.255.255.255	Small networks
D	224.0.0.0 to 239.255.255.255	Multicast
E	240.0.0.0 to 255.255.255.255	Experimental

Public vs Private IP Address

Type	Purpose	Private IP Ranges
Public IP	Unique globally, accessible from Internet	Not applicable
Private IP	Used within local networks, not routable on Internet	10.0.0.0 – 10.255.255.255 172.16.0.0 – 172.31.255.255 192.168.0.0 – 192.168.255.255

6.3.5 ISP and Role of ISP

ISP (Internet Service Provider) is a company that provides individuals and organizations with access to the Internet.

ISP	Description
Public ISPs	BSNL, Airtel, Jio, Vodafone Idea, ACT Fibernet

Roles of ISP

Role	Description
Provide Internet Access	Delivers connectivity through various technologies
Assign IP Address	Provides unique IP to each subscriber
Maintain Infrastructure	Manages cables, routers, servers
Email Services	Often provides email accounts

Role	Description
Web Hosting	Some ISPs offer web hosting services
Technical Support	Provides customer support for connectivity issues

---

6.3.6 Internet Protocol

Internet Protocol (IP) is a set of rules governing the format of data sent over the Internet or local network.

TCP/IP Protocol Suite

The Internet uses the **TCP/IP (Transmission Control Protocol/Internet Protocol)** suite.

Protocol	Layer	Function
IP	Network Layer	Addressing and routing packets
TCP	Transport Layer	Reliable, connection-oriented delivery
UDP	Transport Layer	Faster, connectionless delivery
HTTP/HTTPS	Application Layer	Web browsing
FTP	Application Layer	File transfer
SMTP/POP3	Application Layer	Email

Protocol	Layer	Function
DNS	Application Layer	Domain name resolution

6.3.7 Modes of Connecting Internet

Connection Type	Description	Speed	Mobility
LAN Cable (Ethernet)	Wired connection using Ethernet cable	High (100 Mbps – 1 Gbps)	Fixed
Wi-Fi	Wireless local area network	Medium to High	Within range
Broadband	High-speed internet (DSL, Fiber)	High (up to 1 Gbps)	Fixed
Mobile Data (4G/5G)	Cellular network internet	High	Mobile
Hotspot	Sharing mobile data with other devices	Depends on source	Mobile
USB Tethering	Sharing internet via USB cable	Medium	Mobile
Satellite	Internet via satellite connection	Variable	Remote areas

Hotspot

A **hotspot** is a physical location where people can access the Internet, typically via Wi-Fi, using a router connected to an ISP.

Types:

- **Mobile Hotspot**: Using smartphone to share cellular data
- **Public Hotspot**: Free or paid Wi-Fi in public places (cafes, airports, hotels)

Wi-Fi (Wireless Fidelity)

Wi-Fi is a wireless networking technology that allows devices to connect to the Internet without cables. It operates on radio frequencies (2.4 GHz and 5 GHz).

Broadband

Broadband refers to high-speed internet access that is always on and faster than traditional dial-up. Types include:

- **DSL (Digital Subscriber Line)**: Through telephone lines
- **Fiber Optic**: Through fiber cables (fastest)
- **Cable**: Through cable TV lines

USB Tethering

USB tethering allows sharing a smartphone's internet connection with a computer via a USB cable.

6.3.8 Identifying and Uses of IP/MAC/IMEI of Various Devices

IP Address (Internet Protocol Address)

Aspect	Description
Purpose	Identifies device on network, enables routing
Format	IPv4 (192.168.1.1) or IPv6
Change	Can change (dynamic IP) or fixed (static IP)
How to Find	`ipconfig` (Windows), `ifconfig` (Linux)

MAC Address (Media Access Control Address)

Aspect	Description
Purpose	Unique hardware identifier for network interface
Format	12-digit hexadecimal (e.g., 00:1A:2B:3C:4D:5E)
Permanence	Permanent (burned into hardware)
How to Find	`ipconfig /all` (Windows), `ifconfig` (Linux)

Difference between IP and MAC Address:

IP Address	MAC Address
Logical address	Physical address

IP Address	MAC Address
Assigned by network	Burned into hardware
Can change	Permanent
Used for routing	Used within local network
32-bit (IPv4) or 128-bit (IPv6)	48-bit

IMEI Number (International Mobile Equipment Identity)

Aspect	Description
Purpose	Unique identifier for mobile devices
Format	15-digit number
Use	Tracking stolen phones, network authentication
How to Find	Dial `*#06#` on phone, or check phone settings

6.4 Popular Web Browsers

A **web browser** is software application used to access and view websites on the World Wide Web.

Browser	Developer	Features	Default OS
Google Chrome	Google	Fast, extensive extensions, sync across devices	Android, Chrome OS
Mozilla Firefox	Mozilla Foundation	Privacy-focused, open source	Linux (often default)
Microsoft Edge	Microsoft	Modern, integrated with Windows	Windows
Safari	Apple	Optimized for Apple devices	macOS, iOS
Opera	Opera Software	Built-in VPN, ad blocker	Cross-platform
Internet Explorer	Microsoft	Older browser (discontinued)	Legacy Windows

Common Browser Features

Feature	Description
Address Bar	Enter URL to navigate
Bookmarks/Favorites	Save frequently visited websites
Tabs	Open multiple web pages in one window
History	Record of visited websites
Downloads	Manage downloaded files

Feature	Description
Incognito/Private Mode	Browsing without saving history
Extensions/Add-ons	Additional functionality

6.5 Exploring the Internet

6.5.1 Surfing the Web

Web surfing (or browsing) is the activity of navigating through websites by clicking hyperlinks.

Basic Navigation

Action	Method
Go to Website	Type URL in address bar → Press Enter
Click Link	Click on hyperlink (usually underlined/colored text)
Back/Forward	Click back/forward buttons
Refresh	Click refresh button (↻) or press F5
Home	Click home button to go to default page
New Tab	Ctrl + T
Close Tab	Ctrl + W

Action	Method
New Window	Ctrl + N

6.5.2 Popular Search Engines

A **search engine** is a web-based tool that searches for information on the World Wide Web based on user queries.

Search Engine	Developer	Features
Google	Google	Most popular, comprehensive search
Bing	Microsoft	Visual search, rewards program
Yahoo!	Yahoo	Web search, news, email
DuckDuckGo	DuckDuckGo	Privacy-focused, no tracking
Baidu	Baidu	Leading search engine in China

6.5.3 Searching on Internet

Effective Search Techniques

Technique	Example	Purpose
Use Keywords	`CCC exam syllabus`	Search for specific topics
Use Quotes	`"computer concepts"`	Exact phrase search
Use Minus (-)	`apple -fruit`	Exclude terms
Use Site:	`site:gov.in CCC`	Search within specific site
Use Filetype:	`CCC syllabus filetype:pdf`	Search for specific file type
Use OR	`CCC OR NIELIT`	Search for either term
Use Asterisk (*)	`CCC * course`	Wildcard for unknown words

Search Operators

Operator	Function
`" "`	Exact phrase match
`-`	Exclude word
`site:`	Search specific website
`filetype:`	Search specific file format
`OR`	Either term

Operator	Function
`intitle:`	Word in title
`inurl:`	Word in URL

6.5.4 Downloading Web Pages

Download Web Page

Method	Steps
Save as HTML	Right-click on page → Save As → Choose location → Save
Copy Content	Select text → Ctrl+C → Paste into document
Print to PDF	Print → Select "Save as PDF"
Browser Download	Click download link for files

Download Files

File Type	Description
Documents	PDF, DOCX, XLSX
Images	JPG, PNG, GIF
Videos	MP4, AVI

File Type	Description
Software	EXE, MSI, DEB

6.5.5 Printing Web Pages

Print Web Page

Method	Steps
Browser Menu	Click menu (⋮) → Print
Keyboard Shortcut	Ctrl + P
Right-click	Right-click on page → Print

Print Settings

Setting	Description
Printer	Select printer or "Save as PDF"
Copies	Number of copies
Pages	All, current page, or custom range
Layout	Portrait or Landscape
Color	Color or Black & White

Setting	Description
Simplify Page	Remove ads, optimize for printing
Headers/Footers	Include date, URL, page numbers

6.6 Summary

- **Computer Network**: Interconnected computers sharing resources.
- **LAN**: Local area network (small area, high speed).
- **WAN**: Wide area network (large area, lower speed).
- **Internet**: Global network of networks using TCP/IP.
- **WWW (World Wide Web)** : System of interlinked hypertext documents accessed via browsers.
- **URL**: Web address with protocol, domain name, path, and query string.
- **IP Address**: Unique numerical identifier for devices (IPv4, IPv6).
- **ISP**: Internet Service Provider (BSNL, Airtel, Jio, etc.).
- **Connection Modes**: LAN cable, Wi-Fi, broadband, mobile data, hotspot, USB tethering.
- **IP Address**: Logical address (can change).
- **MAC Address**: Physical hardware address (permanent).
- **IMEI**: Unique 15-digit identifier for mobile devices.
- **Web Browsers**: Chrome, Firefox, Edge, Safari, Opera.
- **Search Engines**: Google, Bing, Yahoo, DuckDuckGo.
- **Search Techniques**: Quotes, site:, filetype:, minus operator.

6.7 Model Questions and Answers

Multiple Choice Questions (MCQs)

1. What does LAN stand for?

- a) Large Area Network
- b) Local Area Network
- c) Long Area Network
- d) Logical Area Network

Answer: b) Local Area Network

---

2. Which of the following is a search engine?

- a) Google Chrome
- b) Mozilla Firefox
- c) Google
- d) Microsoft Edge

Answer: c) Google (Chrome, Firefox, Edge are browsers)

---

3. The protocol used for web browsing is:

- a) FTP
- b) SMTP
- c) HTTP/HTTPS
- d) TCP

Answer: c) HTTP/HTTPS

---

4. What is the full form of URL?

- a) Uniform Resource Locator
- b) Universal Resource Link
- c) Uniform Response Locator
- d) Universal Response Link

Answer: a) Uniform Resource Locator

---

5. Which of the following is a private IP address range?

- a) 192.168.1.1
- b) 8.8.8.8
- c) 1.1.1.1
- d) 172.32.1.1

Answer: a) 192.168.1.1

---

6. The shortcut key to refresh a web page is:

- a) F1
- b) F3
- c) F5
- d) F7

Answer: c) F5

---

7. Which of the following is a web browser?

- a) Google
- b) Bing
- c) Mozilla Firefox
- d) Yahoo

Answer: c) Mozilla Firefox

---

8. What is the full form of ISP?

- a) Internet Service Protocol
- b) Internet Service Provider
- c) International Service Provider
- d) Internet System Protocol

Answer: b) Internet Service Provider

---

9. Which device uses IMEI number?

- a) Laptop
- b) Desktop
- c) Mobile Phone
- d) Printer

Answer: c) Mobile Phone

---

10. The keyboard shortcut to open a new tab in a browser is:

- a) Ctrl + N
- b) Ctrl + T
- c) Ctrl + W
- d) Ctrl + Shift + T

Answer: b) Ctrl + T

True/False Questions

Statement	Answer
LAN covers a large geographical area.	**False** (LAN covers small area; WAN covers large area)
WWW and Internet are the same thing.	**False** (WWW is a service on the Internet)
MAC address is a permanent hardware identifier.	**True**
Google Chrome is a search engine.	**False** (It is a web browser)
URL is the address of a web page.	**True**

Short Answer Questions

Q1: Differentiate between LAN and WAN.
Ans:

- **LAN**: Covers small area (building/campus), high speed, private ownership.
- **WAN**: Covers large area (city/country), lower speed, public/shared ownership.

Q2: What is the difference between Internet and WWW?

Ans:

- **Internet**: Physical infrastructure of networks (hardware, cables, routers).
- **WWW**: Service running on Internet providing access to web pages (software).

Q3: Explain the structure of a URL with an example.

Ans: URL = Protocol + Domain Name + Path + Query String

Example: `https://www.google.com/search?q=ccc`

- `https://` = Protocol
- `www.google.com` = Domain Name
- `/search` = Path
- `?q=ccc` = Query String

Q4: What is the difference between IP address and MAC address?

Ans:

- **IP Address**: Logical address, assigned by network, can change.
- **MAC Address**: Physical address, burned into hardware, permanent.

Q5: List five applications of the Internet.

Ans:

1. Communication (Email, WhatsApp)

2. Information Search (Google, Wikipedia)
3. E-Commerce (Amazon, Flipkart)
4. E-Governance (UMANG, DigiLocker)
5. Education (SWAYAM, ePathshala)

Practical Lab Exercises

Exercise No.	Task	Purpose
1	Identify LAN and WAN connections in your computer lab	Understand network types
2	Find IP address of your computer using `ipconfig` command	IP address identification
3	Find MAC address using `ipconfig /all`	MAC address identification
4	Find IMEI number of mobile phone by dialing `*#06#`	IMEI identification
5	Open Chrome/Firefox and navigate to https://nielit.gov.in	Browser navigation
6	Bookmark the NIELIT website	Bookmarking practice
7	Search for "CCC syllabus" on Google using different search techniques	Search engine usage
8	Download a PDF file from a website	Downloading practice
9	Save a web page as HTML file	Web page saving

Exercise No.	Task	Purpose
10	Print a web page (or save as PDF)	Printing practice
11	Open multiple tabs and switch between them	Tab management
12	Use incognito/private browsing mode	Privacy features

Important Points for UP CCC Exam

Topic	Key Points to Remember
LAN vs WAN	LAN small area, high speed; WAN large area, lower speed
Internet	Network of networks
WWW	Service for web pages (HTTP/HTTPS)
URL	Web address: protocol://domain/path
IP Address	Logical identifier (192.168.1.1)
MAC Address	Physical hardware identifier (00:1A:2B:3C:4D:5E)
IMEI	Mobile phone identifier (dial `*#06#`)
ISP	BSNL, Airtel, Jio, etc.

Topic	Key Points to Remember
Browser	Chrome, Firefox, Edge, Safari
Search Engine	Google, Bing, Yahoo
Shortcut	Ctrl+T (New tab), Ctrl+W (Close tab), F5 (Refresh)

Chapter 7: E-mail, Social Networking and eGovernance Services

7.0 Introduction

In today's digital world, **electronic mail (E-mail)** has become one of the most essential communication tools. It allows instant transmission of messages, documents, and multimedia across the globe. Along with email, **social networking platforms** have revolutionized how people connect, share information, and build professional networks. The Government of India has also embraced digital transformation through various **e-Governance services** that bring public services directly to citizens' devices.

For CCC exam candidates in Uttar Pradesh, understanding these digital tools is crucial as they form the backbone of modern office communication and government service delivery.

7.1 Objectives

After studying this chapter, you will be able to:

- Understand the structure of an email address
- Create and manage an email account
- Perform email operations: compose, send, reply, forward
- Attach files to emails
- Manage mailbox (Inbox, Outbox, Sent, Spam)
- Use email signature
- Understand social networking platforms
- Identify netiquettes for online communication
- Access various e-Governance services
- Use UMANG app for government services

- Understand Digital Locker

7.2 Structure of E-mail

An **email address** uniquely identifies a mailbox where messages can be delivered.

Email Address Format

```text
username@domainname.com
    │          │
    │          └── Domain Name (mail server)
    └──────────── User/Account Name
```

Parts of an Email Address

Component	Description	Example
Username	Unique identifier chosen by user	`ram.sharma`
@ Symbol	Separator (pronounced "at")	`@`
Domain Name	Mail service provider or organization	`gmail.com`, `yahoo.com`, `gov.in`

Email Header Fields

When composing an email, the following fields are essential:

Field	Purpose
To	Primary recipient(s) email address

Field	Purpose
Cc	Carbon Copy – sends copy to additional recipients (visible to all)
Bcc	Blind Carbon Copy – sends copy to recipients (others cannot see)
Subject	Brief summary of email content
Body	Main message content
Attachments	Files sent along with email

7.3 Using E-mails

7.3.1 Opening Email Account

Email accounts can be created using **web-based email services** (like Gmail, Yahoo, Outlook) or through **organization-provided accounts**.

Steps to Create a Gmail Account

Step	Action
1	Go to https://mail.google.com or https://accounts.google.com
2	Click **"Create account"** → **"For myself"**
3	Enter **First name** and **Last name**
4	Create **Username** (email address before @gmail.com)

Step	Action
5	Create a strong **Password** and confirm
6	Enter **Phone number** for verification (optional but recommended)
7	Enter **Recovery email** (optional)
8	Enter **Date of birth** and **Gender**
9	Review and accept **Terms of Service**
10	Click **"Create account"**

Common Free Email Service Providers

Provider	Website	Storage	Features
Gmail	mail.google.com	15 GB shared	Integrated with Google services
Outlook/Hotmail	outlook.com	15 GB	Microsoft integration
Yahoo Mail	mail.yahoo.com	1 TB	Large storage
Rediffmail	rediffmail.com	1 GB	Indian service

7.3.2 Mailbox: Inbox and Outbox

Email accounts have various folders to organize messages.

Folder	Purpose
Inbox	Received emails (messages sent to you)
Sent	Copies of emails you have sent
Drafts	Saved emails not yet sent
Spam/Junk	Suspicious or unwanted emails automatically filtered
Trash/Bin	Deleted emails (temporarily stored)
Outbox	Temporary storage for emails waiting to be sent (in email clients)

Understanding Mailbox Structure

```text
┌──────────────────────────────────────────────────────────────┐
│                        MY MAILBOX                            │
├──────────────────────────────────────────────────────────────┤
│   Inbox         (Unread messages in bold)                   │
│   Sent          (Emails you've sent)                        │
│   Drafts        (Incomplete emails)                         │
│  ⭑ Starred       (Important messages)                       │
│  □ Trash         (Deleted messages - 30 days retention)    │
│   Spam          (Junk emails)                               │
└──────────────────────────────────────────────────────────────┘
```

7.3.3 Creating and Sending a New E-mail

Steps to Compose and Send Email

Step	Action
1	Click **"Compose"** or **"New Message"** button
2	Enter recipient's email address in **"To"** field
3	(Optional) Add email addresses in **"Cc"** or **"Bcc"** fields
4	Enter a meaningful **Subject**
5	Type your message in the **Body** area
6	(Optional) Click **"Attach"** to add files
7	Click **"Send"** button

Example Email

```text
To: employer@company.com
Cc: hr@company.com
Bcc: myself@gmail.com
Subject: Application for Computer Operator Position

Dear Sir/Madam,

I am writing to apply for the Computer Operator position advertised on your website.

I have completed CCC certification and have 2 years of experience in office administration.

My resume is attached for your reference.
```

```
Thank you for your time and consideration.

Sincerely,
Ram Sharma
```

7.3.4 Replying to an E-mail Message

When you receive an email, you can reply to the sender.

Reply Option	Purpose
Reply	Sends response only to the original sender
Reply All	Sends response to sender and all recipients in To/Cc fields

Steps to Reply

Step	Action
1	Open the email message
2	Click **"Reply"** (↩□) or **"Reply All"**
3	Type your response in the message area
4	Click **"Send"**

7.3.5 Forwarding an E-mail Message

Forwarding sends a received email to a new recipient.

Forward Option	Purpose
Forward	Send the email to someone else
Forward as Attachment	Forward as attached file (preserves original format)

Steps to Forward

Step	Action
1	Open the email message
2	Click **"Forward"** (↪□)
3	Enter recipient email address(es) in **"To"** field
4	Add any additional comments
5	Click **"Send"**

7.3.6 Searching Emails

Most email services provide powerful search capabilities to find specific messages.

Search Techniques

Search Method	Example	Purpose
Keyword Search	`CCC exam`	Find emails containing words
From:	`from:ram@gmail.com`	Emails from specific sender
To:	`to:employer@company.com`	Emails sent to specific recipient
Subject:	`subject:application`	Emails with specific subject
Has:attachment	`has:attachment`	Emails with attachments
Before/After	`after:2024/01/01`	Emails within date range
Label:	`label:work`	Emails in specific folder

7.3.7 Attaching Files with Email

Attachments allow you to send files (documents, images, videos) along with your email.

Attachment Process

Step	Action
1	Click **"Attach"** or **"Attach files"** icon (📎)
2	Browse and select file from computer

Step	Action
3	Wait for upload to complete
4	Add more files if needed
5	Click **"Send"**

Attachment Size Limits

Email Service	Maximum Attachment Size
Gmail	25 MB (can send larger via Google Drive)
Outlook	20 MB
Yahoo Mail	25 MB

Common Attachable File Types

Category	Extensions
Documents	.pdf, .doc, .docx, .xls, .xlsx, .txt
Images	.jpg, .jpeg, .png, .gif
Compressed	.zip, .rar, .7z
Presentations	.ppt, .pptx, .odp

7.3.8 Email Signature

An **email signature** is a block of text automatically appended at the end of email messages. It typically contains the sender's contact information.

Common Signature Elements

Element	Example
Name	Ram Sharma
Designation	Computer Operator
Organization	ABC Enterprises
Phone	+91-9876543210
Email	ram.sharma@abcenterprises.com
Website	www.abcenterprises.com
Disclaimer	Confidentiality notice (for corporate)

Setting Up Signature in Gmail

Step	Action
1	Click **Settings** (⚙) → **See all settings**
2	Scroll to **Signature** section
3	Select **"No signature"** or **"Create new"**

Step	Action
4	Enter signature text (format using toolbar)
5	Choose whether to include signature on new emails and replies
6	Scroll down and click **"Save Changes"**

Example Professional Signature

```text
Regards,
Ram Sharma
Computer Operator
Ministry of Electronics & IT
Phone: +91-98765-43210
Email: ram.sharma@meity.gov.in
```

7.4 Social Networking & e-Commerce

7.4.1 Facebook, Twitter, LinkedIn, Instagram

Platform	Type	Primary Use	Key Features
Facebook	Social Network	Connecting with friends, family, communities	Posts, photos, groups, pages, marketplace
Twitter (X)	Microblogging	Short updates, news, public discussions	Tweets (280 chars), hashtags, trending topics

Platform	Type	Primary Use	Key Features
LinkedIn	Professional Network	Career networking, job search, professional branding	Profile, connections, jobs, recommendations
Instagram	Visual Platform	Photo and video sharing	Stories, reels, posts, direct messaging

Facebook

- **Founder**: Mark Zuckerberg (2004)
- **Key Terms**: Profile, Friend, Like, Comment, Share, Page, Group, Timeline
- **Uses**: Social connections, business pages, event promotion, marketplace

Twitter (X)

- **Founded**: 2006 (rebranded as X in 2023)
- **Key Terms**: Tweet, Retweet, Hashtag (#), Follow, Trending, DM (Direct Message)
- **Uses**: News updates, public announcements, customer service

LinkedIn

- **Founded**: 2003
- **Key Terms**: Profile, Connection, Recommendation, Endorsement, Job Posting
- **Uses**: Professional networking, job search, skill development, business networking

Instagram

- **Founded**: 2010 (acquired by Facebook/ Meta)
- **Key Terms**: Post, Story, Reel, Follow, Like, DM, Hashtag
- **Uses**: Visual storytelling, brand promotion, influencer marketing

---

7.4.2 Instant Messaging (WhatsApp, Facebook Messenger, Telegram)

Instant Messaging (IM) allows real-time text communication between users.

Platform	Developer	Features	Encryption
WhatsApp	Meta	Text, voice/video calls, group chats, status, file sharing	End-to-end
Facebook Messenger	Meta	Text, voice/video calls, group chats, games, payments	Not fully E2E
Telegram	Telegram FZ-LLC	Text, voice/video, channels, bots, large groups	Optional secret chats
Signal	Signal Foundation	Text, voice/video, privacy-focused	End-to-end

WhatsApp Key Features

- **Founded**: 2009 (acquired by Facebook/Meta in 2014)
- **Key Terms**: Status, Broadcast, Group, Voice Note, End-to-end Encryption
- **Uses**: Personal communication, business communication (WhatsApp Business), group coordination

Netiquettes for Instant Messaging

- Don't message during odd hours unless urgent
- Avoid sending unsolicited forwards
- Respect privacy – don't add people to groups without permission
- Use appropriate language
- Don't share sensitive information without verification

7.4.3 Introduction to Blogs

A **blog** (short for "weblog") is a regularly updated website or web page, typically run by an individual or small group, that is written in an informal or conversational style.

Term	Description
Blog	Website with chronological posts
Blogger	Person who writes blog
Blog Post	Individual article
Comments	Reader feedback section
Categories/Tags	Organization of posts
RSS Feed	Subscribe to updates

Popular Blogging Platforms

Platform	Type	Features
Blogger	Free (Google)	Easy to use, integrates with Google
WordPress	Free/Paid	Most popular, highly customizable
Medium	Free/Paid	Clean interface, built-in audience
Tumblr	Free	Microblogging, multimedia focus

7.4.4 Basics of E-commerce

E-commerce (Electronic Commerce) refers to buying and selling goods and services over the Internet.

Types of E-commerce

Type	Description	Example
B2B	Business to Business	Alibaba, IndiaMART
B2C	Business to Consumer	Amazon, Flipkart
C2C	Consumer to Consumer	OLX, eBay
C2B	Consumer to Business	Freelancer, Upwork
G2C	Government to Citizen	IRCTC, Passport Seva

E-commerce Components

Component	Purpose
Website/App	Online storefront
Product Catalog	List of products with descriptions, prices
Shopping Cart	Temporary storage of selected items
Payment Gateway	Process online payments (credit/debit, UPI, net banking)

Component	Purpose
Order Management	Track orders, delivery status
Customer Support	Handle queries, returns, refunds

Popular E-commerce Platforms in India

Platform	Type	Specialization
Amazon India	B2C	Wide range of products
Flipkart	B2C	Wide range, electronics focus
Myntra	B2C	Fashion and apparel
Zomato/Swiggy	B2C	Food delivery
IRCTC	G2C	Railway ticket booking
OLX	C2C	Classifieds, used goods

7.4.5 Netiquettes

Netiquettes (Network Etiquette) are the code of acceptable behavior for online communication.

Email Netiquettes

Rule	Description
Use meaningful subject lines	Subject should summarize content
Keep messages concise	Be clear and to the point
Use professional language	Avoid slang, ALL CAPS, excessive exclamation marks
Reply promptly	Respond within reasonable time
Proofread before sending	Check spelling and grammar
Use Bcc for mass emails	Protect recipient privacy
Don't share sensitive info	Avoid passwords, financial details in email

General Online Netiquettes

Rule	Description
Be respectful	Treat others as you would like to be treated
Think before posting	Content posted online is permanent
Respect privacy	Don't share others' personal information
Don't spam	Avoid sending unsolicited messages
Use appropriate language	Avoid offensive or abusive language

Rule	Description
Give credit	Acknowledge sources when sharing content
Don't cyberbully	Harassment online is unacceptable

7.5 Overview of e-Governance Services

e-Governance (Electronic Governance) refers to the use of information technology by government agencies to deliver services to citizens.

Key e-Governance Services

Railway Reservation (IRCTC)

Aspect	Details
Website	www.irctc.co.in
Services	Train ticket booking, cancellation, PNR status, food ordering
Registration	Requires mobile verification
Payment	Credit/Debit card, UPI, Net Banking, IRCTC wallet

Passport Seva

Aspect	Details
Website	www.passportindia.gov.in
Services	Apply for new passport, renewal, appointment booking, status tracking
Process	Online registration → Payment → Appointment → Visit PSK → Police verification

eHospital [ORS] (Online Registration System)

Aspect	Details
Website	https://ors.gov.in
Services	Hospital appointment booking, lab reports, health records
Hospitals	AIIMS, Safdarjung, RML, and other government hospitals
Features	Patient registration, OPD appointment, lab report download

Other Important e-Governance Services

Service	Purpose
DigiLocker	Digital document storage
e-District	District-level services (certificates, licenses)
Income Tax e-Filing	File income tax returns online

Service	Purpose
MCA 21	Company registration and compliance
National Scholarship Portal	Apply for scholarships
Jeevan Pramaan	Digital life certificate for pensioners

7.6 Accessing e-Governance Services on Mobile Using "UMANG APP"

UMANG (Unified Mobile Application for New-age Governance) is a single platform that integrates multiple government services for citizens.

Key Features

Feature	Description
Single Platform	Access over 1,200+ government services
Multi-lingual	Available in English, Hindi, and regional languages
Secure	Aadhaar and mobile authentication
Multi-service	Central, State, and Local government services

Available Services on UMANG

Category	Services
Agriculture	Soil health card, crop insurance, MSP
Education	DigiLocker, ePathshala, NCERT
Finance	EPFO, NPS, Income Tax
Healthcare	Ayushman Bharat, eHospital
Transport	Driving license, vehicle registration
Utilities	Electricity bill, water bill, LPG
Pension	Jeevan Pramaan, pension details
Travel	IRCTC, Passport Seva

Steps to Use UMANG App

Step	Action
1	Download **UMANG** app from Google Play Store or Apple App Store
2	Open app and **register** using mobile number
3	Verify OTP received on mobile
4	Select **service** from categories or search

Step	Action
5	Authenticate using **Aadhaar** or **Mobile OTP** (as required)
6	Access service and complete transaction

7.7 Digital Locker

DigiLocker is a digital document wallet that allows citizens to store, share, and verify documents electronically.

Key Features

Feature	Description
Digital Storage	1 GB free storage per account
Issued Documents	Government-issued documents (Aadhaar, driving license, etc.) directly in DigiLocker
Uploaded Documents	Upload personal documents
e-Sign	Digitally sign documents
Sharing	Share documents securely with service providers
Authentication	Aadhaar-based authentication

Benefits of DigiLocker

Benefit	Description
Paperless	Reduces need for physical documents
Secure	Documents stored securely in cloud
Easy Access	Access anytime, anywhere
Legal Validity	Digital documents are legally valid as per IT Act
Environment Friendly	Reduces paper usage

Documents Available on DigiLocker

Category	Documents
Identity	Aadhaar, PAN, Driving License, Voter ID
Education	Marksheets, Certificates, Degrees
Vehicle	RC Book, Insurance
Health	Health records, Vaccination certificates
Other	Passport, Birth Certificate, Caste Certificate

How to Access DigiLocker

Method	Steps
Website	https://digilocker.gov.in → Sign up with Aadhaar → Login
Mobile App	Download DigiLocker app → Register with Aadhaar → Login

---

7.8 Summary

- **Email Structure**: username@domainname.com (e.g., ram@gmail.com)
- **Email Operations**: Compose, Send, Reply, Reply All, Forward, Attach files
- **Mailbox Folders**: Inbox (received), Sent, Drafts, Spam, Trash
- **Email Signature**: Automatic signature appended to messages
- **Social Media**: Facebook (social), Twitter (microblogging), LinkedIn (professional), Instagram (visual)
- **Instant Messaging**: WhatsApp, Messenger, Telegram – real-time text/voice/video
- **Blogs**: Regularly updated websites with chronological posts
- **E-commerce**: Buying/selling online (Amazon, Flipkart, IRCTC)
- **Netiquettes**: Online etiquette – be respectful, concise, professional
- **e-Governance**: Government services online (IRCTC, Passport, eHospital)
- **UMANG App**: Single platform for multiple government services
- **DigiLocker**: Digital document storage and verification

7.9 Model Questions and Answers

Multiple Choice Questions (MCQs)

1. What does the "@" symbol in an email address represent?

- a) At the rate
- b) Domain separator
- c) Both a and b
- d) None of the above

Answer: c) Both a and b

---

2. Which folder contains emails that have been received?

- a) Sent
- b) Drafts
- c) Inbox
- d) Spam

Answer: c) Inbox

---

3. What is the purpose of Bcc in an email?

- a) Send to primary recipient
- b) Send copy to additional recipients (visible to all)
- c) Send copy to additional recipients (not visible to others)
- d) Send without subject

Answer: c) Send copy to additional recipients (not visible to others)

---

4. Which of the following is a professional networking platform?

- a) Facebook
- b) Instagram
- c) LinkedIn
- d) WhatsApp

Answer: c) LinkedIn

---

5. Which app provides a single platform for multiple government services?

- a) WhatsApp
- b) DigiLocker
- c) UMANG
- d) Facebook

Answer: c) UMANG

---

6. What is the maximum attachment size in Gmail?

- a) 10 MB
- b) 15 MB
- c) 20 MB
- d) 25 MB

Answer: d) 25 MB

---

7. Which of the following is an e-commerce platform?

- a) Gmail
- b) Facebook
- c) Flipkart
- d) UMANG

Answer: c) Flipkart

---

8. What does CC stand for in email?

- a) Confidential Copy
- b) Carbon Copy
- c) Computer Copy
- d) Closed Copy

Answer: b) Carbon Copy

---

9. Which of the following is a blogging platform?

- a) WordPress
- b) WhatsApp
- c) Gmail
- d) IRCTC

Answer: a) WordPress

---

10. What is the primary purpose of DigiLocker?

- a) Send emails
- b) Social networking
- c) Store digital documents
- d) Online shopping

Answer: c) Store digital documents

True/False Questions

Statement	Answer
The Bcc field recipients are visible to all email recipients.	**False** (Bcc recipients are hidden)
Reply All sends response only to the original sender.	**False** (Reply All sends to all recipients)
LinkedIn is used for professional networking.	**True**
UMANG app provides access to multiple government services.	**True**
Netiquettes refer to network etiquette rules.	**True**

Short Answer Questions

Q1: Write the structure of an email address with an example.

Ans: An email address consists of username, @ symbol, and domain name.

Example: `ram.sharma@gmail.com`

- `ram.sharma` = username
- `@` = separator

- `gmail.com` = domain name

Q2: Differentiate between Reply and Reply All.
Ans:

- **Reply**: Sends response only to the original sender
- **Reply All**: Sends response to sender and all recipients in To and Cc fields

---

Q3: What is the difference between Cc and Bcc?
Ans:

- **Cc (Carbon Copy)** : Recipients visible to all email recipients
- **Bcc (Blind Carbon Copy)** : Recipients hidden from other recipients

---

Q4: List any five e-Governance services available in India.
Ans:

1. IRCTC – Railway ticket booking
2. Passport Seva – Passport application
3. eHospital – Hospital appointment
4. DigiLocker – Digital document storage
5. Income Tax e-Filing

Q5: What is UMANG app? List its benefits.

Ans: UMANG (Unified Mobile Application for New-age Governance) is a single platform for accessing multiple government services.

Benefits:

- Single platform for 1,200+ services
- Multi-lingual support
- Aadhaar-based authentication
- Secure access

Practical Lab Exercises

Exercise No.	Task	Purpose
1	Create a Gmail account	Email account creation
2	Compose and send an email to classmate	Compose and send practice
3	Reply to a received email	Reply practice
4	Forward an email with comment	Forward practice
5	Send an email with attachment (resume)	Attachment practice
6	Create and set up email signature	Signature setup
7	Search for a specific email using search operators	Search practice
8	Explore Facebook/LinkedIn and create profile	Social networking

Exercise No.	Task	Purpose
9	Download and install UMANG app	UMANG usage
10	Explore services on UMANG app	e-Governance services
11	Create DigiLocker account and upload document	DigiLocker practice
12	Visit IRCTC website and explore ticket booking process	e-Governance practice

---

Important Points for UP CCC Exam

Topic	Key Points to Remember
Email Structure	username@domainname.com
Email Fields	To, Cc (visible), Bcc (hidden), Subject
Mailbox Folders	Inbox, Sent, Drafts, Spam, Trash
Reply vs Reply All	Reply to sender only; Reply All to all recipients
Forward	Send received email to new recipient
Attachments	Max 25 MB in Gmail; use Google Drive for larger
Signature	Auto-appended text at end of emails
Social Media	Facebook (social), LinkedIn (professional), Twitter (microblogging), Instagram (visual)

Topic	Key Points to Remember
IM Platforms	WhatsApp, Messenger, Telegram
Netiquettes	Be respectful, concise, professional
e-Governance	IRCTC, Passport, eHospital
UMANG	Single platform for government services
DigiLocker	Digital document storage (1 GB free)

Chapter 8: Digital Financial Tools and Applications

8.0 Introduction

Digital Financial Tools have revolutionized the way we handle money and conduct transactions. India has witnessed a massive transformation in the financial sector with the introduction of digital payment systems, making transactions faster, safer, and more convenient. The government's push for a **Digital India** and **cashless economy** has led to widespread adoption of digital payment methods across the country.

For CCC exam candidates in Uttar Pradesh, understanding these digital financial tools is essential as they are now integral to both personal finance and government office operations.

8.1 Objectives

After studying this chapter, you will be able to:

- Understand various digital financial tools and their applications
- Explain the concept of OTP and QR Code
- Use UPI (Unified Payment Interface) for transactions
- Understand AEPS and USSD banking
- Differentiate between Credit and Debit Cards
- Use eWallets for digital payments
- Understand PoS (Point of Sale) terminals
- Perform Internet Banking operations
- Differentiate between NEFT, RTGS, and IMPS
- Make online bill payments

8.2 Digital Financial Tools

8.2.1 Understanding OTP [One Time Password] and QR [Quick Response] Code

OTP (One Time Password)

OTP (One Time Password) is a unique, temporary code used to authenticate digital transactions. It is valid for a single login session or transaction and expires after a short period (typically 60-300 seconds).

Feature	Description
Purpose	Two-factor authentication for secure transactions
Delivery	SMS, Email, or Authenticator App
Validity	Single use; expires in 3-5 minutes
Length	4-6 digits typically
Security	Prevents unauthorized access even if password is compromised

Types of OTP

Type	Description
SMS OTP	Sent via text message to registered mobile number
Email OTP	Sent via email
App-based OTP	Generated by authenticator apps (Google Authenticator, etc.)

Type	Description
Hardware Token	Physical device generating OTP

QR Code (Quick Response Code)

QR Code is a two-dimensional barcode that can store various types of information. It can be scanned using a smartphone camera to initiate payments, access websites, or share information.

Feature	Description
Invented By	Denso Wave (Japan, 1994)
Capacity	Can store up to 4,296 alphanumeric characters
Types	Static QR (fixed amount) and Dynamic QR (variable amount)
Uses	Payments, website links, contact sharing, Wi-Fi login

QR Code in Digital Payments

Component	Information Stored
UPI QR	UPI ID, merchant name, amount (optional)
Bharat QR	Payment information (interoperable across banks)
Merchant QR	Merchant VPA or bank account details

8.2.2 UPI [Unified Payment Interface]

UPI (Unified Payments Interface) is a real-time payment system developed by **NPCI (National Payments Corporation of India)** that allows users to transfer money between bank accounts using a mobile phone.

Feature	Description
Launched By	NPCI
Launch Year	2016
24x7 Availability	Yes
Transaction Limit	Up to ₹1,00,000 per transaction (varies by bank)
Charges	No additional charges for customers

UPI Components

Component	Description	Example
VPA (Virtual Payment Address)	Unique identifier instead of bank details	`ram@okhdfcbank`
UPI PIN	4-6 digit password for authorizing transactions	`1234`
UPI Apps	Applications enabling UPI payments	BHIM, Google Pay, PhonePe, Paytm

Popular UPI Apps in India

App	Developer	Features
BHIM	NPCI	Government-backed, simple interface
Google Pay	Google	Rewards, bill payments
PhonePe	Walmart	Wide merchant network, bill payments
Paytm	Paytm	Wallet + UPI, bill payments
Amazon Pay	Amazon	Integrated with Amazon shopping

UPI Transaction Types

Transaction Type	Description
Peer-to-Peer (P2P)	Transfer money to another individual
Peer-to-Merchant (P2M)	Pay at shops, restaurants, online stores
Collect Request	Request money from another user
Scan & Pay	Scan QR code to pay
Bill Payments	Pay utility bills, recharge mobile

8.2.3 AEPS [Aadhaar Enabled Payment System]

AEPS (Aadhaar Enabled Payment System) is a payment service that allows customers to perform banking transactions using their **Aadhaar number** and **biometric authentication**.

Feature	Description
Launched By	NPCI
Authentication	Biometric (fingerprint) on micro-ATM
Requirement	Aadhaar number linked to bank account
No Card Required	No debit card or PIN needed

AEPS Services

Service	Description
Cash Withdrawal	Withdraw money from bank account using Aadhaar
Balance Inquiry	Check account balance
Fund Transfer	Transfer funds between Aadhaar-linked accounts
Mini Statement	Get last 5-10 transactions

Who Can Use AEPS?

User	Description
Banked Customers	Must have Aadhaar linked to bank account
Bank Mitras	Banking correspondents providing service in rural areas
Merchants	Accept payments via AEPS

Difference: AEPS vs UPI

Parameter	AEPS	UPI
Authentication	Biometric (fingerprint)	UPI PIN (numeric)
Device Required	Micro-ATM / PoS device	Smartphone
Identifier	Aadhaar number	VPA / QR Code
Internet Required	Yes (for micro-ATM)	Yes

8.2.4 USSD [Unstructured Supplementary Service Data]

USSD (Unstructured Supplementary Service Data) is a mobile banking service that allows users to perform banking transactions without internet connectivity, using **mobile number** and **MMID (Mobile Money Identifier)** .

Feature	Description
Service Name	*99# (National Unified USSD Platform - NUUP)
Internet Required	No (works on any mobile phone)
Language Support	12+ Indian languages
Transaction Limit	Up to ₹5,000 per day
Availability	24x7

*USSD Banking Services (*99#)*

Service	Code to Dial
Send Money	*99# → Select "Send Money"
Request Money	*99# → Select "Request Money"
Check Balance	*99# → Select "Check Balance"
Change MPIN	*99# → Select "Change MPIN"

Requirements for USSD Banking

Requirement	Description
Mobile Number	Registered with bank
MMID	7-digit Mobile Money Identifier provided by bank

Requirement	Description
MPIN	Mobile PIN for authentication

8.2.5 Card [Credit / Debit]

Debit Card

A **Debit Card** is linked directly to the cardholder's bank account. When used, funds are immediately deducted from the account.

Feature	Description
Source	Own bank account
Spending Limit	Up to available balance in account
Interest	No interest charged
Usage	ATM withdrawals, PoS, online shopping

Credit Card

A **Credit Card** allows the cardholder to borrow money from the issuing bank up to a pre-approved limit, to be repaid later (with interest if not paid in full by due date).

Feature	Description
Source	Credit limit from bank

Feature	Description
Billing Cycle	30-50 days interest-free period
Interest	Charged on unpaid balance (typically 2-3% per month)
Benefits	Rewards points, cashback, offers

Comparison: Debit Card vs Credit Card

Parameter	Debit Card	Credit Card
Source	Own money	Borrowed money
Interest	No interest	Interest on unpaid balance
Credit Score	No impact	Affects credit score
Eligibility	Anyone with bank account	Income and credit check
Fees	Annual fees (minimal)	Annual fees, interest

Card Security Features

Feature	Description
CVV/CVC	3-digit code on back for online transactions
PIN	4-digit number for ATM/PoS transactions
EMV Chip	Secure chip replacing magnetic stripe

Feature	Description
OTP	One-time password for online transactions

8.2.6 eWallet

An **eWallet (Electronic Wallet)** is a digital version of a physical wallet that stores money and allows users to make payments digitally.

Feature	Description
Storage	Stores money digitally (prepaid)
Loading Money	Add money from bank account, debit card, or credit card
Usage	Online shopping, bill payments, merchant payments

Types of eWallets

Type	Description	Examples
Closed Wallet	Issued by a company for purchases from that company only	Amazon Pay balance (for Amazon purchases)
Semi-Closed Wallet	Can be used at listed merchants	Paytm, PhonePe, Mobikwik
Open Wallet	Can be used for payments and ATM withdrawals	Bank-issued wallets (limited)

Popular eWallets in India

eWallet	Features
Paytm Wallet	Payments, recharges, bill payments, shopping
PhonePe Wallet	Integrated with UPI, payments
Mobikwik	Payments, recharges, loans
Amazon Pay	Amazon shopping, bill payments
Freecharge	Recharges, bill payments

8.2.7 PoS [Point of Sale]

PoS (Point of Sale) refers to the place or device where a retail transaction is completed. In digital payments, PoS terminals are used to accept card payments.

Feature	Description
Purpose	Accept card payments at merchant locations
Types	Physical PoS, Mobile PoS (mPoS), Soft PoS

Types of PoS Terminals

Type	Description
Physical PoS	Traditional machine at stores accepting cards

Type	Description
Mobile PoS (mPoS)	Portable device using SIM/Bluetooth for payments
Soft PoS	Smartphone-based card acceptance (using NFC)
QR Code PoS	Static/dynamic QR for UPI payments

PoS Transaction Process

Step	Action
1	Merchant enters transaction amount on PoS device
2	Customer swipes/inserts/taps card
3	Customer enters PIN
4	Device connects to bank for authorization
5	Transaction approved or declined
6	Receipt printed (optional)

8.3 Internet Banking

Internet Banking (also called Net Banking or Online Banking) allows customers to perform financial transactions through a bank's website or mobile app.

Features of Internet Banking

Feature	Description
Account Information	View account balance, transaction history, statement download
Fund Transfer	Transfer money within own accounts, to other accounts
Bill Payment	Pay utility bills, credit card bills
Fixed Deposit	Open FD/RD online
Cheque Services	Request cheque book, stop cheque payment
Profile Management	Change password, update contact details

8.3.1 National Electronic Fund Transfer (NEFT)

NEFT (National Electronic Fund Transfer) is a nationwide payment system facilitating one-to-one funds transfer between bank accounts.

Feature	Description
Settlement	Batch processing (hourly batches)
Availability	24x7 (since December 2019)
Transaction Limit	No upper limit
Transfer Speed	Up to 2 hours (settled in batches)

Feature	Description
Charges	Up to ₹25 (for transactions above ₹10,000)

NEFT Transaction Limits

Transaction Amount	Maximum Charge
Up to ₹10,000	₹2.50 + GST
₹10,001 to ₹1,00,000	₹5 + GST
₹1,00,001 to ₹2,00,000	₹15 + GST
Above ₹2,00,000	₹25 + GST

NEFT Process

Step	Action
1	Login to Internet Banking
2	Select NEFT fund transfer
3	Enter beneficiary details (name, account number, IFSC)
4	Add beneficiary (takes 24-48 hours for activation)
5	Transfer amount
6	Funds transferred in next available batch

8.3.2 Real Time Gross Settlement (RTGS)

RTGS (Real Time Gross Settlement) is a funds transfer system where transactions are processed individually and in real-time.

Feature	Description
Settlement	Real-time (immediate)
Minimum Amount	₹2,00,000
Maximum Amount	No upper limit
Availability	24x7
Charges	Up to ₹55

RTGS Charges

Transaction Amount	Maximum Charge
₹2,00,000 to ₹5,00,000	₹25 + GST
Above ₹5,00,000	₹55 + GST

8.3.3 Immediate Payment Service (IMPS)

IMPS (Immediate Payment Service) is an instant interbank electronic funds transfer system available 24x7.

Feature	Description
Settlement	Real-time (immediate)
Minimum Amount	No minimum
Maximum Amount	Up to ₹5,00,000 (varies by bank)
Availability	24x7 (including holidays)
Charges	Variable (₹5 to ₹15 typically)

IMPS vs NEFT vs RTGS

Parameter	IMPS	NEFT	RTGS
Settlement	Real-time	Batch (hourly)	Real-time
Minimum	No minimum	No minimum	₹2,00,000
Maximum	₹5,00,000	No limit	No limit
Availability	24x7	24x7	24x7
Speed	Instant	Up to 2 hours	Instant
Best For	Small to medium amounts	Any amount (no urgency)	Large amounts

8.4 Online Bill Payment

Online Bill Payment allows consumers to pay utility bills (electricity, water, gas, mobile, etc.) using digital payment methods.

Common Bill Categories

Category	Examples
Electricity	UPPCL, BSES, TPDDL
Mobile	Jio, Airtel, Vi, BSNL
DTH	Tata Play, Airtel Digital, Dish TV
Water	Jal Board, Municipal water bills
Gas	Indane, HP Gas, Bharat Gas
Credit Card	Various bank credit cards
Education	School/college fees
Insurance	Premium payments
Loan EMI	Home, auto, personal loans

Methods of Online Bill Payment

Method	Description
BBPS (Bharat Bill Payment System)	NPCI's unified bill payment platform
Bank Internet Banking	Pay via bank's net banking portal
UPI Apps	Pay via BHIM, Google Pay, PhonePe
eWallet	Pay via Paytm, Mobikwik
Merchant Website	Pay directly on service provider's website

BBPS (Bharat Bill Payment System)

BBPS is an integrated bill payment system operated by NPCI that provides a single platform for all bill payments.

Feature	Description
Operated By	NPCI
Services	Electricity, water, gas, mobile, DTH, etc.
Channels	Banks, UPI apps, eWallets, agents
Features	Auto-pay, recurring payments

8.5 Summary

- **OTP (One Time Password)** : Temporary code for secure authentication (valid for single use, 3-5 minutes)
- **QR Code (Quick Response)** : Two-dimensional barcode used for payments and information sharing
- **UPI (Unified Payment Interface)** : Real-time payment system using VPA (e.g., ram@okhdfcbank)
- **AEPS (Aadhaar Enabled Payment System)** : Biometric authentication using Aadhaar number
- **USSD (*99#)** : Mobile banking without internet (up to ₹5,000 per day)
- **Debit Card** : Linked to bank account; money deducted immediately
- **Credit Card** : Borrowed money; repayment required with interest
- **eWallet** : Digital wallet storing prepaid money (Paytm, PhonePe)
- **PoS (Point of Sale)** : Terminal for card payments at merchant locations
- **NEFT** : Batch processing; no minimum limit; up to 2 hours
- **RTGS** : Real-time; minimum ₹2,00,000; large transactions
- **IMPS** : Real-time; no minimum; up to ₹5,00,000; 24x7
- **BBPS** : Unified bill payment system for utilities

8.6 Model Questions and Answers

Multiple Choice Questions (MCQs)

1. What is the full form of UPI?

- a) Unified Payment Interface
- b) Unified Payment Integration
- c) Unified Payment Interchange
- d) Uniform Payment Interface

Answer: a) Unified Payment Interface

---

2. Which of the following is a government-backed UPI app?

- a) Google Pay
- b) PhonePe
- c) BHIM
- d) Paytm

Answer: c) BHIM

---

3. What is the transaction limit for USSD (*99#) banking?

- a) ₹1,000 per day
- b) ₹5,000 per day
- c) ₹10,000 per day
- d) ₹25,000 per day

Answer: b) ₹5,000 per day

---

4. Which payment system uses biometric authentication with Aadhaar?

- a) UPI
- b) AEPS
- c) NEFT
- d) RTGS

Answer: b) AEPS

---

5. What is the minimum amount for RTGS transaction?

- a) ₹1,00,000
- b) ₹2,00,000
- c) ₹5,00,000
- d) No minimum

Answer: b) ₹2,00,000

---

6. Which payment system settles transactions in real-time individually?

- a) NEFT
- b) RTGS
- c) Both a and b
- d) Neither

Answer: b) RTGS

---

7. What does OTP stand for?

- a) One Time Password
- b) One Time Payment
- c) Online Transaction Password
- d) One Time PIN

Answer: a) One Time Password

---

8. Which of the following is NOT a digital wallet?

- a) Paytm
- b) PhonePe
- c) Debit Card
- d) Mobikwik

Answer: c) Debit Card

---

9. What is the full form of PoS?

- a) Point of Sale
- b) Point of Service
- c) Payment on Sale
- d) Purchase of Sale

Answer: a) Point of Sale

---

10. Which system provides unified bill payment services?

- a) UPI
- b) BBPS
- c) NEFT
- d) IMPS

Answer: b) BBPS

True/False Questions

Statement	Answer
OTP is valid for multiple transactions.	**False** (Valid for single transaction)
UPI allows fund transfer using VPA.	**True**
AEPS requires internet connection on mobile.	**False** (Works on micro-ATM device)
RTGS has no minimum transaction limit.	**False** (Minimum ₹2,00,000)
NEFT operates 24x7.	**True**

Short Answer Questions

Q1: What is UPI? List its advantages.

Ans: UPI (Unified Payments Interface) is a real-time payment system developed by NPCI.

Advantages:

- 24x7 availability
- No need to share bank details (use VPA)
- No charges for customers
- Single app for multiple bank accounts

Q2: Differentiate between NEFT, RTGS, and IMPS.

Ans:

Parameter	NEFT	RTGS	IMPS
Settlement	Batch	Real-time	Real-time
Minimum	No min	₹2,00,000	No min
Speed	Up to 2 hrs	Instant	Instant

Q3: What is AEPS? How does it work?
Ans: AEPS (Aadhaar Enabled Payment System) allows banking transactions using Aadhaar number and biometric authentication at micro-ATM devices. The user provides Aadhaar number, selects transaction type, and authenticates with fingerprint.

Q4: What is the difference between a debit card and a credit card?
Ans:

- **Debit Card**: Linked to bank account; uses own money; no interest
- **Credit Card**: Credit limit from bank; borrowed money; interest on unpaid balance

---

Q5: Explain USSD banking (*99#).
Ans: USSD banking (*99#) allows mobile banking without internet. Users can send money, check balance, and perform basic transactions using any mobile phone. Daily limit is ₹5,000.

Practical Lab Exercises

Exercise No.	Task	Purpose
1	Create UPI ID on BHIM/PhonePe app	UPI setup practice
2	Perform a small UPI transaction (₹10)	UPI payment practice
3	Scan QR code and pay	QR payment practice
4	Check balance using USSD (*99#)	USSD banking practice
5	Identify difference between credit and debit cards	Card types identification
6	Load money into Paytm/PhonePe wallet	eWallet practice
7	Explore Internet Banking portal (if available)	Net banking practice
8	Add beneficiary for NEFT transfer	NEFT practice
9	Pay an electricity bill online	Online bill payment
10	Explore BBPS on UPI app	BBPS practice

Important Points for UP CCC Exam

Topic	Key Points to Remember
UPI	VPA format: `username@bankhandle`; 24x7; no charges
BHIM	Government UPI app by NPCI
AEPS	Aadhaar + biometric; no card/PIN
USSD	*99#; no internet; ₹5,000/day limit
NEFT	No minimum; batch processing; up to 2 hours
RTGS	Minimum ₹2,00,000; real-time
IMPS	No minimum; real-time; up to ₹5,00,000
OTP	Single use; expires in 3-5 minutes
QR Code	Two-dimensional barcode for payments
BBPS	Unified bill payment platform

Chapter 9: Overview of Futureskills & Cyber Security

9.0 Introduction to Futureskills

Futureskills refer to the emerging technologies and competencies that are shaping the future of work, industry, and society. As technology evolves rapidly, understanding these futuristic concepts is essential for staying relevant in the digital age. The Government of India, through NIELIT and other agencies, promotes Futureskills to prepare the workforce for Industry 4.0 and beyond.

This chapter introduces key emerging technologies that are transforming various sectors and also covers essential **cyber security** practices to protect yourself and your organization from online threats.

9.1 Objectives

After studying this chapter, you will be able to:

- Understand the concept of Futureskills and emerging technologies
- Explain Internet of Things (IoT) and its applications
- Understand Big Data Analytics
- Describe Cloud Computing and its service models
- Understand Virtual Reality and Augmented Reality
- Explain Artificial Intelligence and Machine Learning
- Understand Blockchain Technology
- Describe 3D Printing and Additive Manufacturing
- Understand Robotics Process Automation
- Recognize the need for cyber security
- Implement best practices for securing PC and smartphone

9.1 Introduction to Emerging Technologies

9.1.1 Internet of Things (IoT)

Internet of Things (IoT) refers to the network of physical objects embedded with sensors, software, and connectivity that enables them to collect and exchange data over the internet.

Feature	Description
Definition	Network of interconnected devices that communicate without human interventi
Key Components	Sensors, connectivity, data processing, user interface
Communication	Machine-to-Machine (M2M) communication

IoT Architecture

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┌──────────────────────────────────────────────────────────────────────┐
│                          IoT ECOSYSTEM                               │
├──────────────────────────────────────────────────────────────────────┤
│  ┌──────────────┐    ┌──────────────┐    ┌──────────────────────┐   │
│  │   Devices    │──▶│   Gateway    │──▶│   Cloud/Platform    │   │
│  │  (Sensors)   │    │ (Connectivity)│    │   (Data Storage)    │   │
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│                                       │   Applications       │       │
│                                       │   (Analytics, UI)    │       │
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Applications of IoT

Domain	Applications
Smart Home	Smart lights, thermostats, security cameras, voice assistants
Healthcare	Wearable health monitors, remote patient monitoring, smart pills
Agriculture	Soil moisture sensors, automated irrigation, crop monitoring
Manufacturing	Predictive maintenance, industrial automation, smart factories
Transportation	Connected vehicles, fleet management, traffic management
Smart Cities	Smart parking, waste management, smart street lighting

IoT Examples in India

Initiative	Description
Smart City Mission	IoT-based solutions for urban infrastructure
Digital Agriculture	IoT sensors for soil monitoring and irrigation
Wearable Devices	Fitness trackers, smartwatches for health monitoring

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9.1.2 Big Data Analytics

Big Data refers to extremely large and complex datasets that cannot be processed using traditional data processing methods. **Big Data Analytics** involves examining these datasets to uncover patterns, correlations, and insights.

Characteristics of Big Data (5 V's)

V	Description
Volume	Massive amount of data generated (terabytes to petabytes)
Velocity	High speed of data generation and processing (real-time)
Variety	Different types of data (structured, unstructured, semi-structured)
Veracity	Uncertainty and quality of data
Value	Extracting meaningful insights from data

Types of Data

Type	Description	Examples
Structured	Organized in rows and columns	Databases, Excel sheets
Unstructured	No predefined format	Text documents, images, videos
Semi-structured	Partial organization	JSON, XML files

Big Data Analytics Tools

Tool	Purpose
Hadoop	Distributed storage and processing
Spark	Fast data processing engine
Tableau	Data visualization
Python/R	Data analysis and machine learning

Applications of Big Data

Domain	Applications
Healthcare	Patient outcome prediction, drug discovery
Banking	Fraud detection, risk analysis
Retail	Customer behavior analysis, personalized recommendations
Government	Policy making, citizen service improvement

9.1.3 Cloud Computing

Cloud Computing is the delivery of computing services (servers, storage, databases, networking, software) over the internet ("the cloud") on a pay-as-you-go basis.

Feature	Description
On-demand	Self-service access to resources
Scalability	Scale up/down as needed
Pay-per-use	Pay only for resources consumed
Accessibility	Access from anywhere with internet

Cloud Service Models

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┌──────────────────────────────────────────────────────────────┐
│                    CLOUD SERVICE MODELS                      │
├──────────────────────────────────────────────────────────────┤
│                                                              │
│  ┌──────────────────────────────────────────────────────┐   │
│  │                  Software as a Service (SaaS)        │   │
│  │         Applications delivered over internet         │   │
│  │            Examples: Gmail, Google Drive             │   │
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│                                                              │
│  ┌──────────────────────────────────────────────────────┐   │
│  │              Platform as a Service (PaaS)            │   │
│  │      Platform to develop, run, manage applications   │   │
│  │             Examples: Google App Engine              │   │
│  └──────────────────────────────────────────────────────┘   │
│                                                              │
│  ┌──────────────────────────────────────────────────────┐   │
│  │              Infrastructure as a Service (IaaS)      │   │
│  │        Virtualized computing resources over internet│   │
│  │               Examples: AWS EC2, Google Compute      │   │
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Cloud Deployment Models

Model	Description	Users
Public Cloud	Services offered over public internet	General public
Private Cloud	Dedicated to single organization	Large enterprises, government
Hybrid Cloud	Combination of public and private	Organizations with specific security needs
Community Cloud	Shared by multiple organizations	Government agencies, research institutions

Cloud Computing Examples in India

Service	Description
Meghraj	Government of India cloud initiative
Google Drive	Cloud storage
Microsoft OneDrive	Cloud storage
Amazon Web Services (AWS)	Cloud infrastructure

9.1.4 Virtual Reality

Virtual Reality (VR) is a computer-generated simulation of a three-dimensional environment that users can interact with using special electronic equipment (VR headsets, gloves, etc.).

Feature	Description
Immersion	Complete immersion in digital environment
Interaction	User can interact with virtual objects
Equipment	VR headset, controllers, tracking sensors

Types of Reality Technologies

Technology	Description	Example
Virtual Reality (VR)	Fully digital environment	VR gaming, virtual tours
Augmented Reality (AR)	Digital overlay on real world	Pokémon Go, Snapchat filters
Mixed Reality (MR)	Digital and real-world interaction	Microsoft HoloLens

Applications of VR/AR

Domain	Applications
Education	Virtual classrooms, interactive learning
Healthcare	Surgical simulation, therapy
Entertainment	VR gaming, 360° videos
Training	Flight simulators, military training
Retail	Virtual try-on, product visualization

9.1.5 Artificial Intelligence

Artificial Intelligence (AI) is the simulation of human intelligence in machines programmed to think, learn, and make decisions. **Machine Learning (ML)** is a subset of AI that enables systems to learn from data without explicit programming.

Types of AI

Type	Description	Examples
Narrow AI (Weak AI)	Designed for specific tasks	Voice assistants, facial recognition
General AI (Strong AI)	Human-like intelligence	Still theoretical
Super AI	Surpasses human intelligence	Futuristic concept

Key AI Technologies

Technology	Description
Machine Learning (ML)	Systems learn from data to improve performance
Deep Learning	Neural networks with multiple layers
Natural Language Processing (NLP)	Understanding and generating human language
Computer Vision	Analyzing and interpreting visual information
Robotics	Intelligent machines performing physical tasks

Applications of AI

Domain	Applications
Healthcare	Disease diagnosis, drug discovery, medical imaging
Banking	Fraud detection, algorithmic trading
Transportation	Self-driving cars, traffic prediction
Customer Service	Chatbots, virtual assistants
Education	Personalized learning, automated grading

AI Examples in India

Initiative	Description
NITI Aayog AI Strategy	National strategy for AI adoption
Aarogya Setu	AI-powered COVID-19 tracking app
DigiLocker	AI for document verification

9.1.6 Social & Mobile

Social & Mobile refers to the convergence of social media platforms and mobile technology, enabling anytime, anywhere connectivity and communication.

Social Media Trends

Trend	Description
Mobile-first	Platforms designed for mobile devices
Short-form Video	TikTok, Instagram Reels, YouTube Shorts
Social Commerce	Buying/selling through social platforms
Live Streaming	Real-time video content
Influencer Marketing	Brand promotion through social influencers

Mobile Technology Evolution

Generation	Features
4G	High-speed internet, video streaming
5G	Ultra-fast speeds, low latency, IoT connectivity
6G (Future)	Terabit speeds, AI integration

9.1.7 Blockchain Technology

Blockchain is a distributed, decentralized digital ledger that records transactions across multiple computers securely and transparently.

Feature	Description
Decentralized	No central authority; distributed across network
Immutable	Records cannot be altered once added
Transparent	Transactions visible to all participants
Secure	Cryptographic security

Blockchain Structure

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┌──────────────────────────────────────────────────────────────────────┐
│                      BLOCKCHAIN STRUCTURE                            │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  ┌──────────┐    ┌──────────┐    ┌──────────┐    ┌──────────┐      │
│  │ Block 1 │──▶│ Block 2 │──▶│ Block 3 │──▶│ Block 4 │      │
│  ├──────────┤    ├──────────┤    ├──────────┤    ├──────────┤      │
│  │ Hash     │    │ Hash     │    │ Hash     │    │ Hash     │      │
│  │ Data     │    │ Data     │    │ Data     │    │ Data     │      │
│  └──────────┘    └──────────┘    └──────────┘    └──────────┘      │
│                                                                      │
│  Each block contains:                                                │
│  • Previous block's hash (linking blocks)                            │
│  • Timestamp                                                         │
│  • Transaction data                                                  │
│  • Current block hash                                                │
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Types of Blockchain

Type	Description	Examples
Public	Open to anyone	Bitcoin, Ethereum

Type	Description	Examples
Private	Restricted access	Enterprise blockchains
Consortium	Managed by group	Banking consortiums

Applications of Blockchain

Domain	Applications
Finance	Cryptocurrencies, cross-border payments
Supply Chain	Product tracking, authenticity verification
Healthcare	Patient records, drug traceability
Government	Land records, voting systems
Identity	Digital identity, credential verification

Blockchain Initiatives in India

Initiative	Description
IndiaChain	Government blockchain project
Digital Locker	Blockchain-based document storage
Cryptocurrency Regulation	RBI guidelines for digital currencies

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9.1.8 3D Printing / Additive Manufacturing

3D Printing (Additive Manufacturing) is the process of creating three-dimensional objects by depositing materials layer by layer based on a digital model.

Feature	Description
Process	Additive (layer-by-layer) vs subtractive (cutting)
Materials	Plastics, metals, ceramics, bio-materials
Applications	Prototyping, manufacturing, healthcare

3D Printing Process

Step	Description
1	Create 3D model using CAD software
2	Convert to STL file format
3	Slice model into layers
4	Print layer by layer
5	Post-processing (cleaning, curing)

Applications of 3D Printing

Domain	Applications
Healthcare	Prosthetics, implants, surgical guides

Domain	Applications
Manufacturing	Prototyping, spare parts, tooling
Construction	3D-printed buildings
Education	Learning aids, models
Aerospace	Lightweight components

9.1.9 Robotics Process Automation (RPA)

Robotics Process Automation (RPA) is the use of software robots (bots) to automate repetitive, rule-based business processes.

Feature	Description
Purpose	Automate manual, repetitive tasks
Interaction	Bots interact with applications like humans
Benefits	Efficiency, accuracy, cost reduction

RPA vs Traditional Automation

Parameter	RPA	Traditional Automation
Integration	Works with existing systems	Requires API or integration
Implementation	Quick, non-invasive	Complex, time-consuming
Cost	Lower	Higher
Flexibility	High	Limited

Applications of RPA

Domain	Applications
Banking	Customer onboarding, KYC verification
Insurance	Claims processing
Healthcare	Patient registration, billing
HR	Employee onboarding, payroll
Finance	Invoice processing, reconciliation

9.2 Cyber Security

9.2.1 Need of Cyber Security

Cyber Security is the practice of protecting systems, networks, programs, and data from digital attacks, unauthorized access, and damage.

Why Cyber Security is Important

Reason	Description
Data Protection	Protects sensitive personal and organizational data
Financial Security	Prevents financial fraud and theft
Privacy	Safeguards personal information
Business Continuity	Prevents disruptions to operations
National Security	Protects critical infrastructure
Reputation	Maintains trust and credibility

Common Cyber Threats

Threat	Description
Phishing	Fraudulent attempts to obtain sensitive information
Malware	Malicious software (viruses, worms, trojans, ransomware)
Ransomware	Malware that encrypts data and demands payment

Threat	Description
Hacking	Unauthorized access to systems
Identity Theft	Stealing personal information for fraud
Social Engineering	Manipulating people to reveal information

Cyber Security Tips

Tip	Description
Strong Passwords	Use complex, unique passwords
Two-Factor Authentication	Enable 2FA wherever possible
Regular Updates	Keep software and OS updated
Antivirus Software	Install and update antivirus
Backup Data	Regular backups of important data
Be Cautious	Don't click suspicious links

9.2.2 Securing PC

Steps to Secure Your PC

Step	Action
1. Operating System Updates	Enable automatic updates; install security patches
2. Antivirus/Anti-malware	Install reputable antivirus; keep definitions updated
3. Firewall	Enable built-in firewall (Windows Defender Firewall)
4. Strong Passwords	Use complex passwords; change regularly
5. User Account Control	Use standard account (not administrator) for daily use
6. Backup	Regular backups to external drive or cloud
7. Browser Security	Use secure browsers; enable pop-up blockers
8. Secure Wi-Fi	Use strong encryption (WPA2/WPA3) for Wi-Fi

Windows Security Settings

Setting	Location	Action
Windows Update	Settings → Update & Security	Enable automatic updates
Windows Defender	Settings → Privacy & Security	Keep real-time protection ON
Firewall	Control Panel → Firewall	Ensure firewall is ON
User Account	Settings → Accounts	Use standard user account

Signs Your PC May Be Compromised

Sign	Possible Issue
Slow performance	Malware consuming resources
Pop-up ads	Adware infection
Unusual network activity	Data being sent to unknown locations
Files missing/encrypted	Ransomware attack
Browser redirects	Browser hijacker

9.2.3 Securing Smartphone

Smartphones contain vast amounts of personal data and are increasingly targeted by cyber criminals.

Steps to Secure Smartphone

Step	Action
1. Screen Lock	Use PIN, password, pattern, fingerprint, or face recognition
2. OS Updates	Keep Android/iOS updated
3. App Installation	Download only from official stores (Google Play, Apple App Store)
4. App Permissions	Review and restrict unnecessary permissions

Step	Action
5. Find My Device	Enable device tracking (Find My iPhone / Find My Device)
6. Secure Wi-Fi	Avoid public Wi-Fi; use VPN when necessary
7. Two-Factor Authentication	Enable 2FA for important accounts
8. Remote Wipe	Enable ability to remotely erase data if lost

Android Security Settings

Setting	Location	Action
Screen Lock	Settings → Security → Screen Lock	Set strong PIN/password
Find My Device	Settings → Security → Find My Device	Enable
Google Play Protect	Play Store → Play Protect	Keep enabled
App Permissions	Settings → Apps → Permission manager	Review and restrict

iOS (iPhone) Security Settings

Setting	Location	Action
Face ID/Touch ID	Settings → Face ID & Passcode	Enable
Find My iPhone	Settings → Apple ID → Find My	Enable
App Permissions	Settings → Privacy	Review permissions

Common Smartphone Threats

Threat	Description	Prevention
Malicious Apps	Apps containing malware	Download only from official stores
SMS Phishing (Smishing)	Fraudulent SMS with links	Don't click suspicious links
Fake Wi-Fi	Rogue Wi-Fi networks	Use known networks; use VPN
SIM Swap Fraud	Fraudster gets new SIM with your number	Contact bank immediately

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9.3 Summary

Futureskills

Technology	Key Concept
Internet of Things (IoT)	Network of connected devices with sensors
Big Data Analytics	Analyzing large datasets for insights
Cloud Computing	On-demand computing services over internet
Virtual Reality (VR)	Immersive digital environments

Technology	Key Concept
Artificial Intelligence (AI)	Machines simulating human intelligence
Social & Mobile	Convergence of social media and mobile tech
Blockchain	Decentralized, immutable digital ledger
3D Printing	Additive manufacturing layer by layer
Robotics Process Automation (RPA)	Software bots automating repetitive tasks

Cyber Security

Topic	Key Points
Need of Cyber Security	Protect data, privacy, finances, national security
Common Threats	Phishing, malware, ransomware, hacking
Securing PC	Updates, antivirus, firewall, strong passwords, backup
Securing Smartphone	Screen lock, app permissions, find my device, updates

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9.4 Model Questions and Answers

Multiple Choice Questions (MCQs)

1. What does IoT stand for?

- a) Internet of Technology
- b) Internet of Things
- c) Integration of Things
- d) Interconnection of Things

Answer: b) Internet of Things

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2. Which of the following is a characteristic of Big Data?

- a) Volume
- b) Velocity
- c) Variety
- d) All of the above

Answer: d) All of the above

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3. In cloud computing, what does IaaS stand for?

- a) Internet as a Service
- b) Infrastructure as a Service
- c) Integration as a Service
- d) Information as a Service

Answer: b) Infrastructure as a Service

---

4. Which technology is used for creating digital ledgers that are decentralized and immutable?

- a) Cloud Computing
- b) Big Data
- c) Blockchain
- d) IoT

Answer: c) Blockchain

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5. What is the process of creating objects by depositing materials layer by layer called?

- a) Subtractive Manufacturing
- b) Additive Manufacturing (3D Printing)
- c) Injection Molding
- d) Casting

Answer: b) Additive Manufacturing (3D Printing)

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6. Which of the following is a type of cyber threat?

- a) Phishing
- b) Ransomware
- c) Malware
- d) All of the above

Answer: d) All of the above

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7. What does RPA stand for?

- a) Rapid Process Automation
- b) Robotics Process Automation
- c) Real-time Process Automation
- d) Remote Process Automation

Answer: b) Robotics Process Automation

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8. Which of the following is a cloud storage service?

- a) Google Drive
- b) Microsoft OneDrive
- c) Dropbox
- d) All of the above

Answer: d) All of the above

---

9. What is the best practice for securing a smartphone?

- a) Keep screen lock enabled
- b) Download apps only from official stores
- c) Keep OS updated
- d) All of the above

Answer: d) All of the above

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10. Which AI technology enables machines to understand and generate human language?

- a) Computer Vision
- b) Natural Language Processing (NLP)
- c) Deep Learning
- d) Robotics

Answer: b) Natural Language Processing (NLP)

True/False Questions

Statement	Answer
IoT devices can communicate without human intervention.	**True**
Big Data is characterized only by volume.	**False** (Also velocity, variety, veracity, value)
Blockchain records cannot be altered once added.	**True**
Ransomware encrypts data and demands payment.	**True**
3D printing is also known as subtractive manufacturing.	**False** (Additive manufacturing)

Short Answer Questions

Q1: What is Internet of Things (IoT)? Give two examples.
Ans: IoT is a network of physical devices embedded with sensors and connectivity to exchange data over the internet.

Examples:

1. Smart home devices (smart lights, thermostats)
2. Wearable health monitors (fitness trackers)

Q2: What are the 5 V's of Big Data?
Ans:

1. **Volume** – Large amount of data
2. **Velocity** – High speed of data generation
3. **Variety** – Different types of data
4. **Veracity** – Quality and uncertainty of data
5. **Value** – Extracting meaningful insights

Q3: Differentiate between Public Cloud and Private Cloud.
Ans:

- **Public Cloud**: Services offered over public internet; shared infrastructure; cost-effective
- **Private Cloud**: Dedicated to single organization; higher security; more control

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Q4: What is the difference between VR and AR?
Ans:

- **Virtual Reality (VR)** : Fully immersive digital environment; user isolated from real world
- **Augmented Reality (AR)** : Digital overlay on real world; user still sees surroundings

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Q5: List five tips for securing a PC.

Ans:

1. Enable automatic OS updates
2. Install and update antivirus software
3. Use strong, unique passwords
4. Enable firewall
5. Take regular backups

Practical Lab Exercises

Exercise No.	Task	Purpose
1	Research and list 5 IoT devices used in daily life	Understand IoT applications
2	Identify cloud services you use (Google Drive, etc.)	Cloud computing awareness
3	Explore blockchain applications in India (DigiLocker, etc.)	Blockchain understanding
4	Enable Windows Update and Windows Defender on PC	PC security practice
5	Set up screen lock on smartphone	Smartphone security
6	Review app permissions on smartphone	Permission management
7	Enable Find My Device/Find My iPhone	Device tracking practice
8	Create a strong password using best practices	Password security
9	Identify phishing emails in your inbox	Phishing awareness

Exercise No.	Task	Purpose
10	Research cyber security initiatives in India (CERT-In, etc.)	National cyber security

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Important Points for UP CCC Exam

Topic	Key Points to Remember
IoT	Network of connected devices; M2M communication
Big Data	5 V's: Volume, Velocity, Variety, Veracity, Value
Cloud Computing	SaaS, PaaS, IaaS; Public, Private, Hybrid
Blockchain	Decentralized, immutable, transparent
AI	Machine Learning, NLP, Computer Vision
3D Printing	Additive manufacturing; layer by layer
RPA	Software bots automating repetitive tasks
Cyber Security	Protect data, privacy, prevent attacks
Common Threats	Phishing, malware, ransomware
PC Security	Updates, antivirus, firewall, backup
Smartphone Security	Screen lock, app permissions, find my device

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This concludes Chapter 9 and the complete CCC course outline!

Final Summary - Complete CCC Course

Chapter	Topic	Key Focus
1	Introduction to Computer	Hardware, software, memory, generations
2	Operating System	Windows, Ubuntu, file management, settings
3	Word Processing	Writer/Word, formatting, tables, mail merge
4	Spreadsheet	Calc/Excel, formulas, functions, charts
5	Presentation	Impress/PowerPoint, slides, transitions
6	Internet & WWW	Networks, IP address, browsers, searching
7	Email & e-Governance	Email operations, social media, UMANG, DigiLocker
8	Digital Financial Tools	UPI, AEPS, NEFT, RTGS, IMPS, eWallets
9	Futureskills & Cyber Security	IoT, AI, Blockchain, cyber security practices

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1.1	Generations of Computers	Chapter 1
1.2	Classification of Computers by Size	Chapter 1
1.3	Applications of Computers	Chapter 1
1.4	IT Gadgets and Their Applications	Chapter 1
1.5	Units of Memory	Chapter 1
2.1	Operating System Functions	Chapter 2
2.2	Desktop/Laptop Operating Systems	Chapter 2
2.3	Mobile Operating Systems	Chapter 2
2.4	Common File Extensions	Chapter 2
3.1	Common Font Formatting Shortcuts	Chapter 3
3.2	Word Processing Shortcut Keys	Chapter 3
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6.2	Network Topologies	Chapter 6
6.3	URL Structure	Chapter 6
6.4	IP Address Classes	Chapter 6
6.5	Public vs Private IP	Chapter 6
6.6	Internet Connection Modes	Chapter 6
7.1	Email Folders	Chapter 7
7.2	Social Media Platforms	Chapter 7
7.3	Instant Messaging Apps	Chapter 7
7.4	e-Governance Services	Chapter 7
8.1	UPI Apps in India	Chapter 8
8.2	Debit vs Credit Card	Chapter 8
8.3	NEFT vs RTGS vs IMPS	Chapter 8
9.1	5 V's of Big Data	Chapter 9
9.2	Cloud Service Models	Chapter 9
9.3	Types of AI	Chapter 9
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9.2	Cloud Service Models Diagram	Chapter 9
9.3	Blockchain Structure	Chapter 9
9.4	3D Printing Process	Chapter 9

List of Practical Lab Exercises

Exercise No.	Chapter	Exercise Title
1.1	Chapter 1	Identify Hardware Components
1.2	Chapter 1	Check System Properties
2.1	Chapter 2	Explore Taskbar and Desktop
2.2	Chapter 2	Practice Keyboard Shortcuts
2.3	Chapter 2	Change Date/Time and Display Settings
2.4	Chapter 2	File and Folder Operations
3.1	Chapter 3	Create, Save, and Export PDF
3.2	Chapter 3	Text Formatting Practice
3.3	Chapter 3	Create and Format Table
3.4	Chapter 3	Mail Merge Practice
4.1	Chapter 4	Create Spreadsheet with Formulas
4.2	Chapter 4	AutoFill Practice
4.3	Chapter 4	Sorting and Filtering

Exercise No.	Chapter	Exercise Title
4.4	Chapter 4	Create Charts
5.1	Chapter 5	Create Presentation with Template
5.2	Chapter 5	Master Slide Modification
5.3	Chapter 5	Apply Transitions and Timings
5.4	Chapter 5	Print Handouts
6.1	Chapter 6	Find IP Address (ipconfig)
6.2	Chapter 6	Find MAC Address
6.3	Chapter 6	Browser Navigation and Bookmarks
6.4	Chapter 6	Advanced Search Techniques
7.1	Chapter 7	Create Email Account
7.2	Chapter 7	Compose, Reply, Forward Email
7.3	Chapter 7	Attach Files and Set Signature
7.4	Chapter 7	Explore UMANG and DigiLocker
8.1	Chapter 8	Create UPI ID
8.2	Chapter 8	Perform UPI Transaction
8.3	Chapter 8	Explore USSD Banking (*99#)
8.4	Chapter 8	Pay Bill Online
9.1	Chapter 9	Enable Windows Security Settings
9.2	Chapter 9	Configure Smartphone Security

Exercise No.	Chapter	Exercise Title
9.3	Chapter 9	Research Cyber Security Initiatives

Quick Revision Summary

Chapter	Key Topics	No. of MCQs	Lab Exercises
1	Hardware, Software, Memory	15	2
2	OS, File Management, Settings	15	4
3	Word Processing, Tables, Mail Merge	20	4
4	Spreadsheet, Formulas, Charts	20	4
5	Presentation, Transitions, Master Slide	15	4
6	Internet, Networks, Browsers	15	4
7	Email, Social Media, e-Governance	15	4
8	Digital Payments, NEFT, RTGS, IMPS	15	4
9	Futureskills, Cyber Security	10	3

Important Contact Information

Organization	Purpose	Website
NIELIT	CCC Exam Conducting Body	www.nielit.gov.in
NPCI	UPI, BHIM, BBPS	www.npci.org.in
CERT-In	Cyber Security	www.cert-in.org.in
UMANG	Government Services App	www.umang.gov.in
DigiLocker	Digital Documents	www.digilocker.gov.in

Document End Notes

Field	Details
Document Type	CCC Course Study Material
Based on	NIELIT CCC Syllabus (Latest Pattern)
Target	Uttar Pradesh CCC Exam Candidates
Software Covered	Windows 10/11, Ubuntu Linux, LibreOffice, MS Office
Total Pages	[Auto-calculated by PDF software]
Last Updated	March 2026

Note to the Learner:

This document contains complete theory, practical lab exercises, and model questions for all 9 chapters as per the NIELIT CCC syllabus. For best results:

1. Study each chapter thoroughly
2. Practice all lab exercises
3. Attempt all model questions
4. Focus on Ubuntu/LibreOffice as the exam uses them
5. Pay special attention to Digital Financial Tools (Chapter 8) and e-Governance (Chapter 7) as they are frequently asked in UP exams